8	7	6	5	4	3	2	1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
<REV>	<ECN>	<ECO_DESCRIPTION>	<ECODATE>

J92 MLB NEWARK - DVT

11/21/2014

ALIASES RESOLVED

PRODUCT SAFETY REQUIREMENTS:
PCB, UL RECOGNIZED, MIN. 130-C TEMP. RATING AND V-0 FLAME RATING PER UL 796 & UL 94.
PCB TO BE SILK-SCREENED WITH UL/CUL RECOGNITION MARK, MANUFACTURER'S UL FILE NUMBER, UL PCB MATERIAL DESIGNATION, 130-C TEMP. RATING AND V-0 FLAME RATING.

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-00107	1	SCHEM,MLB-NEWARK,J92	SCH	CRITICAL	
820-00045	1	PCBF,MLB-NEWARK,J92	PCB	CRITICAL	

DRAWING TITLE
<PART_DESCRIPTION>

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 1 OF 130

SHEET 1 OF 75

8	7	6	5	4	3	2	1

BOM Groups

BOM GROUP	BOM OPTIONS
MLB_COMMON	ALTERNATE,COMMON,CCSAK,MLB_MISC,MLB_DEBUG:ENG,MLB_PROGPARTS,EQ:4CH
MLB_MISC	CAM_FREQ:24M,CAM_XTAL:NO,EDP,RSMRST:SMC,PGOOD8:SLP_S4,SSD_LPSR:S3
MLB_DEBUG:ENG	XDP_CONN,XDP,HPMBB:SSP0,HPMVBUS:VDET
MLB_DEBUG:PVT	XDP_CONN,XDP,HPMBB:SSP0,HPMVBUS:VDET
MLB_DEBUG:PROD	XDP

CPU DRAM CFG Chart

VENDOR	CFG 1	CFG 0
HYNIX	0	0
SAMSUNG	1	0
MICRON	0	1
ELPIDA	1	1

SIZE	CFG 3	CFG 2
2GB	0	0
4GB QDP	0	1
4GB DDP	1	0
8GB	1	1

CPU DRAM SPD Straps

BOM GROUP	BOM OPTIONS
DRAM:HYN_2GB	RAMCFG0:L,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,DRAM_TYPE:HYNIX_2GB
DRAM:HYN_4GB_QDP	RAMCFG0:L,RAMCFG1:L,RAMCFG2:H,RAMCFG3:L,DRAM_TYPE:HYNIX_4GB_QDP
DRAM:HYN_4GB	RAMCFG0:L,RAMCFG1:L,RAMCFG2:L,RAMCFG3:H,DRAM_TYPE:HYNIX_4GB
DRAM:HYN_8GB	RAMCFG0:L,RAMCFG1:L,RAMCFG2:H,RAMCFG3:H,DRAM_TYPE:HYNIX_8GB
DRAM:ELP_2GB	RAMCFG0:H,RAMCFG1:H,RAMCFG2:L,RAMCFG3:L,DRAM_TYPE:ELPIDA_2GB
DRAM:ELP_4GB_QDP	RAMCFG0:H,RAMCFG1:H,RAMCFG2:H,RAMCFG3:L,DRAM_TYPE:ELPIDA_4GB_QDP
DRAM:ELP_4GB	RAMCFG0:H,RAMCFG1:H,RAMCFG2:L,RAMCFG3:H,DRAM_TYPE:ELPIDA_4GB
DRAM:ELP_8GB	RAMCFG0:H,RAMCFG1:H,RAMCFG2:H,RAMCFG3:H,DRAM_TYPE:ELPIDA_8GB

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4638	1	IC,MCU,LPC11U37,128KB/12KB,TFBGA48	U4600	CRITICAL	HPM:BLANK
341S00190	1	IC,HPM (VXXX) DVT,X261	U4600	CRITICAL	HPM:PROG
338S1231	1	IC,SMC12-B1,40MHZ/50DMIPS MCU,7X7,168BGA	U5000	CRITICAL	SMC:BLANK
341S00031	1	IC,SMC-B1,EXTERNAL (VXXXX) PROTO2A, J92	U5000	CRITICAL	SMC:PROG
335S1009	1	64 MBIT SPI QUAD I/O FLASH,CSP,3.3V,QUAD IO	U6100	CRITICAL	BOOTROM_WIN:BLANK
335S1010	1	64 MBIT SPI QUAD I/O FLASH,CSP,3.3V,QUAD IO	U6100	CRITICAL	BOOTROM_MAC:BLANK
335S1029	1	64 MBIT SPI QUAD I/O FLASH,CSP,3.3V,QUAD IO	U6100	CRITICAL	BOOTROM_MIC:BLANK
341S00191	1	EFI ROM,MLB (VXXX) DVT,X261	U6100	CRITICAL	BOOTROM:PROG
335S0948	1	IC,SPI SERIAL FLASH,64M BITS,1.8V,WLBGA	U8800	CRITICAL	SSDROM_WIN:BLANK
335S0966	1	IC,SPI SERIAL FLASH,64M BITS,1.8V,WLCSP	U8800	CRITICAL	SSDROM_MAC:BLANK
341S00091	1	S1-X SSD BOOTROM, UI64	U8800	CRITICAL	SSDROM:PROG

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00054	1	IC,CPU,BW,QGKZ,QS,E0,2/2,0.8,4.5W,.8,B1234	U0500	CRITICAL	CPU:0.8GHZ
337S00099	1	IC,CPU,BW,QH2Z,PRQ,F0,2/2,1.1,5W,.85,B1234	U0500	CRITICAL	CPU:1.1GHZ
337S00098	1	IC,CPU,BW,QH2V,PRQ,F0,2/2,1.2,5W,.9,B1234	U0500	CRITICAL	CPU:1.2GHZ
337S00097	1	IC,CPU,BW,QH2R,PRQ,F0,2/2,1.3,5W,.9,B1234	U0500	CRITICAL	CPU:1.3GHZ
946-3892	1	J11/J13 MLB DYMAX ADHESIVE 29993-SC 0.4G	GLUE	CRITICAL	
825-7995	1	LABEL,BARCODE,2D,1D,CONFIG,MLB,X261	LABEL		

SSD POP Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
339S0243	1	POP,S1-X CONTROLLER+4GBITS ELP,841B,FCBGA	U8400	CRITICAL	SSDRAM:A0_ELP
339S0244	1	POP,S1-X CONTROLLER+4GBITS HYN,841B,FCBGA	U8400	CRITICAL	SSDRAM:A0_HYN
685-00003	1	POP,MLB,S1X-A1,ELP-4GBIT,X261	U8400	CRITICAL	SSDRAM:A1_ELP
685-00004	1	POP,MLB,S1X-A1,HYN-4GBIT,X261	U8400	CRITICAL	SSDRAM:A1_HYN
338S1288	1	IC,S1-X,CONTROLLER,A1,841(312)B,FCCSP	SSD_CTRL	CRITICAL	S1X:A1
338S00055	1	IC,S1-X,CONTROLLER,A2,841(312)B,FCBGA	SSD_CTRL	CRITICAL	S1X:A2
333S0733	1	IC,LPDDR2,128MX32,1.2V,ELPIDA 28NM,312B	SSD_DRAM	CRITICAL	S1X_DRAM:ELPIDA
333S0694	1	IC,LPDDR2,128MX32,1.2V,HYNIX 29NM,312B	SSD_DRAM	CRITICAL	S1X_DRAM:HYNIX
870-00878	1	TAPE,CONDUCTIVE,SSD,REEL,X261	SSD_TAPE	CRITICAL	

DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
998-6486	2	IC,SDRAM,LPDDR-1600,8GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S00028	2	IC,SDRAM,LPDDR-1600,16GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:HYNIX_4GB_QDP
998-6453	2	IC,SDRAM,LPDDR-1600,8GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:ELPIDA_2GB
998-6454	2	IC,SDRAM,LPDDR-1600,16GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:ELPIDA_4GB_QDP
333S0730	2	IC,SDRAM,LPDDR-1600,16GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0731	2	IC,SDRAM,LPDDR-1600,32GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:HYNIX_8GB
333S0740	2	IC,SDRAM,LPDDR-1600,16GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0741	2	IC,SDRAM,LPDDR-1600,32GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:ELPIDA_8GB

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S00074	376S0855		ALL	Toshiba alt for Diodes dual
376S1129	376S0855		ALL	NXP alt for Diodes dual
376S1089	376S1128		ALL	NXP alt for Diodes single
138S0703	138S0648		ALL	Murata alt to Taiyo Yuden
128S0393	128S0334		ALL	Kemet alt to Sanyo
372S0186	372S0185		ALL	NXP alt to Diodes
376S1053	376S0604		ALL	Diodes alt to Fairchild
197S0590	197S0588		ALL	Epson alt to TXC
197S0591	197S0588		ALL	NDK alt to TXC
107S0249	107S0251		ALL	TFT alt to Cyntec
138S0789	138S0941		ALL	Samsung alt to Murata
333S0700	333S0704		ALL	Hynix CAM DRAM alt to Elpida
333S00016	333S0704		ALL	Elpida new die CAM DRAM alt
333S00030	333S0704		ALL	Hynix new die CAM DRAM alt
339S0243	339S0244		ALL	Elpida SSD DRAM alt to Hynix
335S0948	335S0966		ALL	Winbond alt to Macronix
128S0631	128S0351		ALL	NEC alt to Sanyo
128S00008	128S0380		ALL	NEC alt to Sanyo
311S00007	311S0426		ALL	Sanyo alt to NEC
311S00008	311S0271		ALL	Samsung alt to Murata
311S00018	311S0409		ALL	Diodes alt to NXP
740S00004	740S0134		ALL	Kemet alt to Sanyo
740S00005	740S0190		ALL	Polytronics alt to Wayon
128S0296	128S0487		ALL	Sanyo alt to NEC
128S00012	128S0487		ALL	ROHM alt to NEC
128S00025	128S0469		ALL	Kemet alt to Sanyo
128S0374	128S0469		ALL	NEC alt to Sanyo
376S00007	376S1179		ALL	AOS alt to Vishay
376S1080	376S0820		ALL	Diodes alt to OnSemi
376S00036	376S1194		ALL	Vishay alt to OnSemi
376S00037	376S1193		ALL	Vishay alt to OnSemi
685-00004	685-00003	SSDRAM:A1_ELP	ALL	Hynix SSD DRAM alt to Elpida
337S00061	337S00054	CPU:0.8GHZ	ALL	2.4G turbo CPU alt to 2.0G

SYNC MASTER=J43 MLB SYNC DATE=10/24/2012

PAGE TITLE

BOM Configuration

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Top level BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-6568	PCBA,MLB,1.1GHZ,EL 8GB,SAND 256G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6569	PCBA,MLB,1.1GHZ,EL 8GB,SAND 512G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6570	PCBA,MLB,1.1GHZ,EL 8GB,TOSH 256G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6571	PCBA,MLB,1.1GHZ,EL 8GB,TOSH 512G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6572	PCBA,MLB,1.2GHZ,EL 8GB,SAND 256G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6573	PCBA,MLB,1.2GHZ,EL 8GB,SAND 512G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6574	PCBA,MLB,1.2GHZ,EL 8GB,TOSH 256G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6575	PCBA,MLB,1.2GHZ,EL 8GB,TOSH 512G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6576	PCBA,MLB,1.3GHZ,EL 8GB,SAND 256G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6577	PCBA,MLB,1.3GHZ,EL 8GB,SAND 512G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6578	PCBA,MLB,1.3GHZ,EL 8GB,TOSH 256G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6579	PCBA,MLB,1.3GHZ,EL 8GB,TOSH 512G,WIFI FCC,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP
639-6580	PCBA,MLB,1.1GHZ,EL 8GB,SAND 256G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6581	PCBA,MLB,1.1GHZ,EL 8GB,SAND 512G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6582	PCBA,MLB,1.1GHZ,EL 8GB,TOSH 256G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6583	PCBA,MLB,1.1GHZ,EL 8GB,TOSH 512G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6584	PCBA,MLB,1.2GHZ,EL 8GB,SAND 256G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6585	PCBA,MLB,1.2GHZ,EL 8GB,SAND 512G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6586	PCBA,MLB,1.2GHZ,EL 8GB,TOSH 256G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6587	PCBA,MLB,1.2GHZ,EL 8GB,TOSH 512G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6588	PCBA,MLB,1.3GHZ,EL 8GB,SAND 256G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6589	PCBA,MLB,1.3GHZ,EL 8GB,SAND 512G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6590	PCBA,MLB,1.3GHZ,EL 8GB,TOSH 256G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6591	PCBA,MLB,1.3GHZ,EL 8GB,TOSH 512G,WIFI ETSI,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:ETSI,SSDRAM:A1_ELP
639-6592	PCBA,MLB,1.1GHZ,EL 8GB,SAND 256G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6593	PCBA,MLB,1.1GHZ,EL 8GB,SAND 512G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6594	PCBA,MLB,1.1GHZ,EL 8GB,TOSH 256G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6595	PCBA,MLB,1.1GHZ,EL 8GB,TOSH 512G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6596	PCBA,MLB,1.2GHZ,EL 8GB,SAND 256G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6597	PCBA,MLB,1.2GHZ,EL 8GB,SAND 512G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6598	PCBA,MLB,1.2GHZ,EL 8GB,TOSH 256G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6599	PCBA,MLB,1.2GHZ,EL 8GB,TOSH 512G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6600	PCBA,MLB,1.3GHZ,EL 8GB,SAND 256G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6601	PCBA,MLB,1.3GHZ,EL 8GB,SAND 512G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6602	PCBA,MLB,1.3GHZ,EL 8GB,TOSH 256G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6603	PCBA,MLB,1.3GHZ,EL 8GB,TOSH 512G,WIFI APAC,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:APAC,SSDRAM:A1_ELP
639-6604	PCBA,MLB,1.1GHZ,EL 8GB,SAND 256G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6605	PCBA,MLB,1.1GHZ,EL 8GB,SAND 512G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6606	PCBA,MLB,1.1GHZ,EL 8GB,TOSH 256G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6607	PCBA,MLB,1.1GHZ,EL 8GB,TOSH 512G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.1GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6608	PCBA,MLB,1.2GHZ,EL 8GB,SAND 256G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6609	PCBA,MLB,1.2GHZ,EL 8GB,SAND 512G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6610	PCBA,MLB,1.2GHZ,EL 8GB,TOSH 256G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6611	PCBA,MLB,1.2GHZ,EL 8GB,TOSH 512G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.2GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6612	PCBA,MLB,1.3GHZ,EL 8GB,SAND 256G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:SAND_256GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6613	PCBA,MLB,1.3GHZ,EL 8GB,SAND 512G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:SAND_512GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6614	PCBA,MLB,1.3GHZ,EL 8GB,TOSH 256G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP
639-6615	PCBA,MLB,1.3GHZ,EL 8GB,TOSH 512G,WIFI IND,J92	ALTERNATE,CMN,CPU:1.3GHZ,DRAM:ELP_8GB,NAND:TOSH_512GB_1Y_128GBIT,WIFI:IND,SSDRAM:A1_ELP

Partial & development BOMs

BOM NUMBER	BOM NAME	BOM OPTIONS
685-00014	CMN PTS,PCBA,MLB-NEWARK,J92	MLB_COMMON
685-00003	POP,MLB,S1X-A2,ELP-4GBIT,X261	S1X:A2,S1X_DRAM:ELPIDA
685-00004	POP,MLB,S1X-A2,HYN-4GBIT,X261	S1X:A2,S1X_DRAM:HYNIX
939-00043	PCBA,MLB,NO CPU,EL 8GB,TOSH 256G,WIFI FCC,X261	ALTERNATE,CMN,DRAM:ELP_8GB,NAND:TOSH_256GB_1Y_128GBIT,WIFI:FCC,SSDRAM:A1_ELP

BOM Groups

BOM GROUP	BOM OPTIONS
MLB_PROGPARTS	BOOTROM:PROG,BT:PROG,SMC:PROG,SSDROM:PROG,HPM:PROG

Common BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-00014	1	CMN PTS,PCBA,MLB-NEWARK,J92	CMNPTS	CRITICAL	CMN

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S00196	1	BT ROM (VXX) DVT,2MBIT,X261	U3570	CRITICAL	BT:PROG
341S00197	1	WIFI ROM (PXXXX) DVT,WW1,X261	U3580	CRITICAL	WIFI:FCC
341S00198	1	WIFI ROM (PXXXX) DVT,WW2,X261	U3580	CRITICAL	WIFI:ETSI
341S00199	1	WIFI ROM (PXXXX) DVT,WW3,X261	U3580	CRITICAL	WIFI:APAC
341S00200	1	WIFI ROM (PXXXX) DVT,IND,X261	U3580	CRITICAL	WIFI:IND

SYNC MASTER=J43 MLB SYNC DATE=10/24/2012

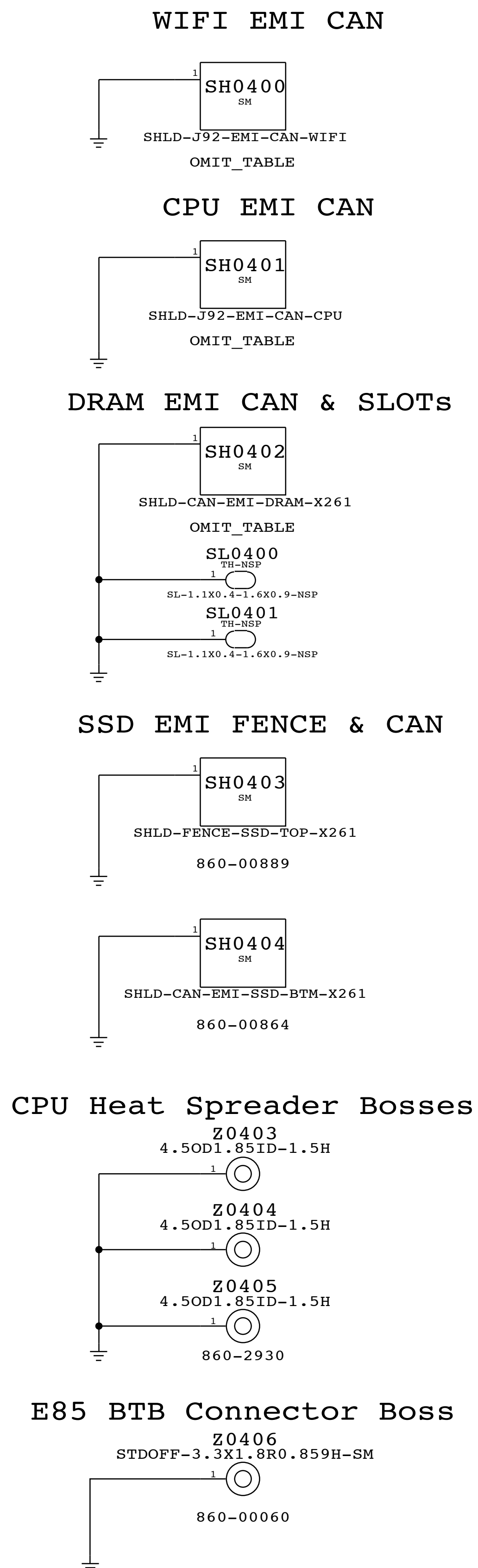
PAGE TITLE
J92 BOM Variants

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-7064	1	CAN,EMI,WIFI,X261	SH0400	CRITICAL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-00112	1	CAN,EMI,CPU,X261	SH0401	CRITICAL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-00400	1	CAN,EMI,DRAM,TALL,X261	SH0402	CRITICAL	

SYNC_MASTER=J43_MLB SYNC_DATE=10/24/2012

PAGE TITLE

PD PARTS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

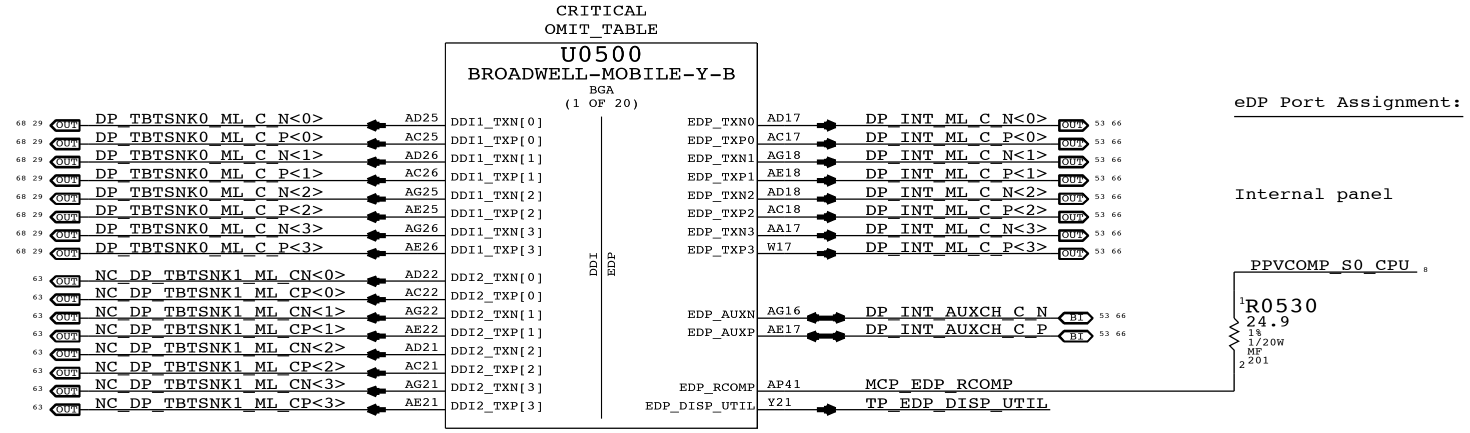
DDI Port Assignments:

TBT Sink 0

TBT Sink 1
(MUXed with HDMI
if necessary)

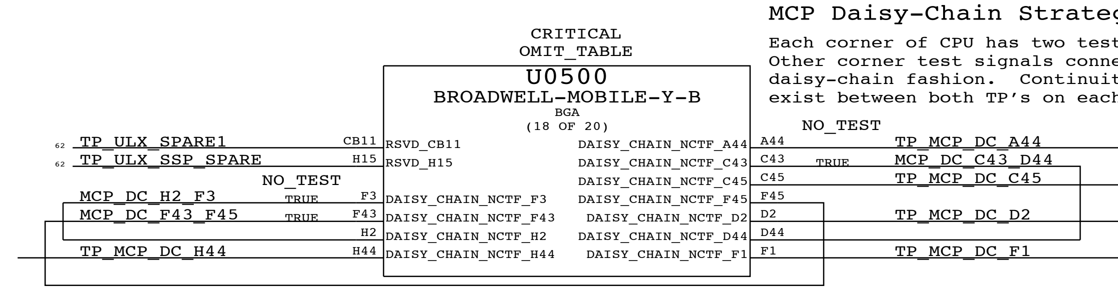
eDP Port Assignment:

Internal panel



MCP Daisy-Chain Strategy:

Each corner of CPU has two testpoints.
Other corner test signals connected in
daisy-chain fashion. Continuity should
exist between both TP's on each corner.



SYNC_MASTER=J92_WILL SYNC_DATE=04/10/2013

PAGE TITLE

CPU GFX/DC_TEST

Apple Inc.

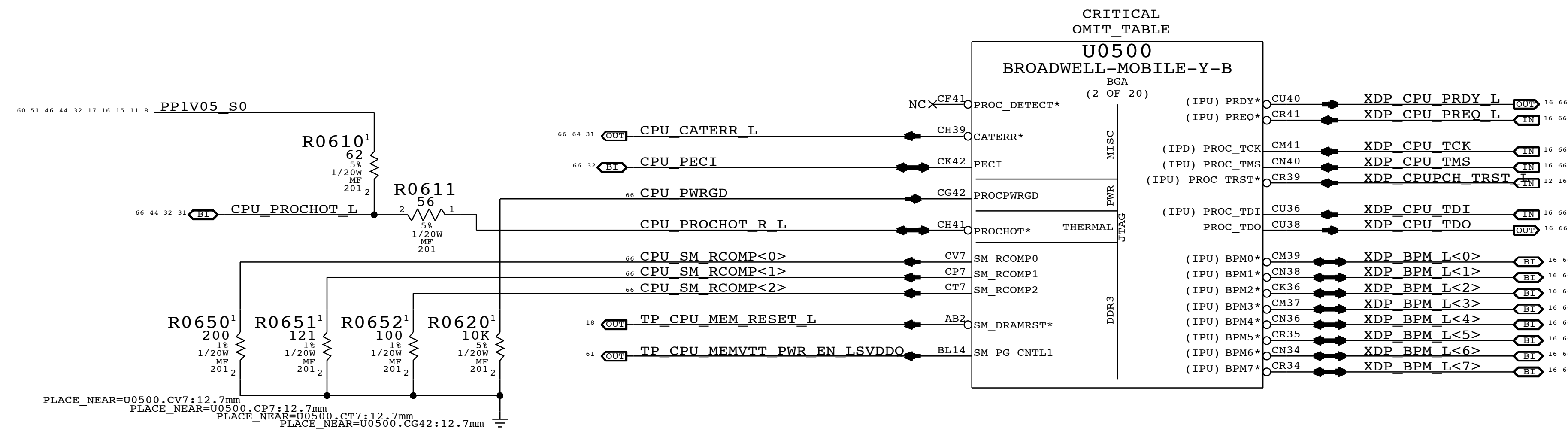
DRAWING NUMBER <SCH_NUM> SIZE D

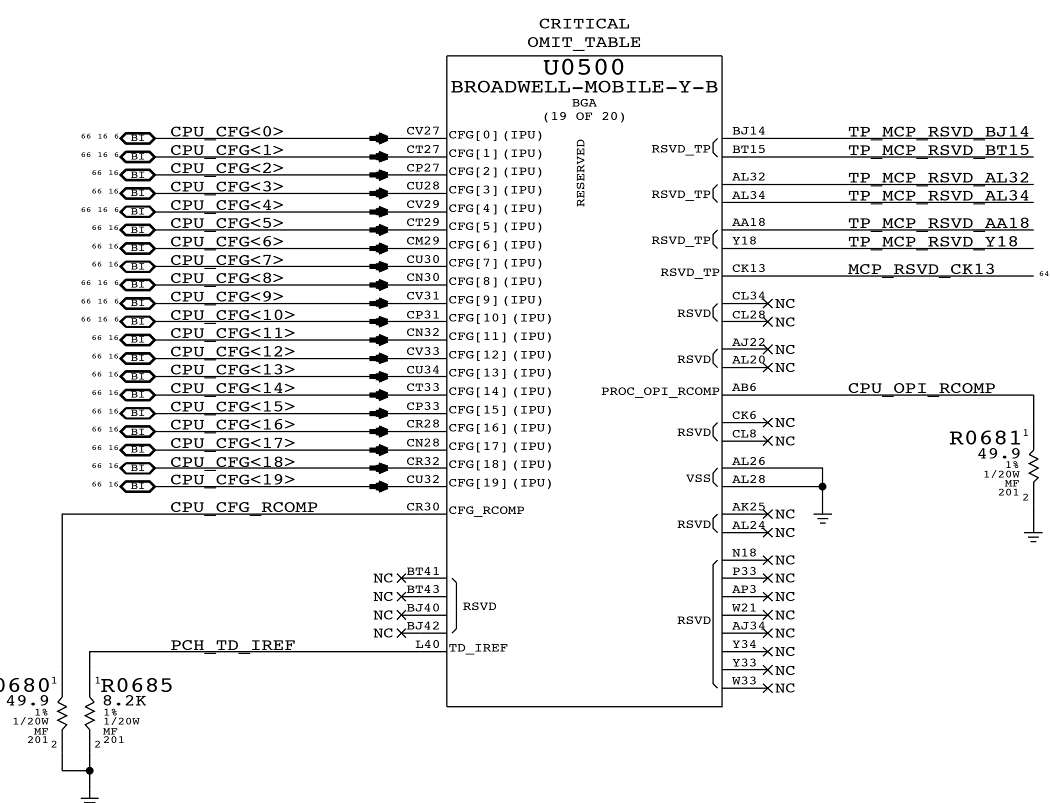
REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 5 OF 130

SHEET 5 OF 75

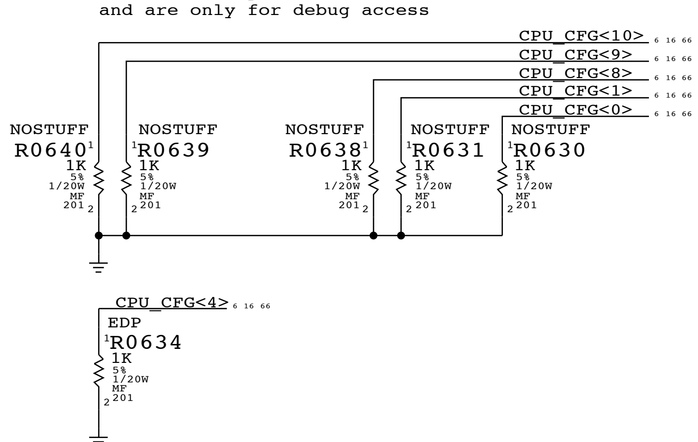




TBD: Confirm w/ Intel which still apply for BDW-Y

CFG<10>:SAFE MODE BOOT	1 = NORMAL OPERATION	0 = POWER FEATURES NOT ACTIVE
CFG<9> :NO SVID-CAPABLE VR	1 = VR SUPPORTS SVID	0 = VR DOES NOT SUPPORT SVID
CFG<8> :ALLOW NOA ON LOCKED UNITS	1 = NORMAL OPERATION	0 = NOA ALWAYS UNLOCKED
CFG<4> :eDP ENABLE/DISABLE	1 = DISABLED	0 = ENABLED
CFG<1> :PCH-LESS MODE	1 = NORMAL OPERATION	0 = PCH-LESS MODE
CFG<0> :RESET SEQUENCE STALL	1 = NORMAL OPERATION	0 = STALL AFTER PCU PLL LOCK

These can be placed close to J1800
and are only for debug access



SYNC_MASTER=J92_WILL SYNC_DATE=04/10/2013

PAGE TITLE
CPU Misc/JTAG/CFG/RSVD

Apple Inc.

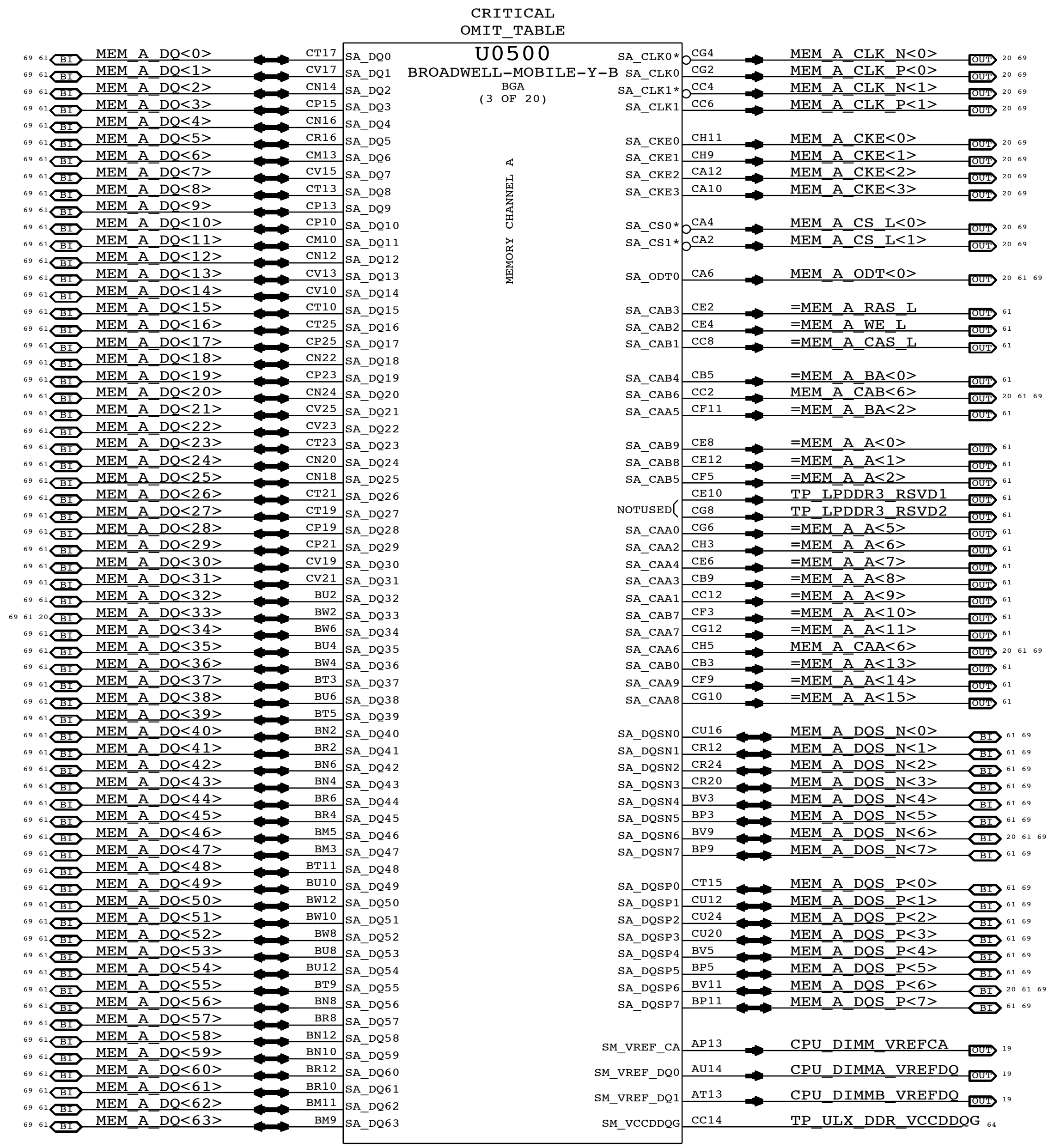
DRAWING NUMBER <SCH_NUM> SIZE D

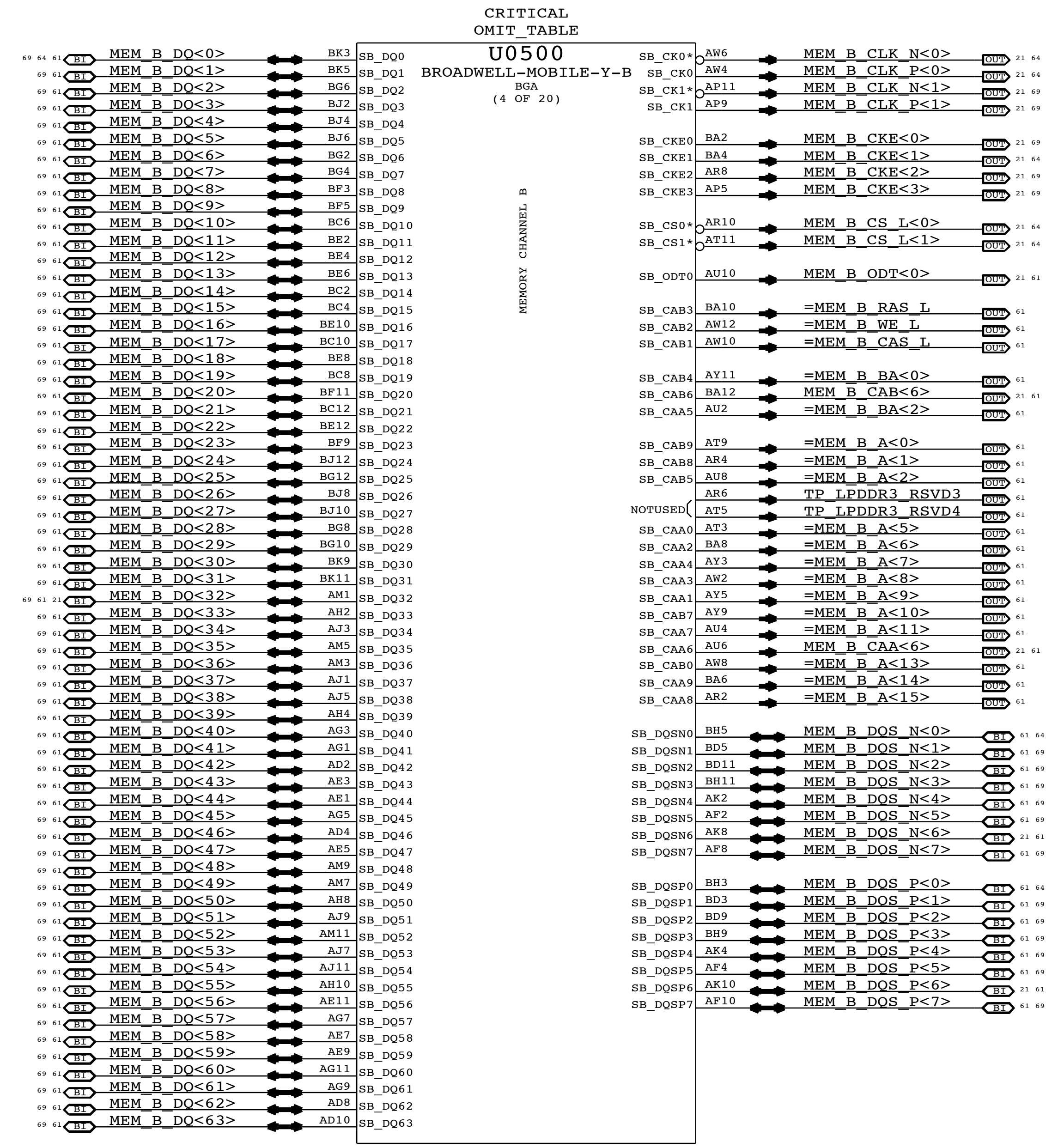
REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 6 OF 130

SHEET 6 OF 75





SYNC_MASTER=J92_LS_MLB SYNC_DATE=07/17/2013

PAGE TITLE

CPU DDR3/LPDDR3 Interfaces

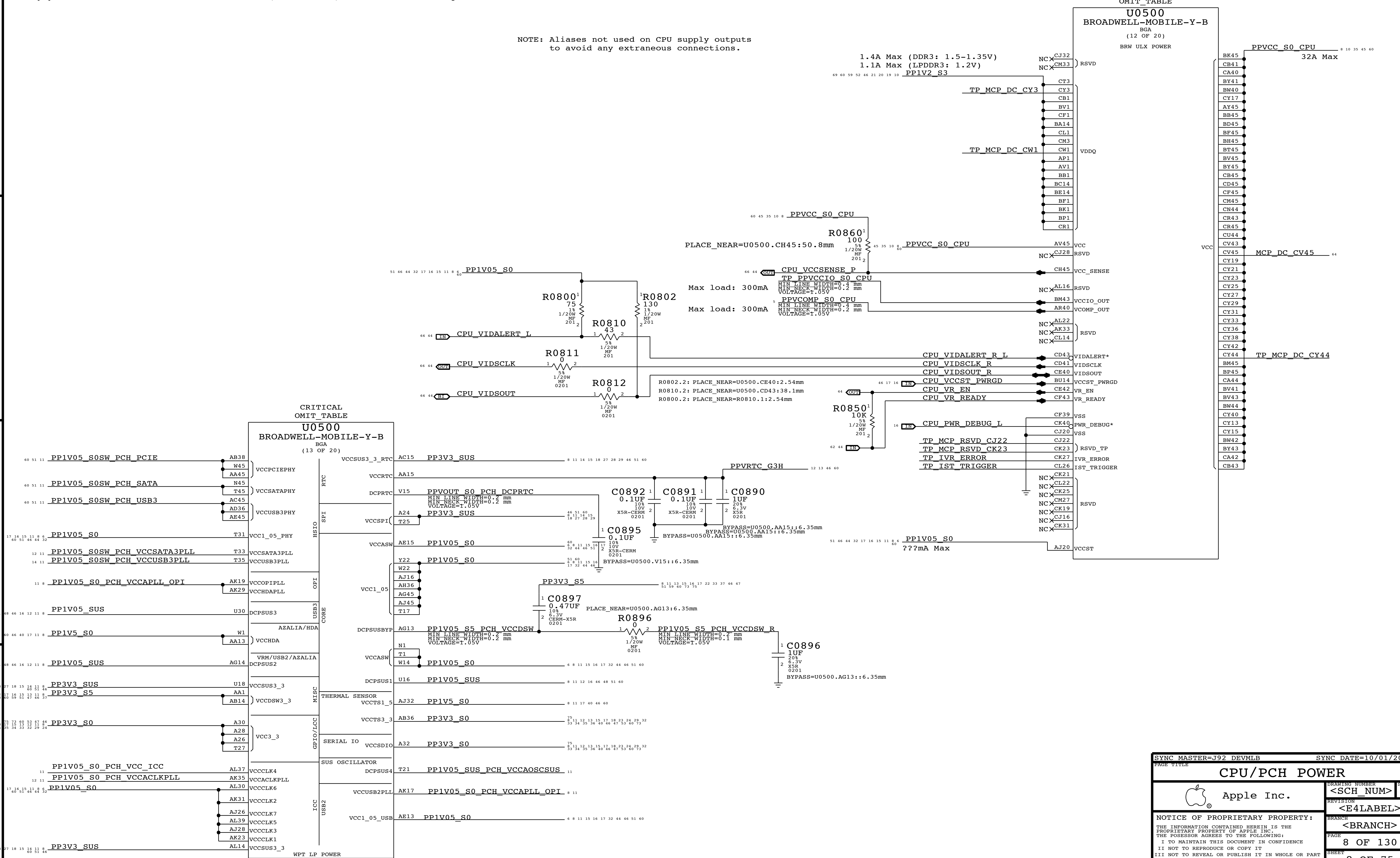
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

HSW-ULT current estimates from Haswell Mobile ULT Processor EDS vol 1, doc #502406, v0.9.
LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0.
Note [1] current numbers from clarification email, from Srini, dated 9/10/2012 2:11pm.



SYNC_MASTER=J92_DEVMLB SYNC_DATE=10/01/2013

PAGE TITLE

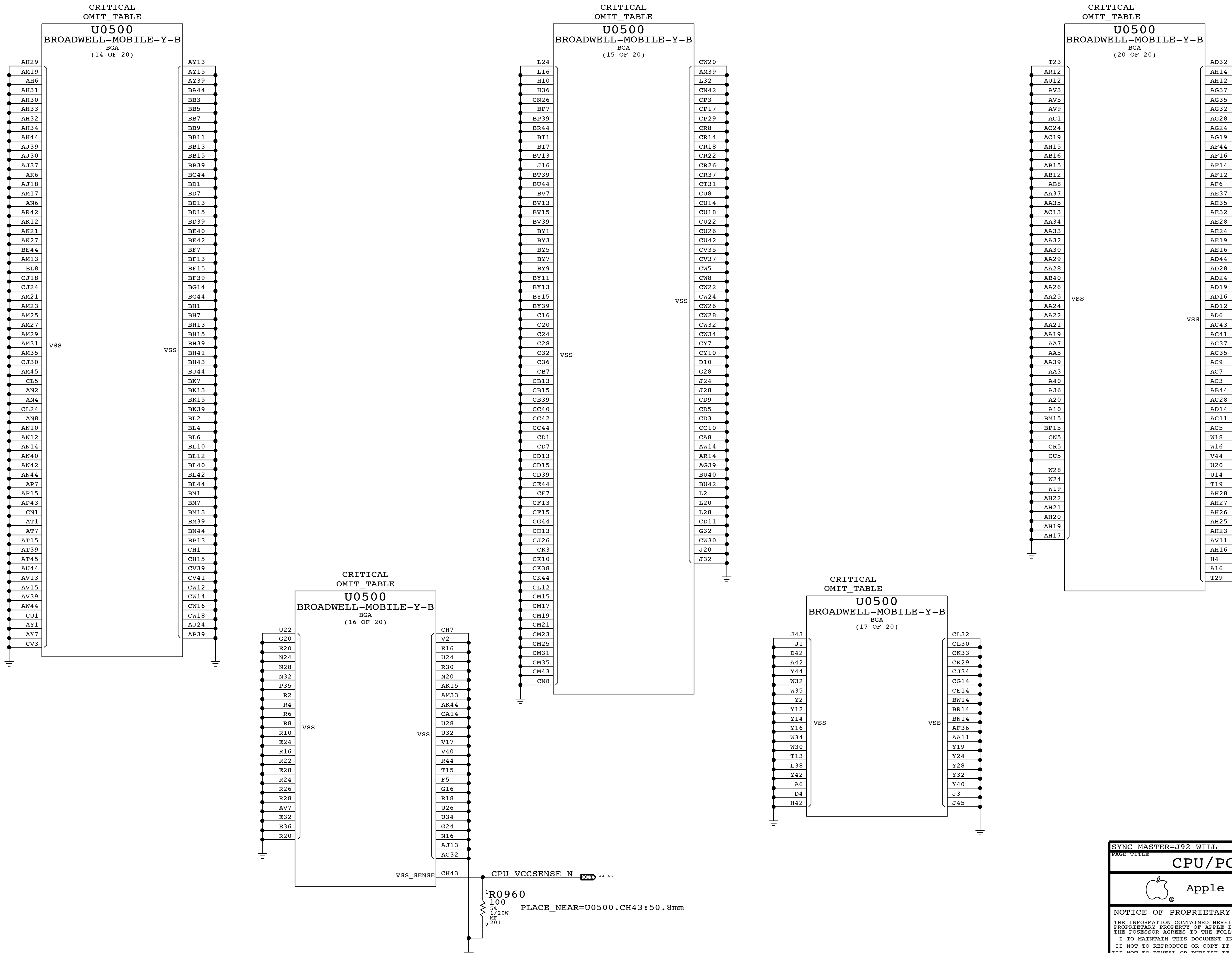
CPU/PCH POWER

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_WILL SYNC_DATE=04/10/2013

PAGE TITLE

CPU/PCH GROUNDS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

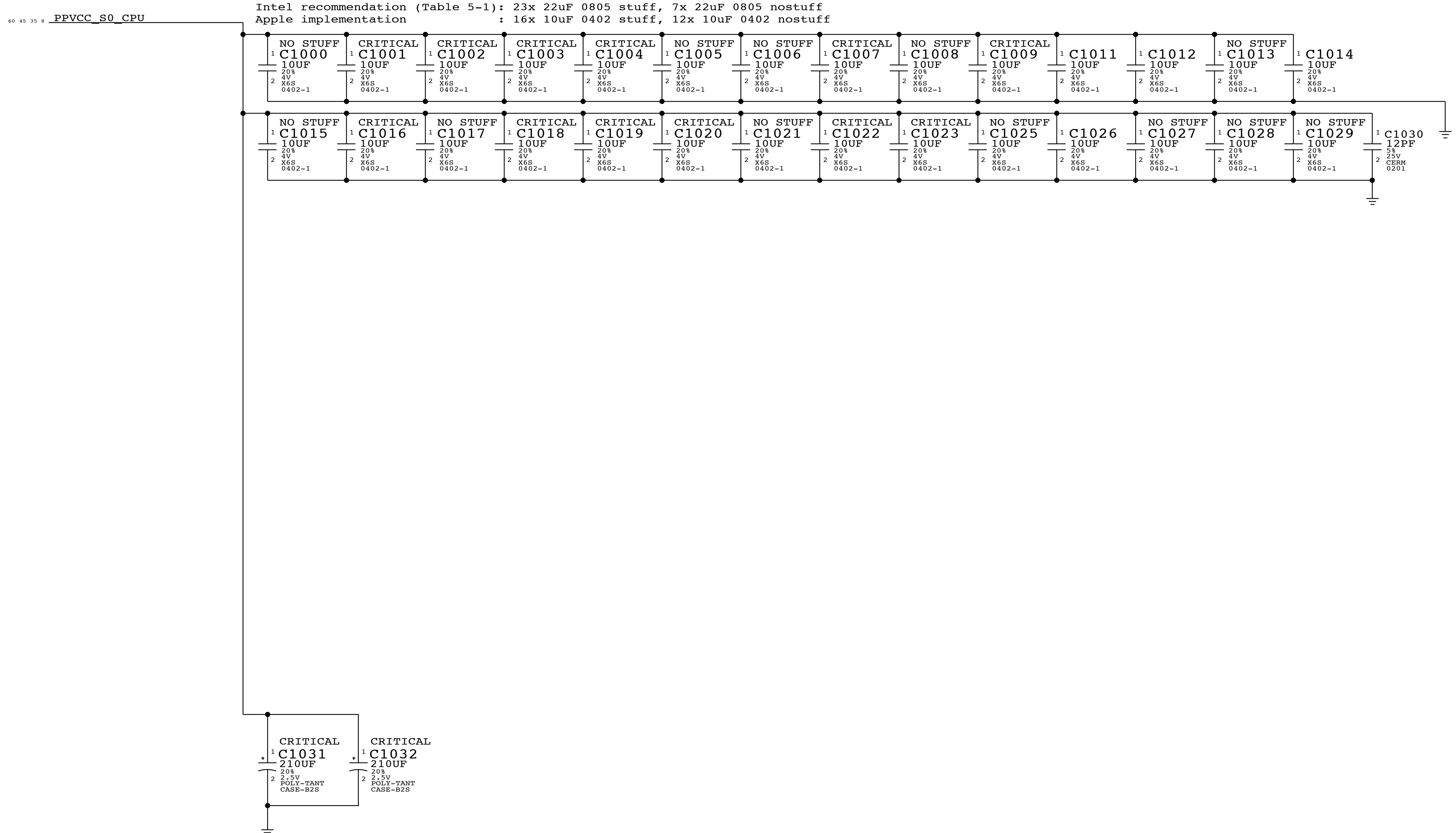
REVISION <E4LABEL>

BRANCH <BRANCH>

All Intel recommendations from Intel doc #503160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 0.9 unless stated otherwise

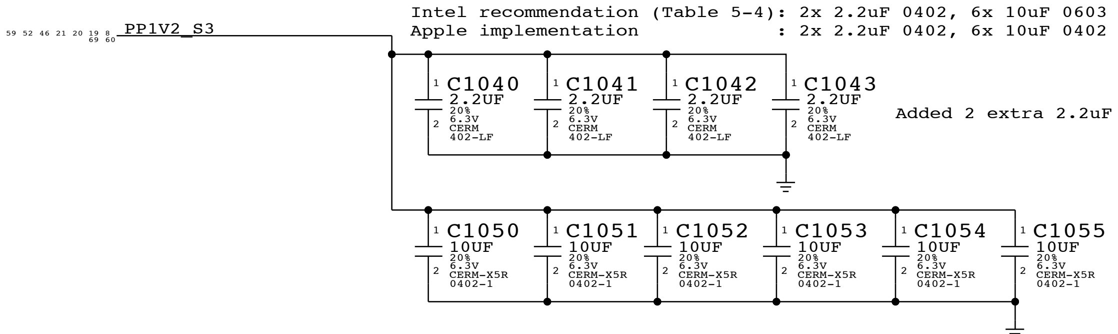
CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff
Apple implementation : 16x 10uF 0402 stuff, 12x 10uF 0402 nostuff



CPU VDDQ DECOUPLING

Intel recommendation (Table 5-4): 2x 2.2uF 0402, 6x 10uF 0603
Apple implementation : 2x 2.2uF 0402, 6x 10uF 0402



Added 2 extra 2.2uF per Harris Beach v0.9 schematic

SYNC_MASTER=J92_DEVMLB SYNC_DATE=10/01/2013

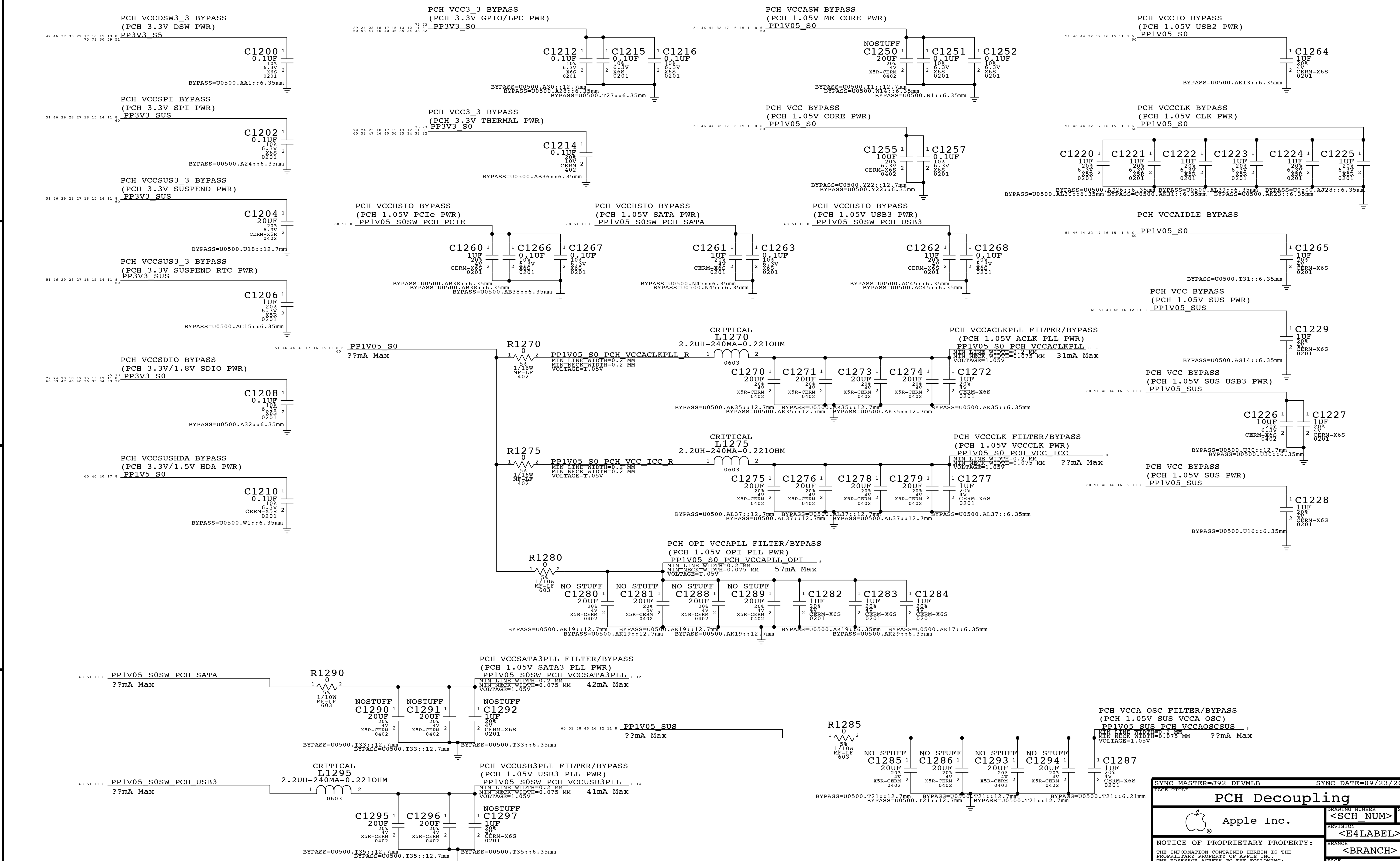
PAGE TITLE: CPU Decoupling

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>



LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0
as well as from clarification email, from Srini, dated 9/10/2012 2:11pm.

SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/23/2013

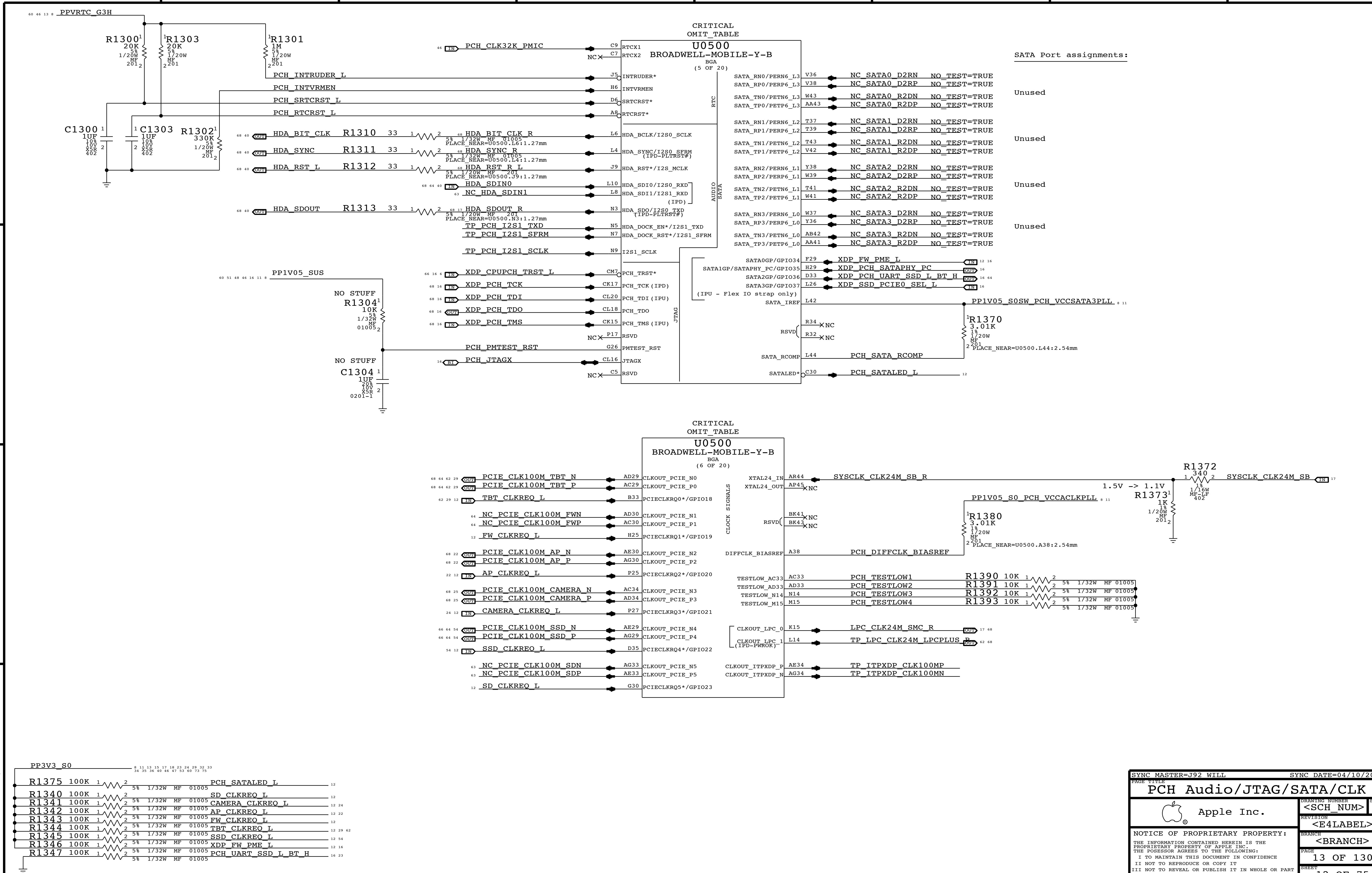
PAGE TITLE
PCH Decoupling

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_WILL SYNC_DATE=04/10/2013

PAGE TITLE

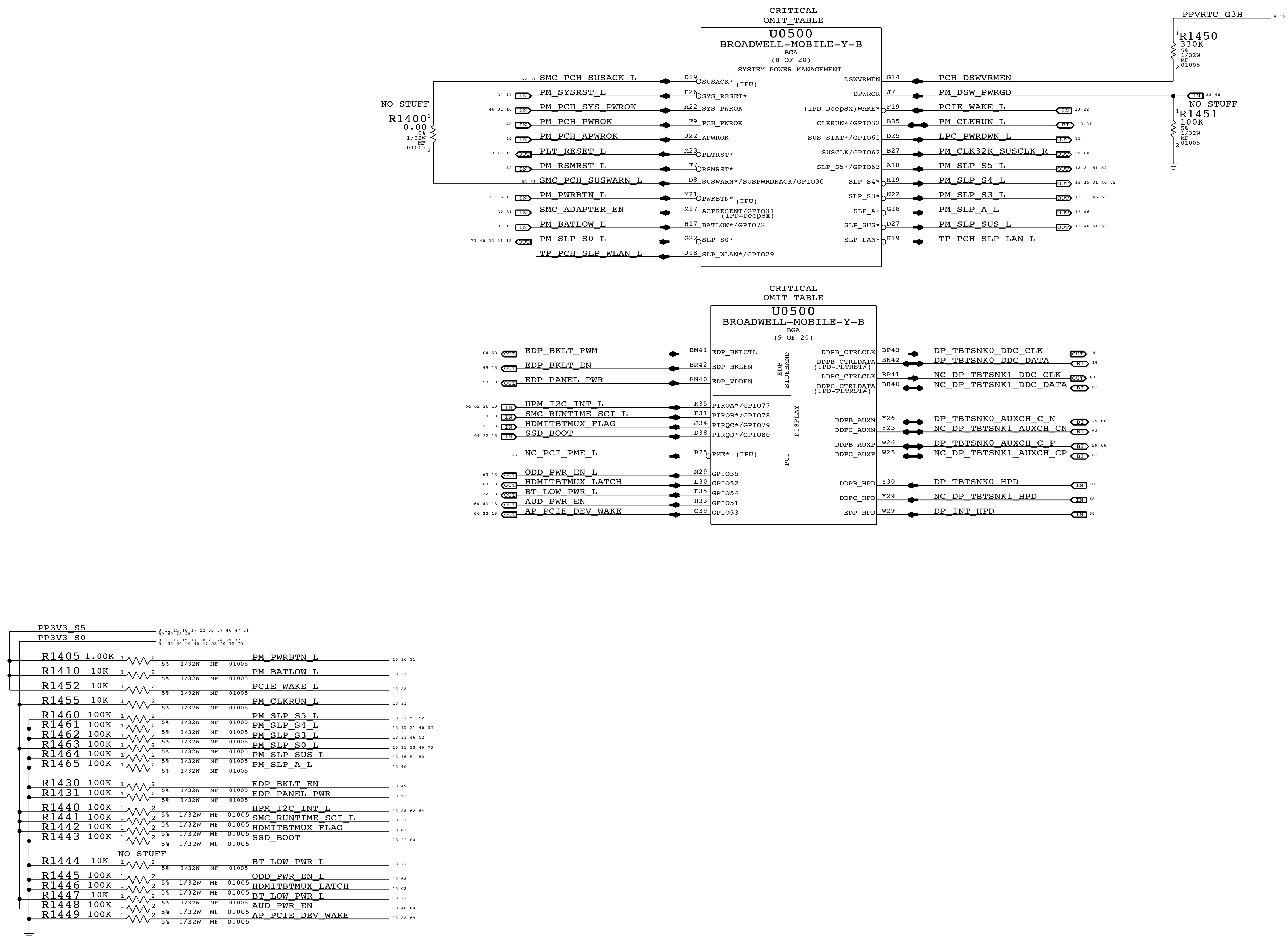
PCH Audio/JTAG/SATA/CLK

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_WILL SYNC_DATE=04/10/2013

PAGE TITLE

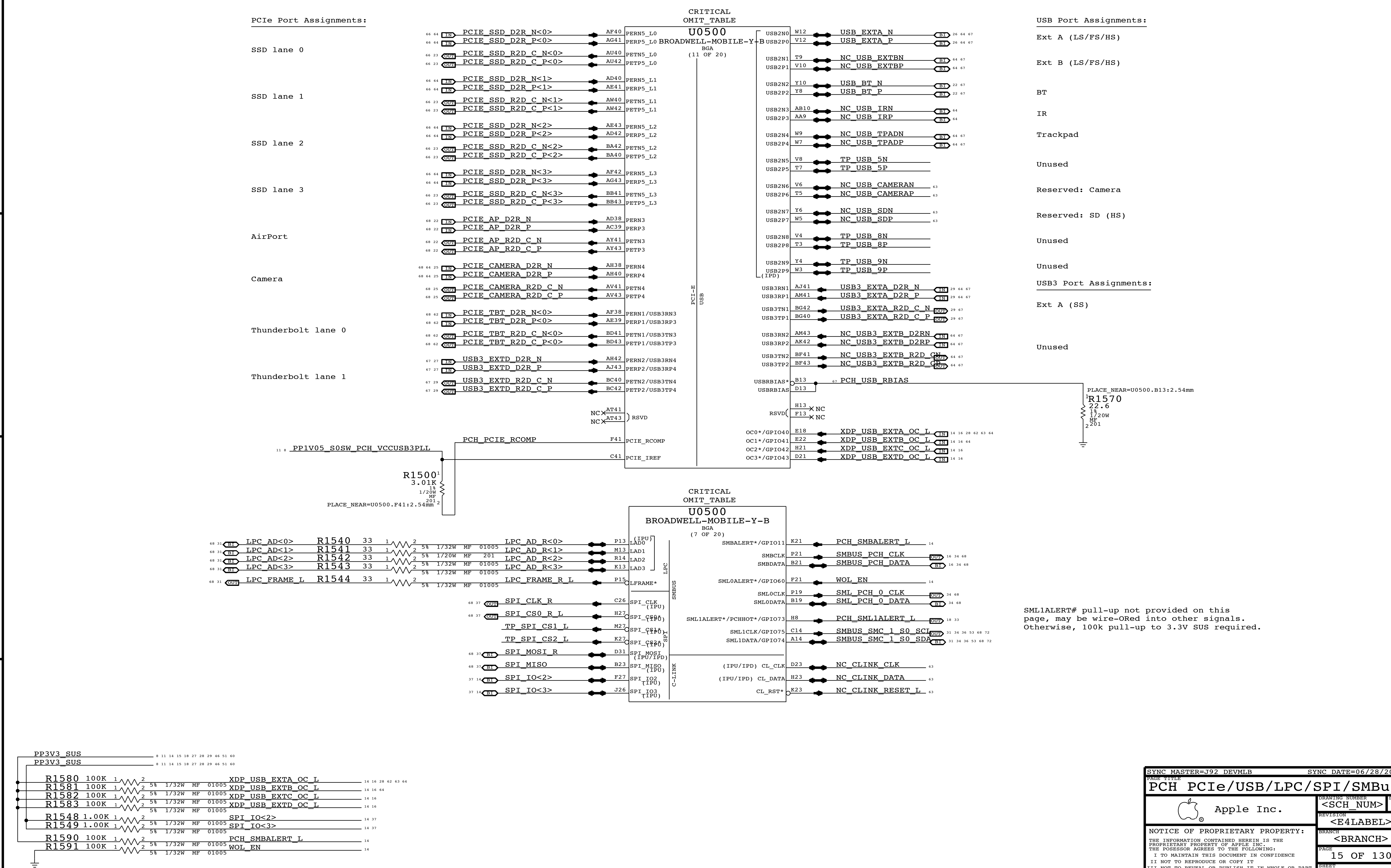
PCH PM/PCI/GFX

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_DEVMLB SYNC_DATE=06/28/2013

PAGE TITLE

PCH PCIe/USB/LPC/SPI/SMBus

Apple Inc.

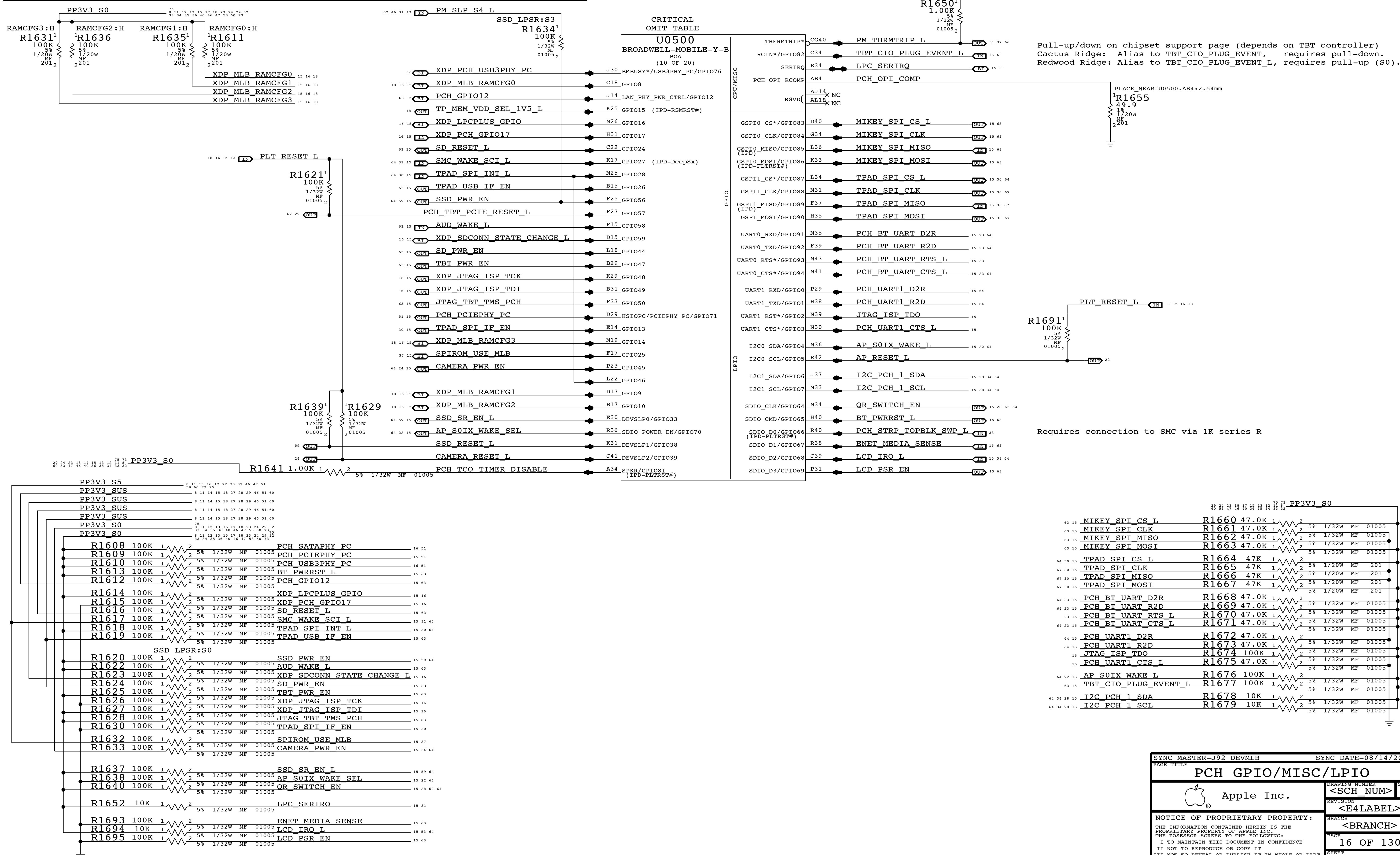
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG3:H,RAMCFG2:H,RAMCFG1:H,RAMCFG0:H

Pull-up/down on chipset support page (depends on TBT controller)
Cactus Ridge: Alias to TBT_CIO_PLUG_EVENT, requires pull-down.
Redwood Ridge: Alias to TBT_CIO_PLUG_EVENT_L, requires pull-up (S0).



SYNC_MASTER=J92_DEVMLB SYNC_DATE=08/14/2013

PAGE TITLE

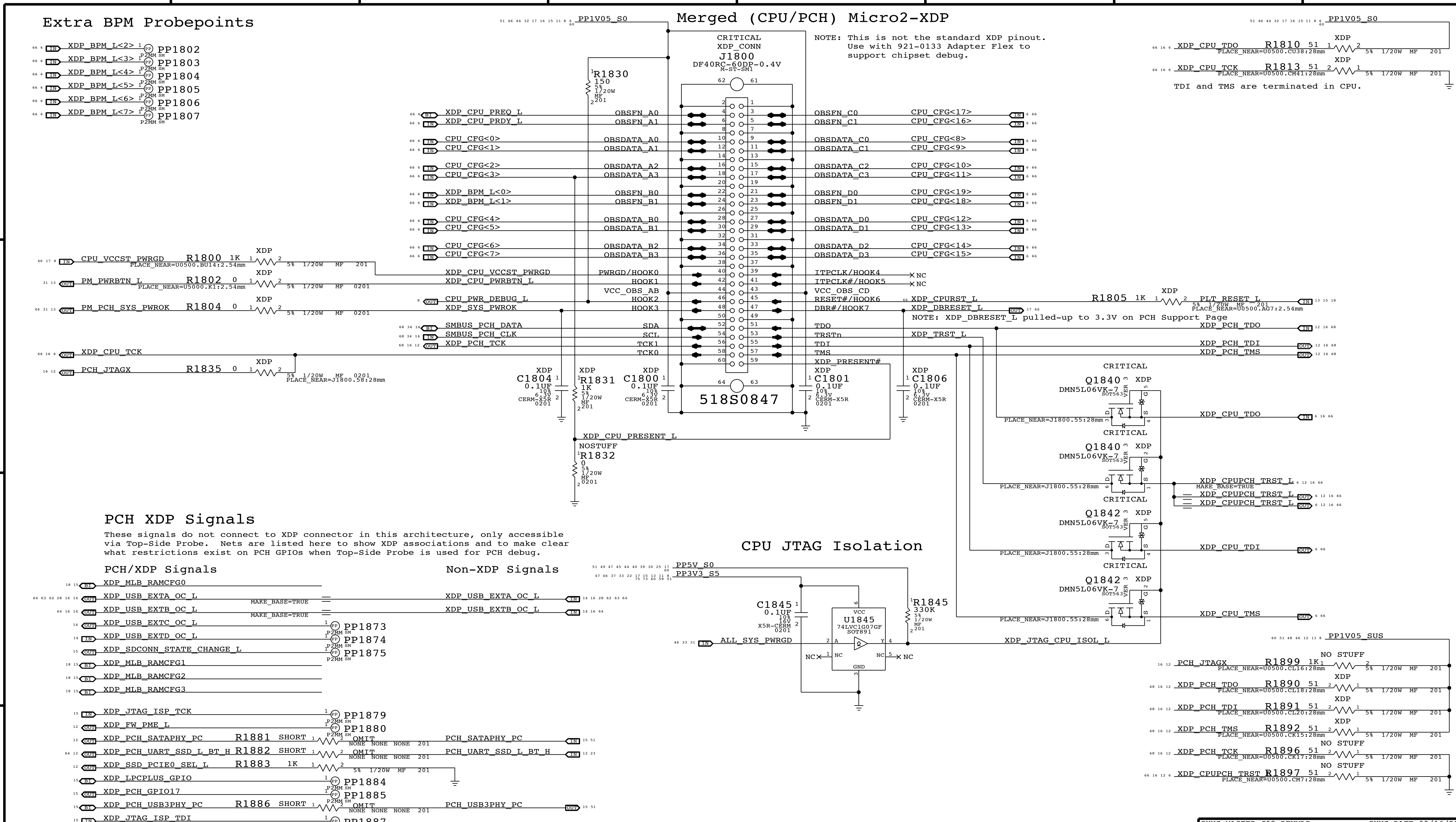
PCH GPIO/MISC/LPIO

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



Unused & MLB_RAMCFGx GPIOs have TPs.

USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.

SDCONN_STATE_CHANGE_L is aliased, do not plug/unplug SD Cards during PCH debug.

JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.

NOTE: Should force PCH GPIO47 high to ensure TBT router powered to avoid leakage/clamping of signals.

SSD_PCIEx_SEL_L straps are connected via 1K to common net.

LPCPLUS_GPIO is aliased, do not attempt use during PCH debug.

SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/16/2013

PAGE TITLE

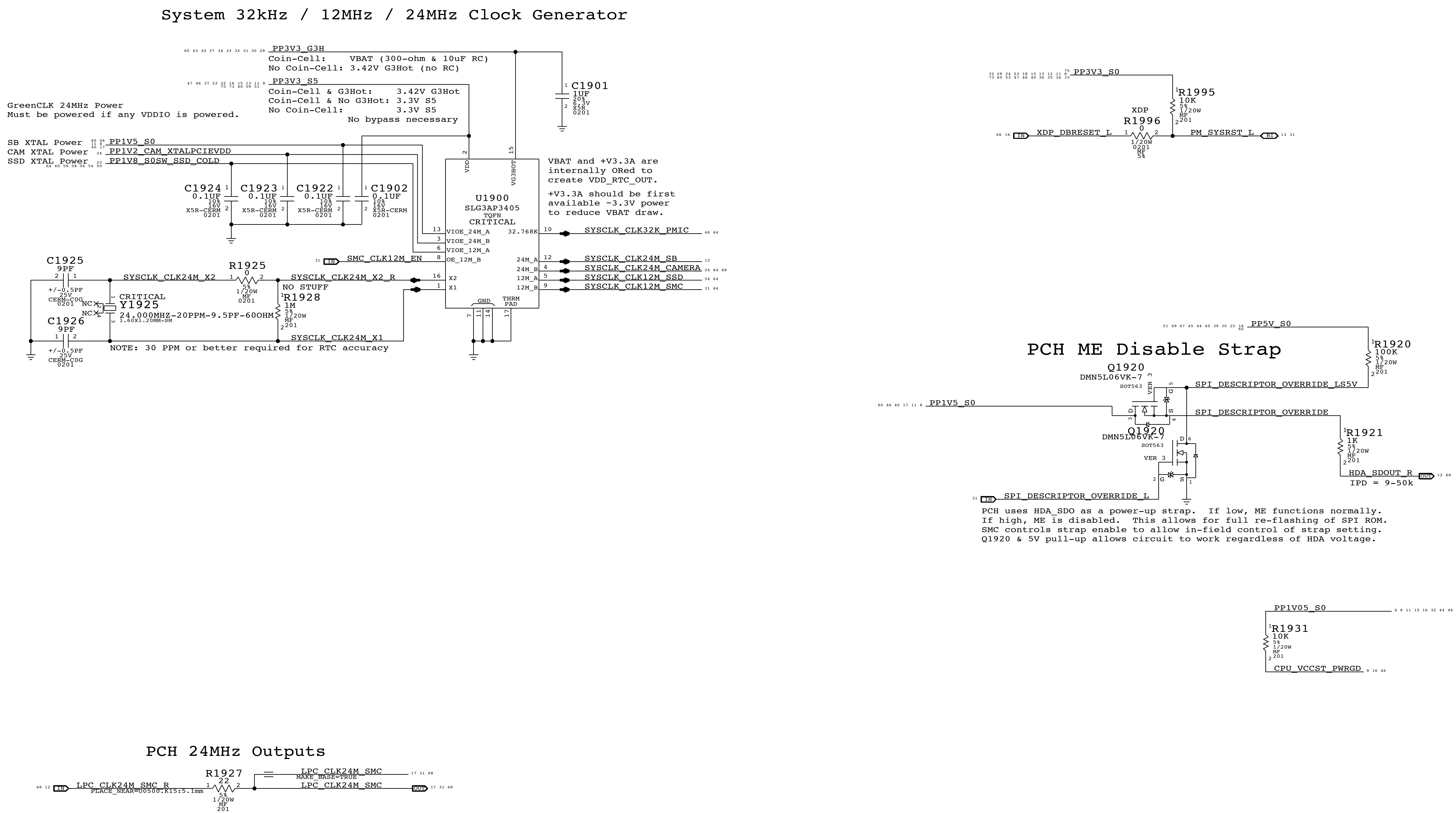
CPU/PCH Merged XDP

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_DEVMLB SYNC_DATE=06/28/2013

PAGE TITLE

Chipset Support

DRAWING NUMBER <SCH_NUM> SIZE D

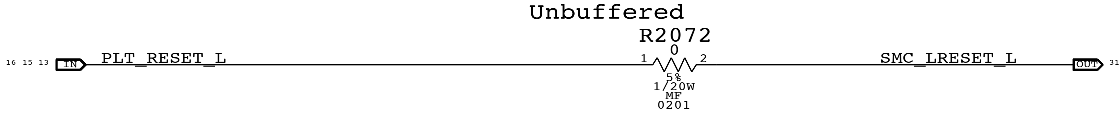
Apple Inc.

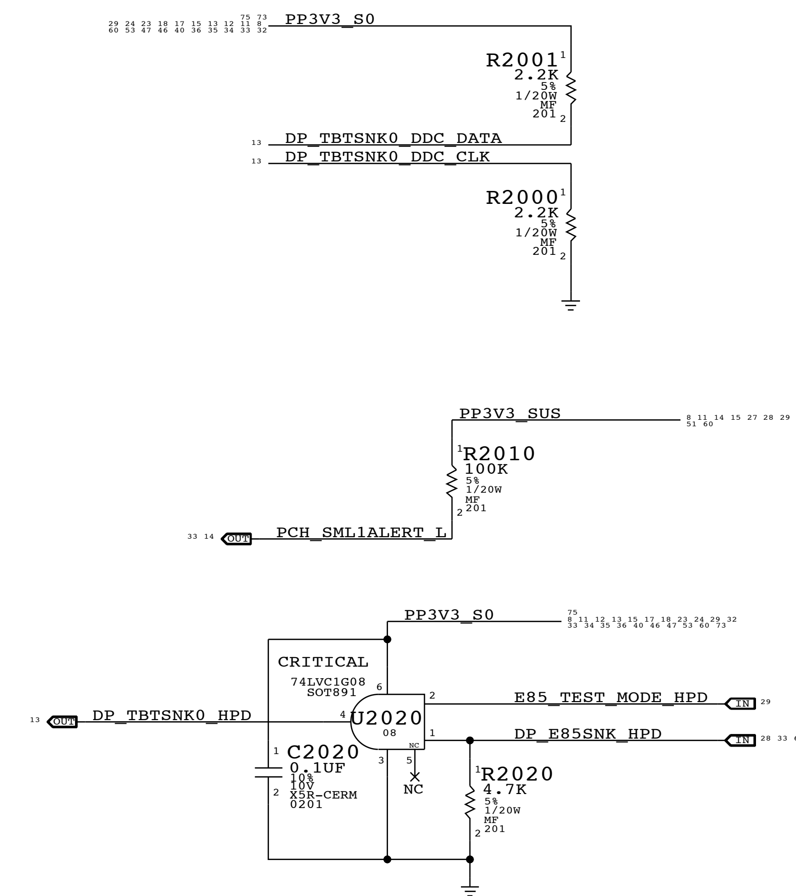
REVISION <E4LABEL>

BRANCH <BRANCH>

Platform Reset Connections

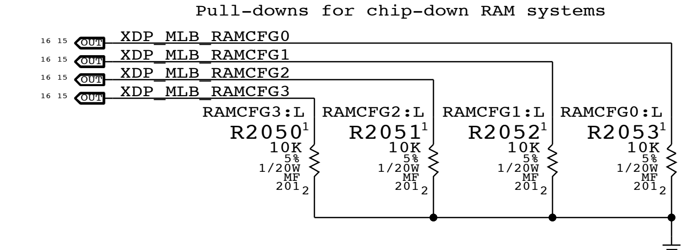
Unbuffered



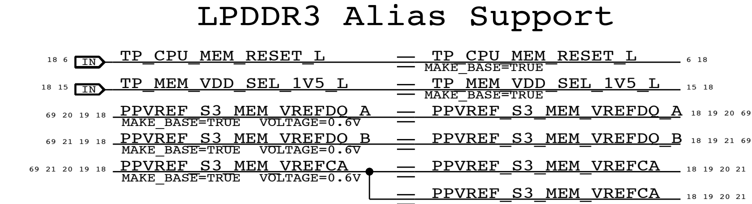


RAM Configuration Straps

Pull-downs for chip-down RAM systems



LPDDR3 Alias Support



SYNC_MASTER=J92_DEVMLB SYNC_DATE=08/01/2013

PAGE TITLE

Project Chipset Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

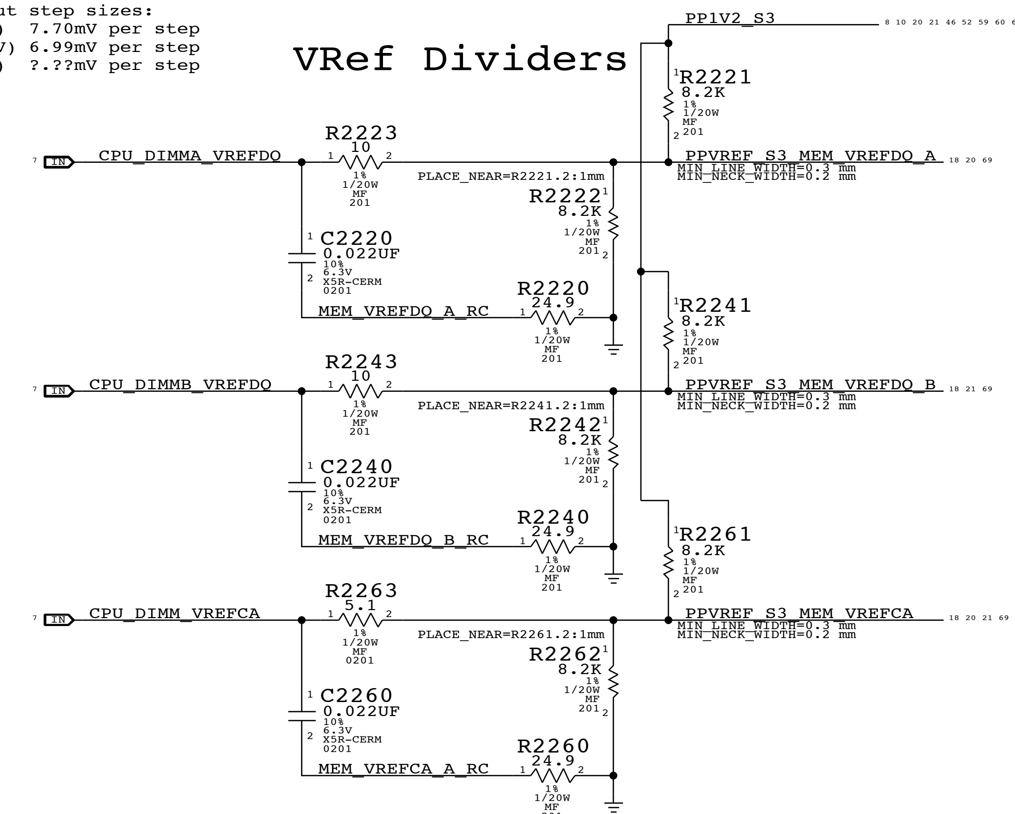
REVISION <E4LABEL>

BRANCH <BRANCH>

CPU-Based Margining

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step
LPDDR3 (1.2V) ?.??mV per step

VRef Dividers



SYNC_MASTER=J92_DEVMLB SYNC_DATE=06/28/2013

PAGE TITLE: LPDDR3 VREF MARGINING

Apple Inc.

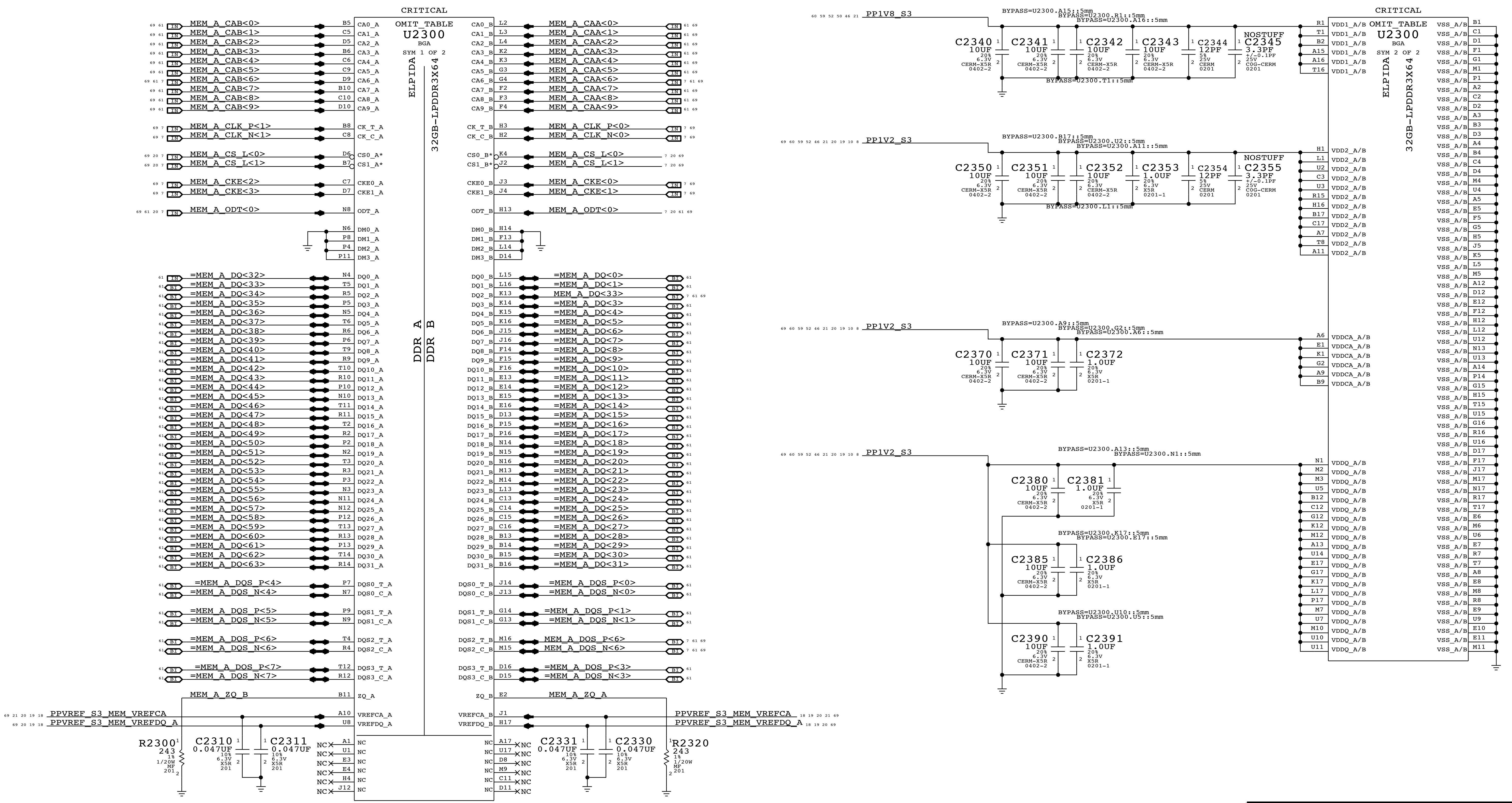
DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

PAGE: 22 OF 130

SHEET: 19 OF 75



SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE

LPDDR3 DRAM Channel A (0-63)

Apple Inc.

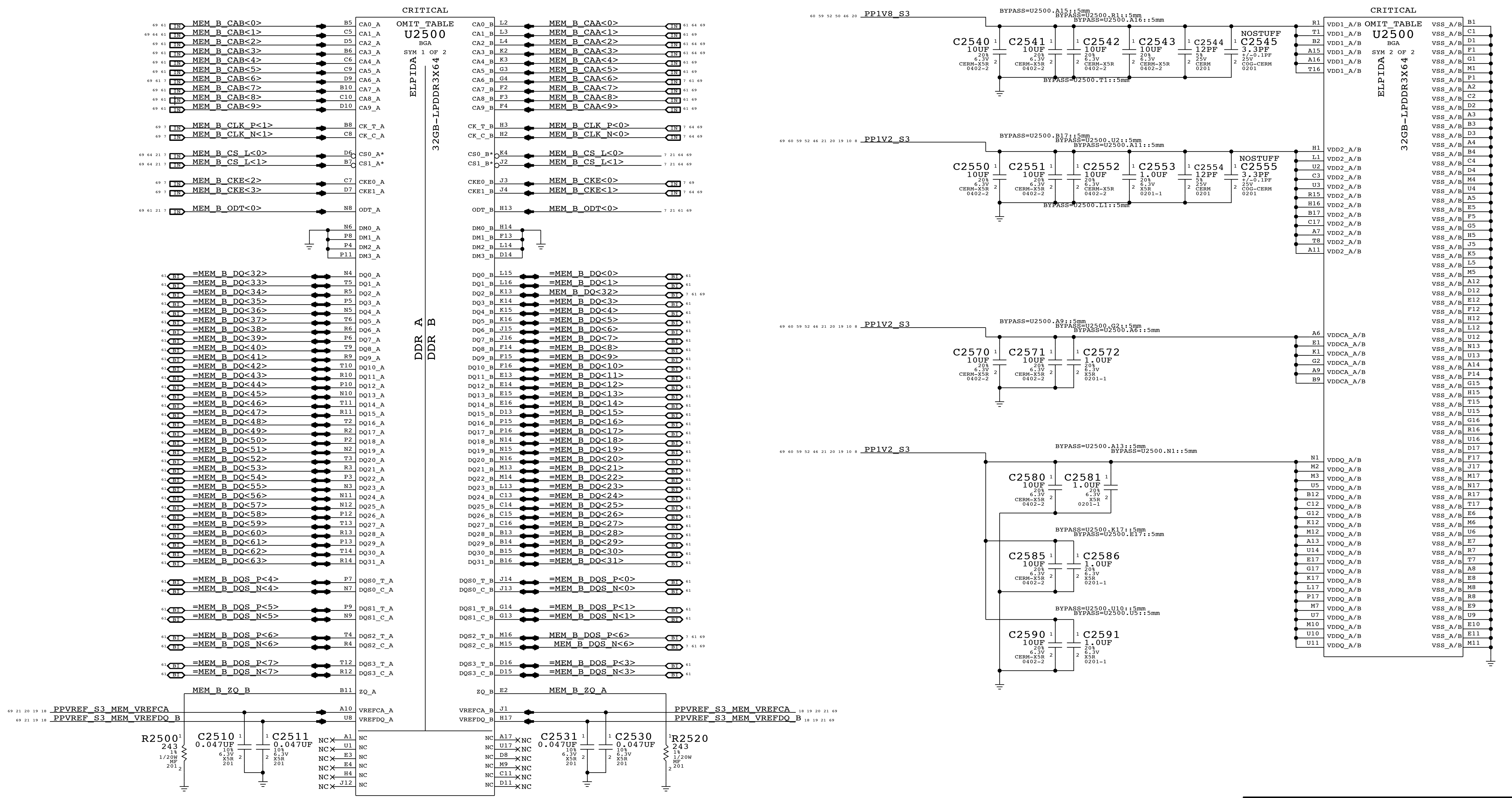
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 23 OF 130

SHEET 20 OF 75



SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE

LPDDR3 DRAM Channel B (0-63)

Apple Inc.

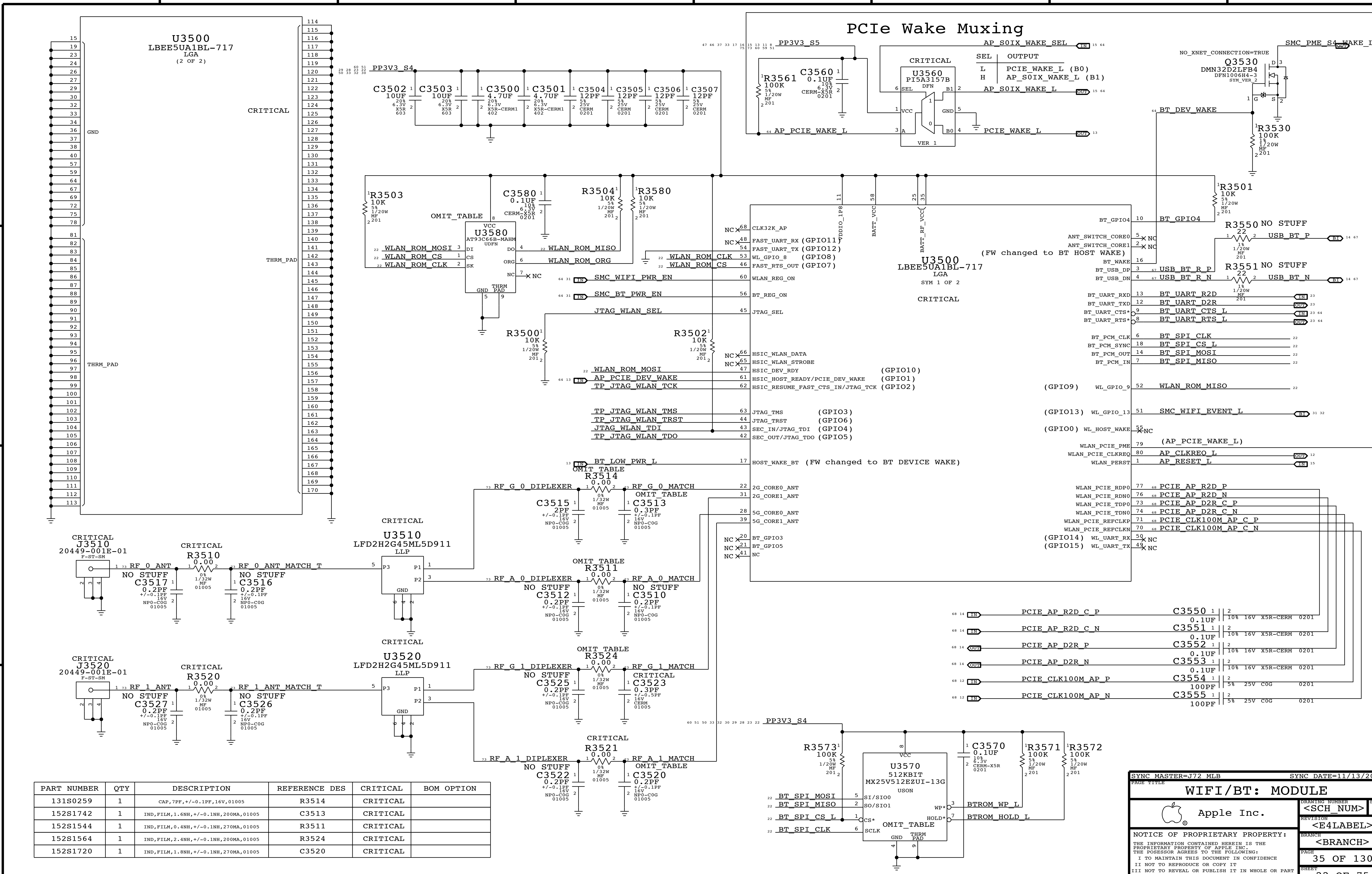
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 25 OF 130

SHEET 21 OF 75



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
131S0259	1	CAP,7PF,+/-0.1PF,16V,01005	R3514	CRITICAL	
152S1742	1	IND,FILM,1.6NH,+/-0.1NH,200MA,01005	C3513	CRITICAL	
152S1544	1	IND,FILM,0.4NH,+/-0.1NH,270MA,01005	R3511	CRITICAL	
152S1564	1	IND,FILM,2.4NH,+/-0.1NH,200MA,01005	R3524	CRITICAL	
152S1720	1	IND,FILM,1.8NH,+/-0.1NH,270MA,01005	C3520	CRITICAL	

SYNC_MASTER=J72_MLB SYNC_DATE=11/13/2012

PAGE TITLE

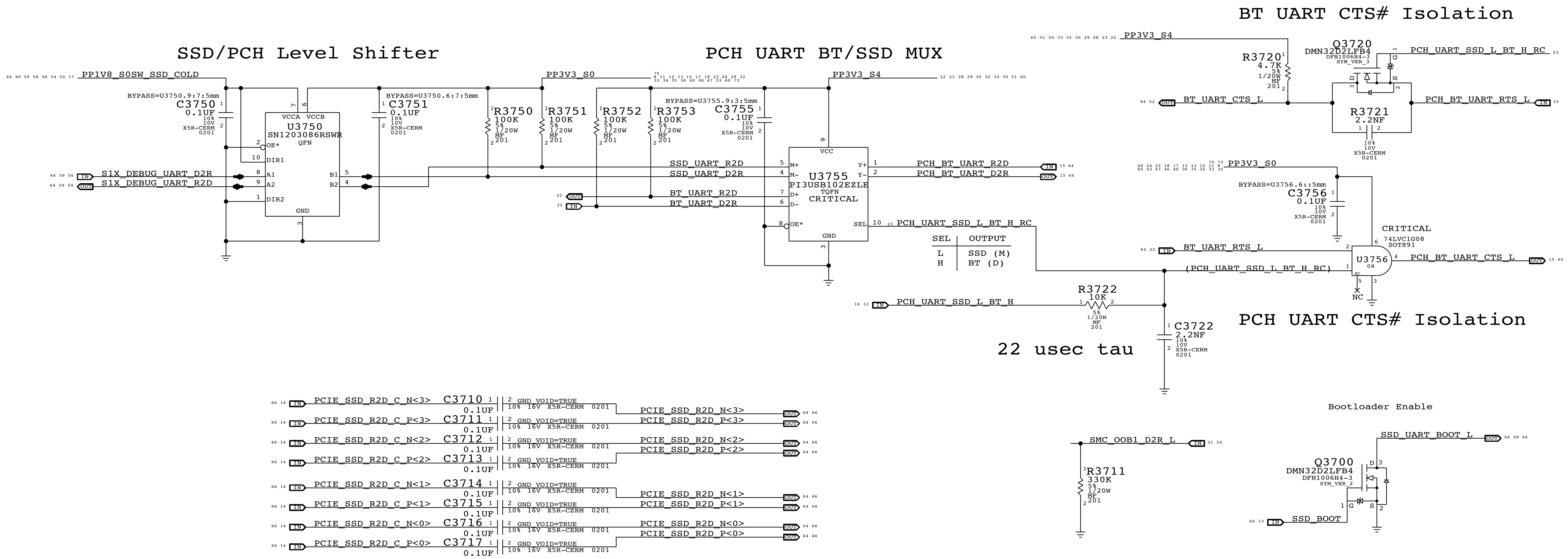
WIFI/BT: MODULE

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/11/2013

PAGE TITLE

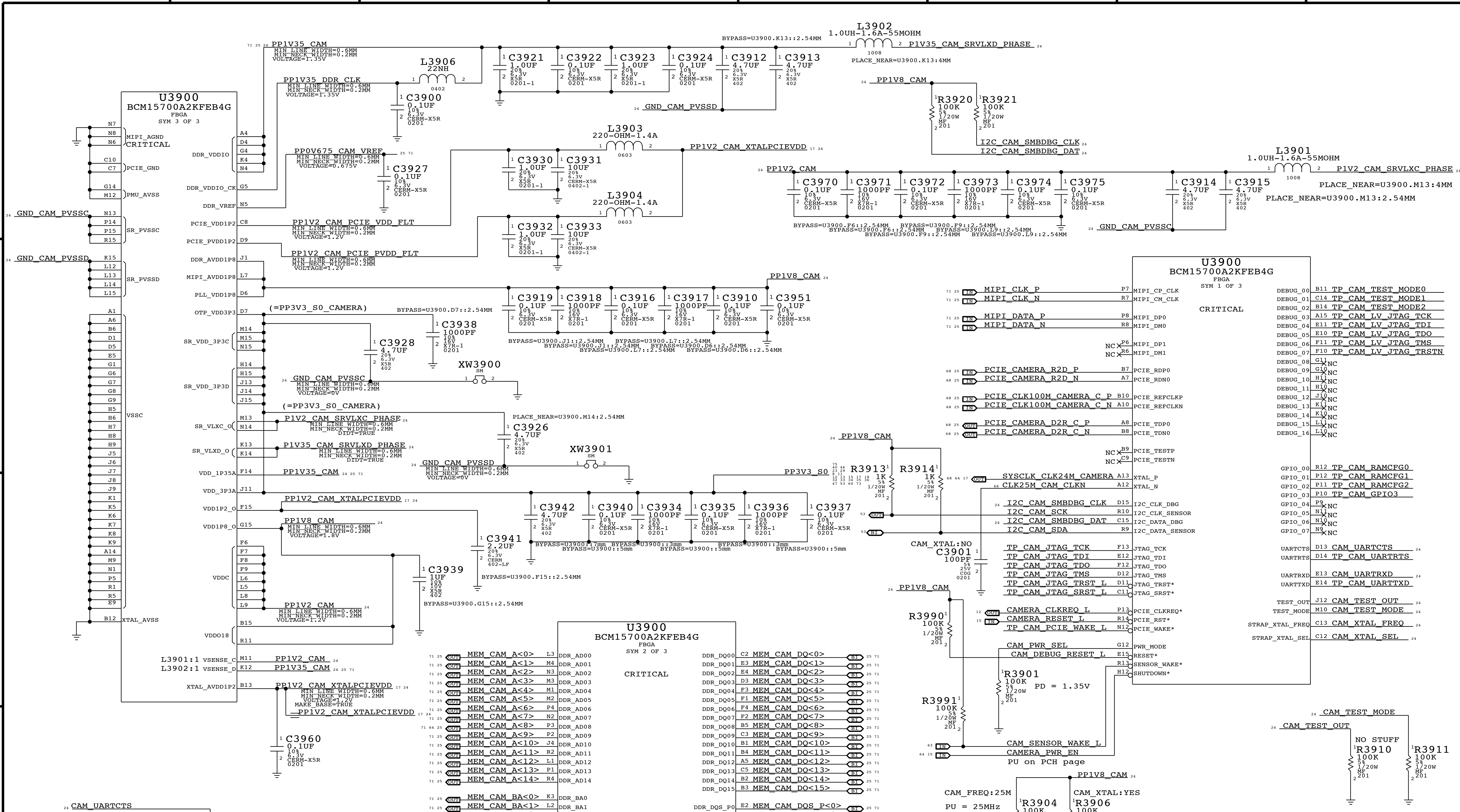
SSD Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_DEVMLB SYNC_DATE=07/24/2013

PAGE TITLE

Camera 1 of 2

Apple Inc.

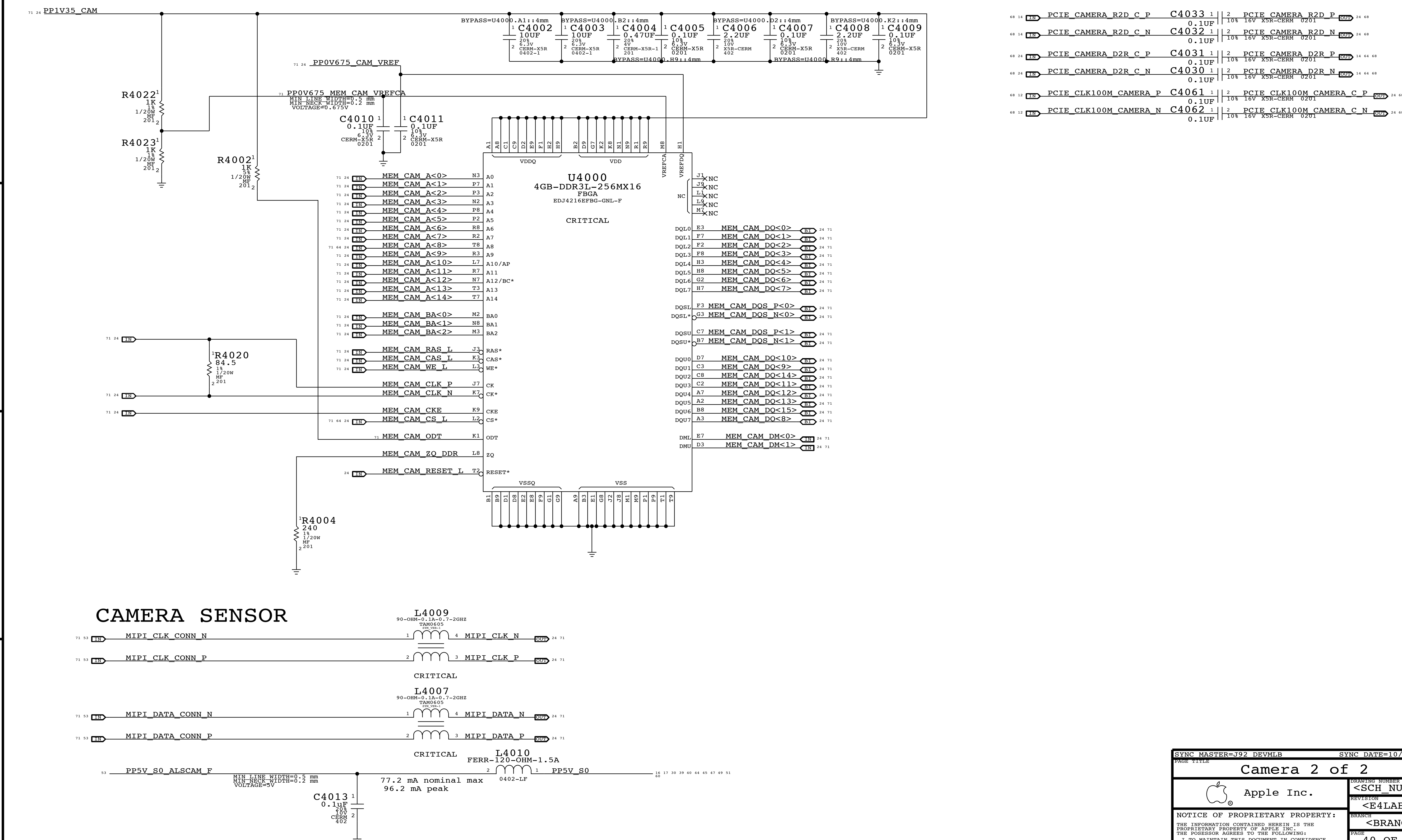
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 39 OF 130

SHEET 24 OF 75



SYNC_MASTER=J92_DEVMLB SYNC_DATE=10/10/2013

PAGE TITLE

Camera 2 of 2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

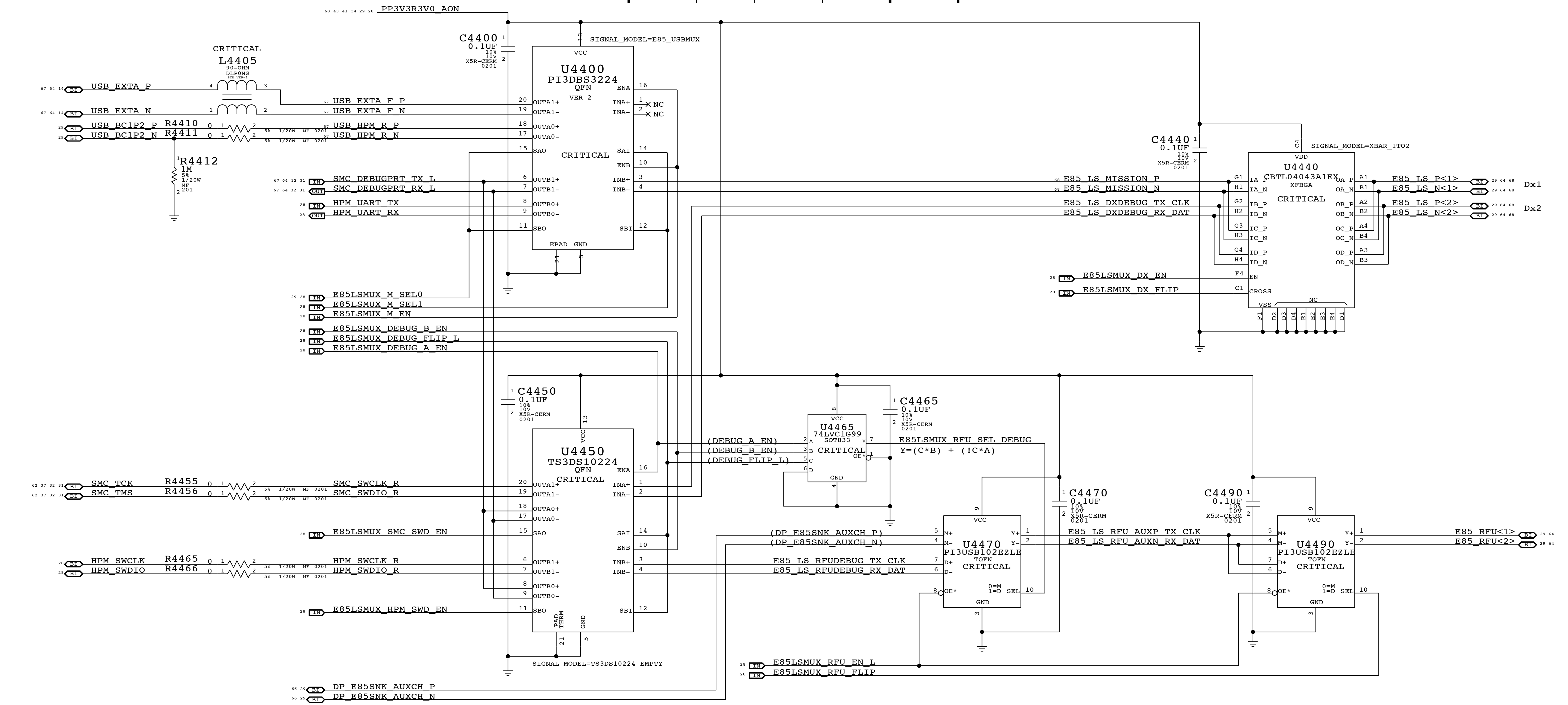
REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 40 OF 130

SHEET 25 OF 75

Mode	DX_EN	M_EN	M_SEL0	M_SEL1	INB	Mission
Empty	0	0	X	X	Hi-Z	Hi-Z
Mojo	1	1	1	1	OUTB1	UART (SMC)
USB (Host)	1	1	1	0	OUTA1	USB (PCH)
USB (Device)	1	1	0	0	OUTA0	USB (HPM)
HPM UART	1	1	0	1	OUTB0	UART (HPM)



Mode	DX_EN	DEBUG_A_EN	DEBUG_B_EN	DEBUG_FLIP_L	SMC_SWD_EN	HPM_SWD_EN	INA	DXDEBUG
Empty	0	0 X	X 0	1 0	X	X	Hi-Z	Hi-Z
No Debug	1	0 X	X 0	1 0	X	X	Hi-Z	Hi-Z
SMC SWD	1	1	X	1	1	X	OUTA1	SWD (SMC)
Mojo	1	1 X	X 1	1 0	0 X	X 0	OUTA0 OUTB0	UART (SMC)
HPM SWD	1	X	1	0	X	1	OUTB1	SWD (HPM)

Mode	RFU_EN_L	DEBUG_A_EN	DEBUG_B_EN	DEBUG_FLIP_L	SMC_SWD_EN	HPM_SWD_EN	INB	RFU
Empty	1	X 0	0 X	1 0	X	X	Hi-Z	Hi-Z
DP	0	X 0	0 X	1 0	X	X	Hi-Z	AUX+/HPD
HPM SWD	0	X	1	1	X	1	OUTB1	SWD (HPM)
Mojo	0	X 1	1 X	1 0	X 0	0 X	OUTB0 OUTA0	UART (SMC)
SMC SWD	0	1	X	0	1	X	OUTA1	SWD (SMC)

SYNC_MASTER=J92_DEVMLB SYNC_DATE=07/08/2014

PAGE TITLE

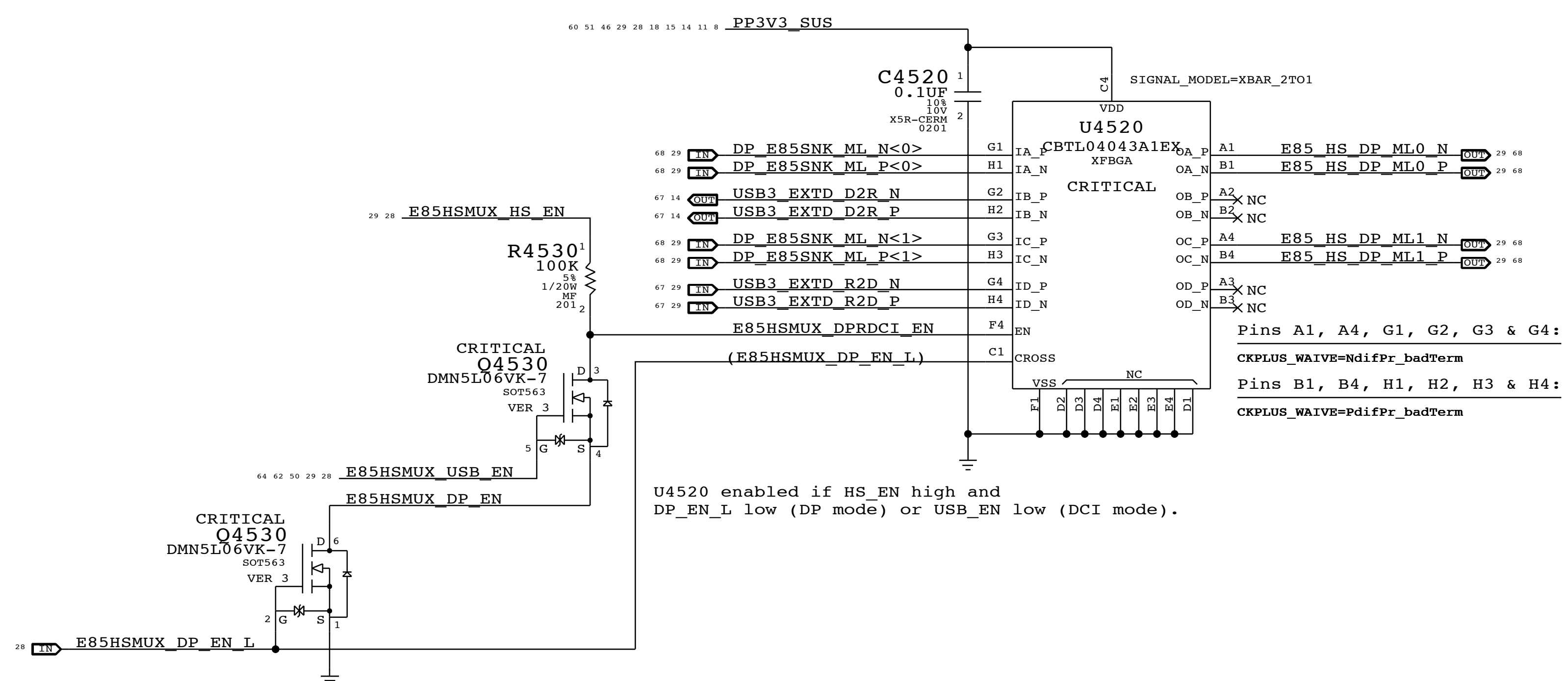
Low Speed MUXing

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



HS_FLIP swaps Tx1/2 and Rx1/2

Mode	HS_EN	DP_EN_L	USB_EN	SSTx1	SSRx1	SSTx2	SSRx2
Empty	0	X	X	Hi-Z	Hi-Z	Hi-Z	Hi-Z
USB3	1	1	1	R2D	D2R	Hi-Z	Hi-Z
USB3/DPx2	1	0	1	R2D	D2R	ML1	ML0
DPx4 (Passive)	1	0	0	ML2	ML3	ML1	ML0
DCI (Debug)	1	1	0	(ML2)	(ML3)	DCI-R2D	DCI-D2R

NOTE: Invert HS_FLIP in DCI mode to put DCI-USB on USB3 pins. USB3 and DP not usable in DCI mode.

SYNC_MASTER=DEV_MLB SYNC_DATE=04/17/2014

PAGE TITLE

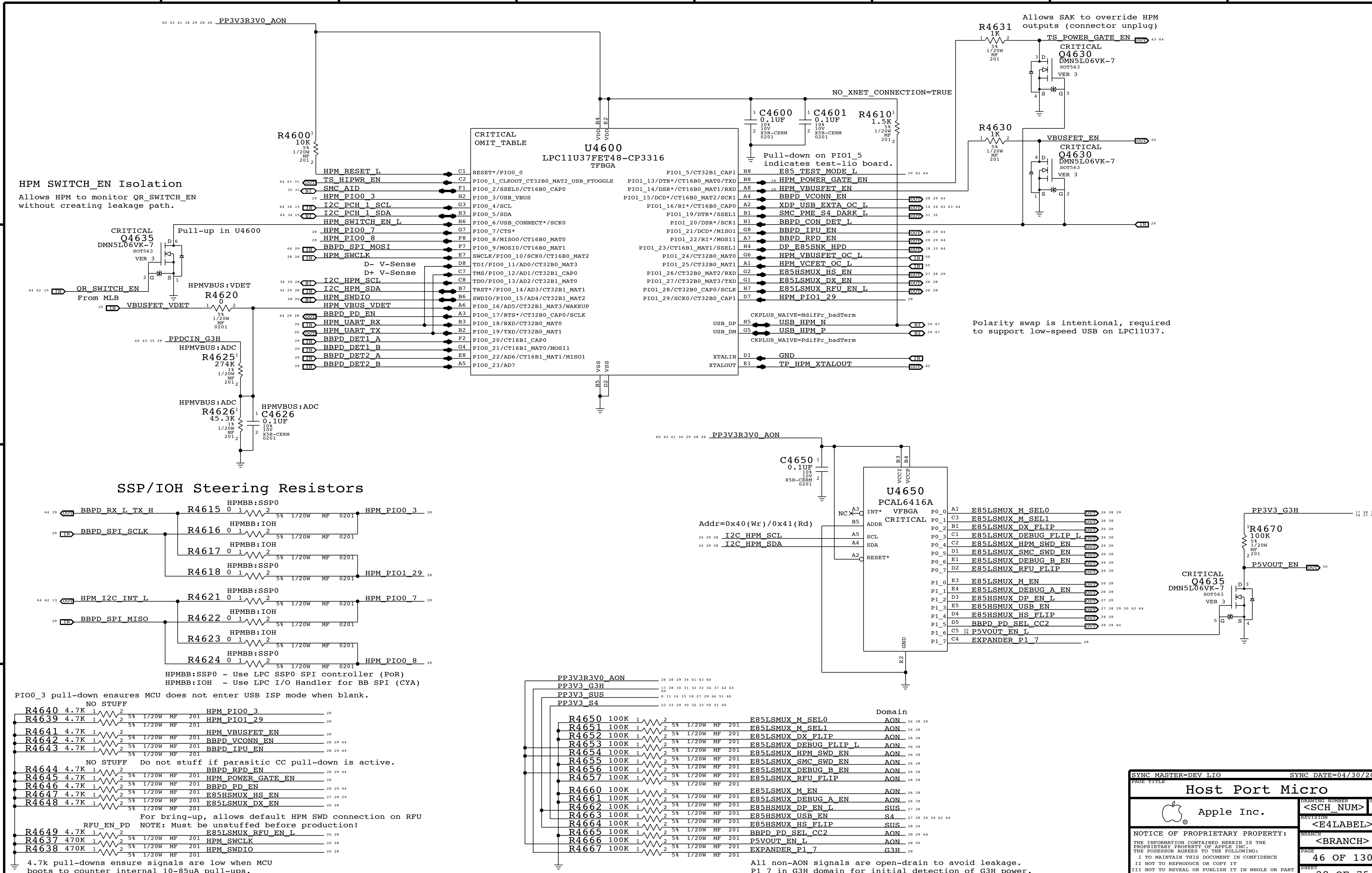
High Speed MUXing

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=DEV_LIO SYNC_DATE=04/30/2014

PAGE TITLE

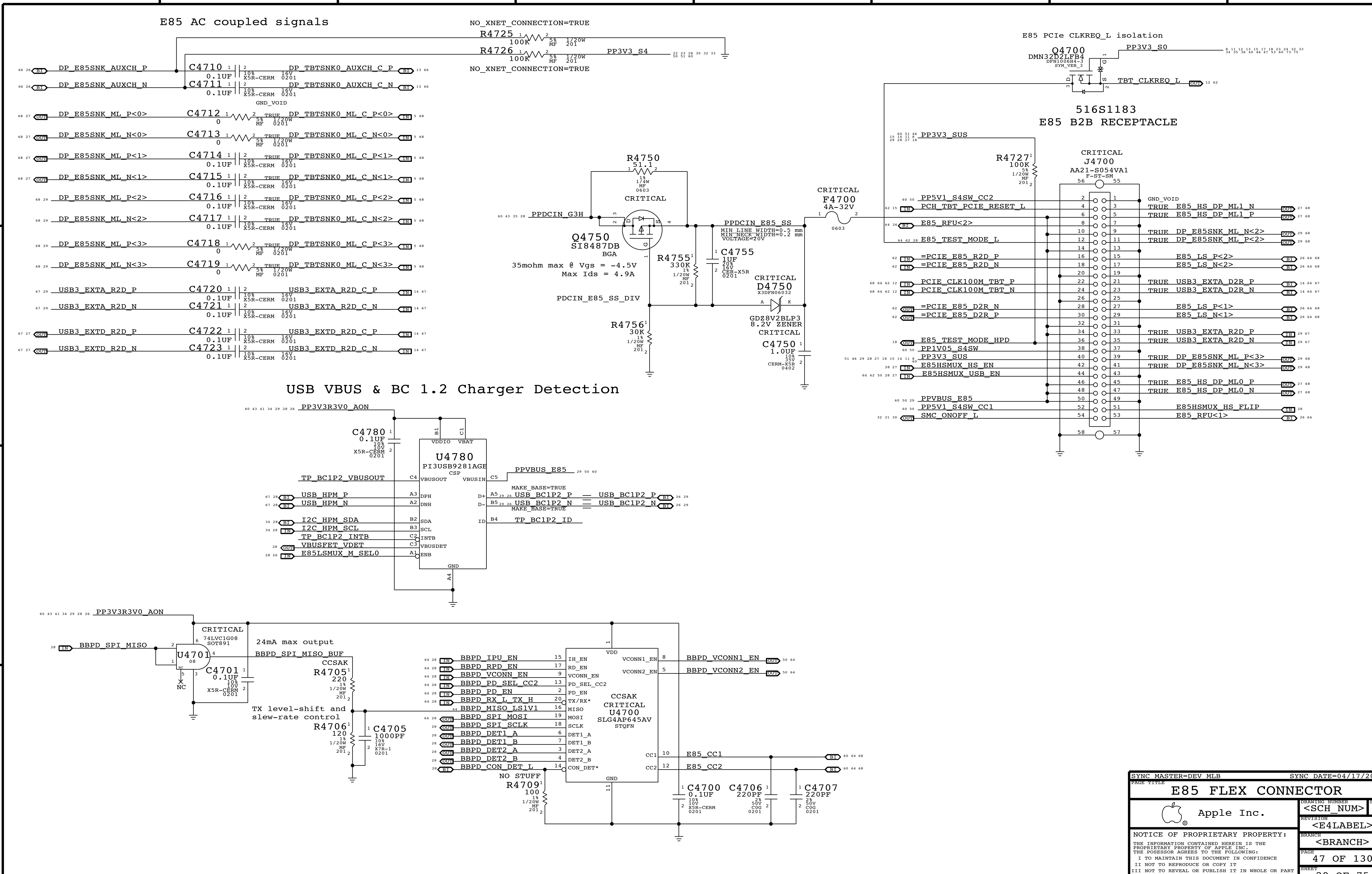
Host Port Micro

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=DEV_MLB SYNC_DATE=04/17/2014

PAGE TITLE

E85 FLEX CONNECTOR

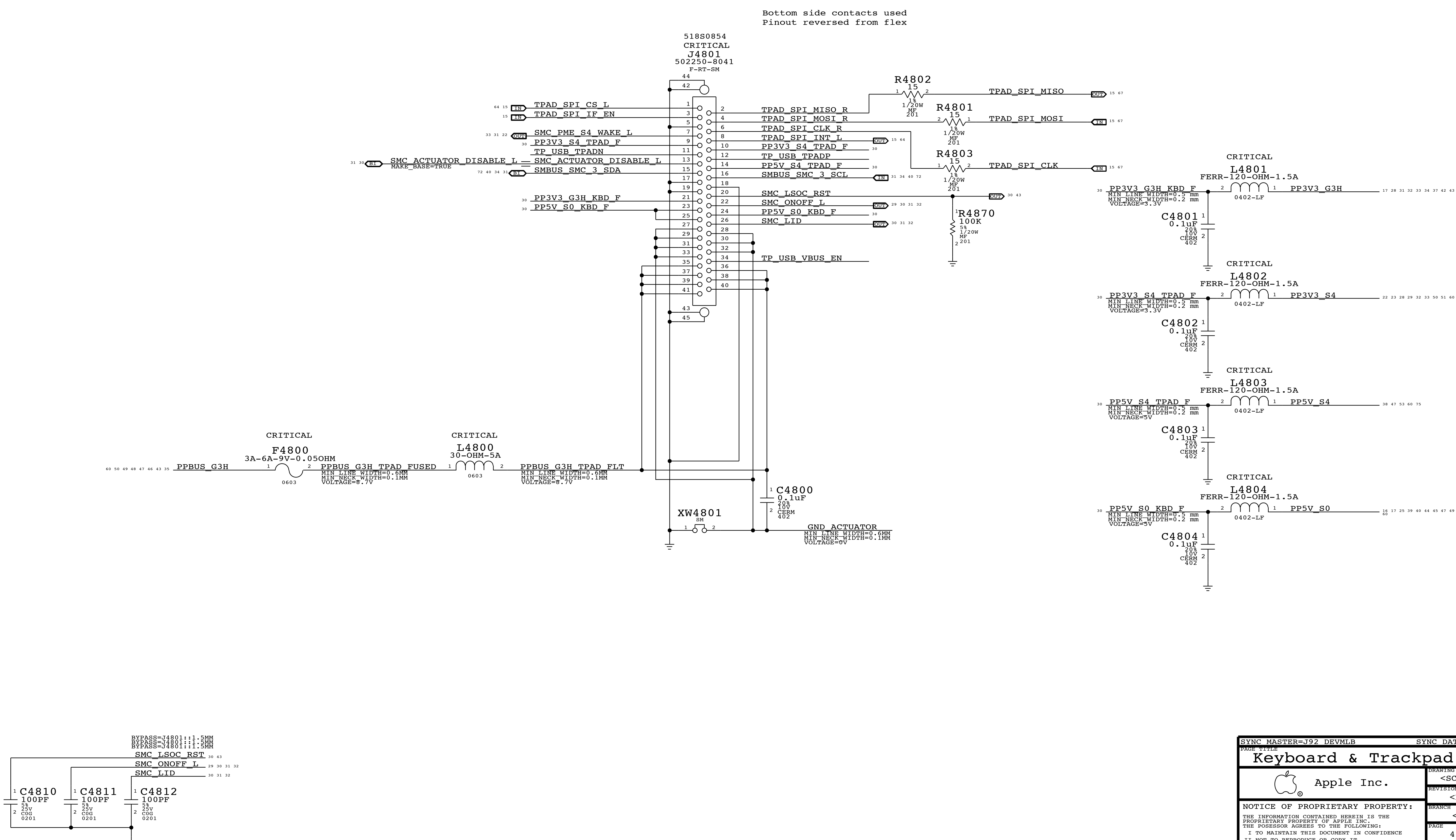
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

IPD ZIF CONNECTOR



SYNC_MASTER=J92_DEVMLB SYNC_DATE=03/26/2014

PAGE TITLE
Keyboard & Trackpad Conn

Apple Inc.

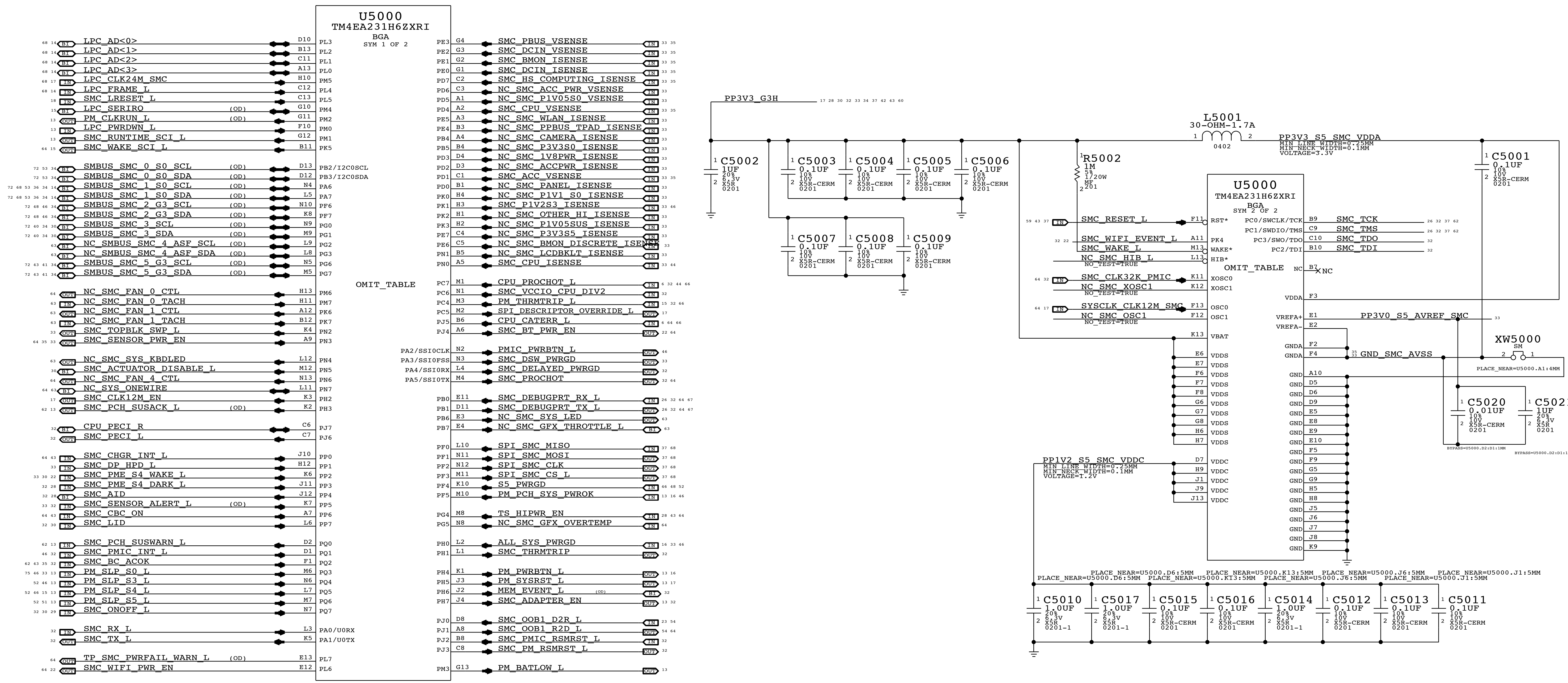
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 48 OF 130

SHEET 30 OF 75



NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

SYNC_MASTER=J92_DEVMLB SYNC_DATE=10/11/2013

PAGE TITLE

SMC

Apple Inc.

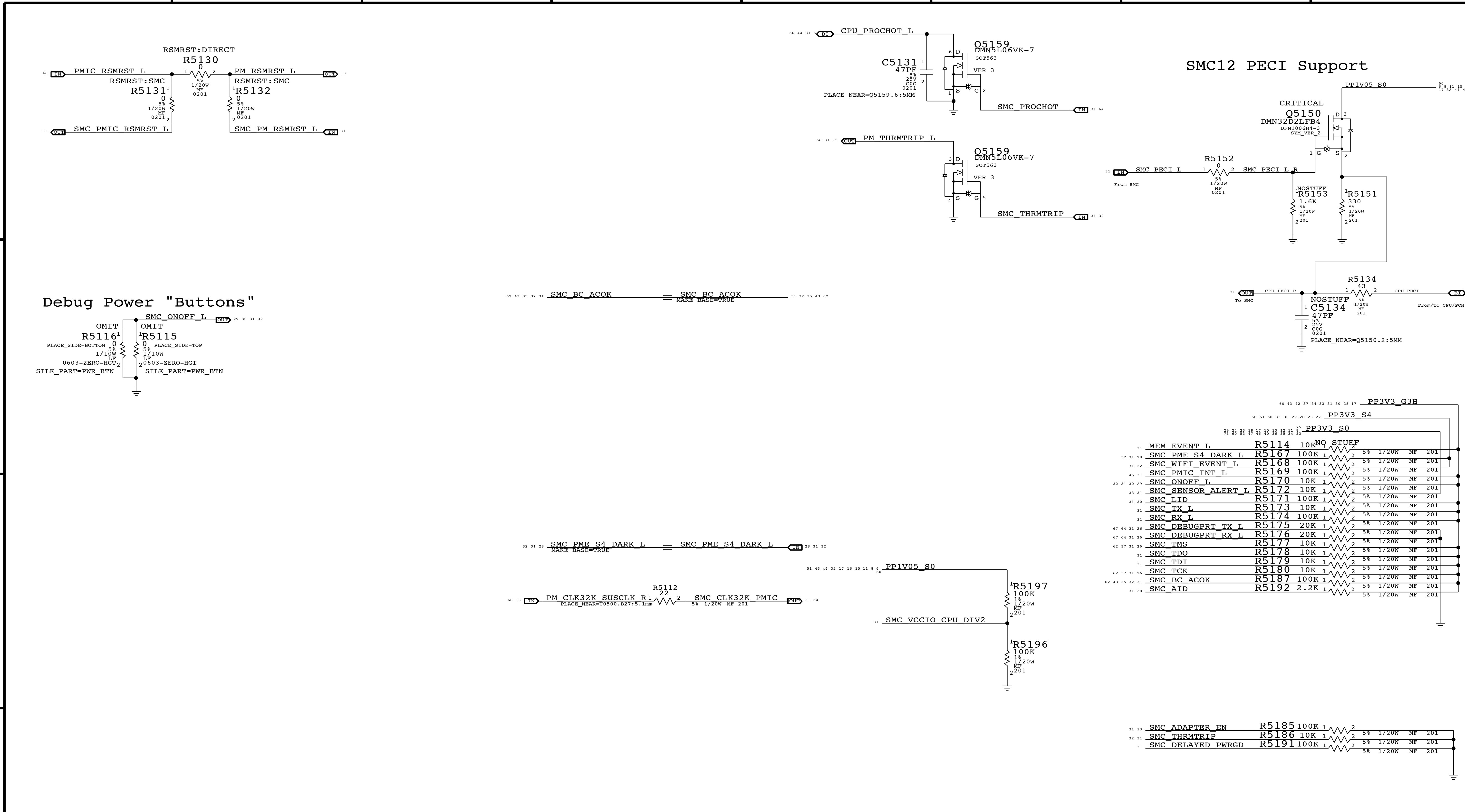
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 50 OF 130

SHEET 31 OF 75



SYNC_MASTER=J92_DEVMLB SYNC_DATE=10/11/2013

PAGE TITLE

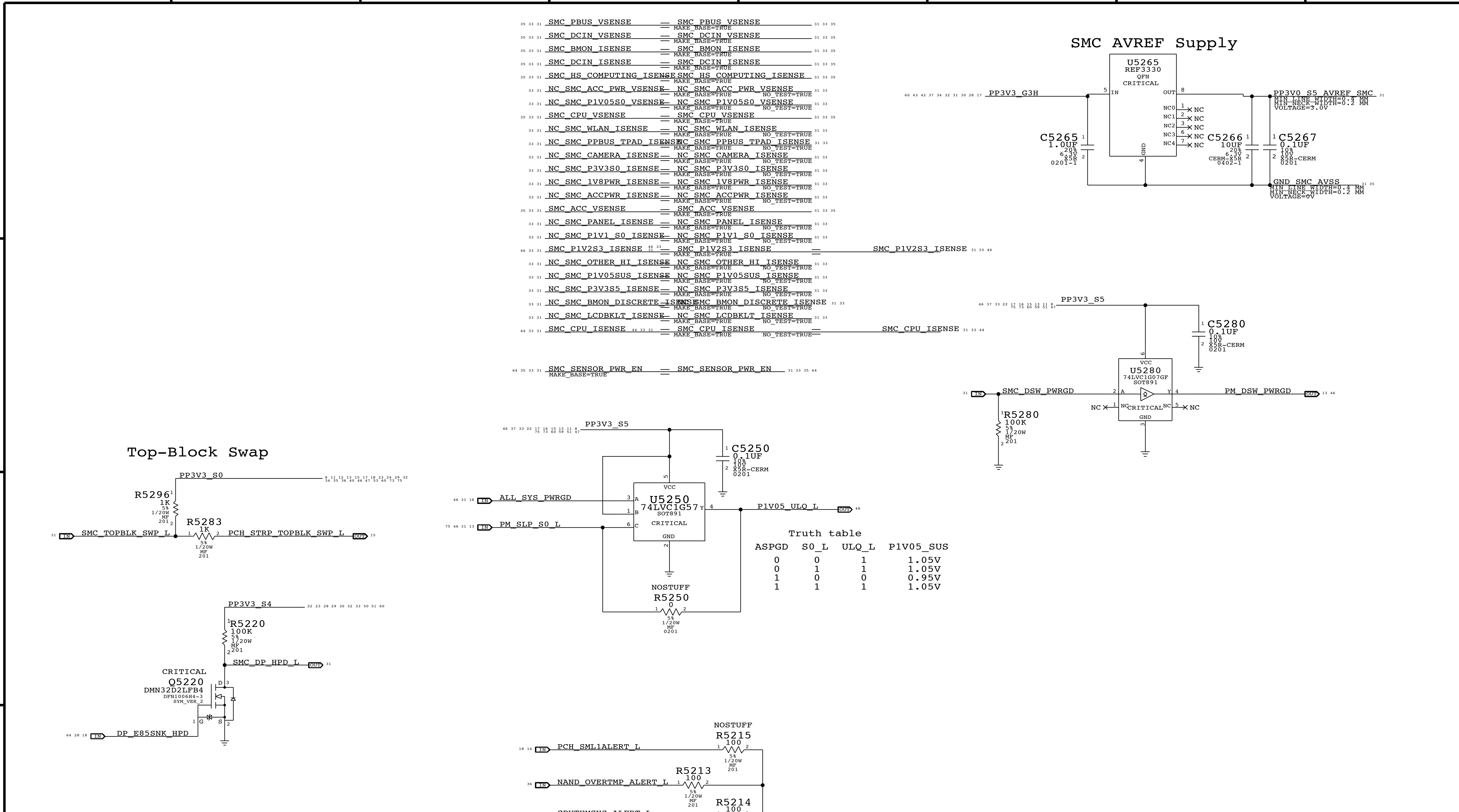
SMC Shared Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



Truth table

ASPGD	S0_L	ULQ_L	P1V05_SUS
0	0	1	1.05V
0	1	1	1.05V
1	0	0	0.95V
1	1	1	1.05V

SYNC_MASTER=J43_MLB SYNC_DATE=10/24/2012

PAGE TITLE

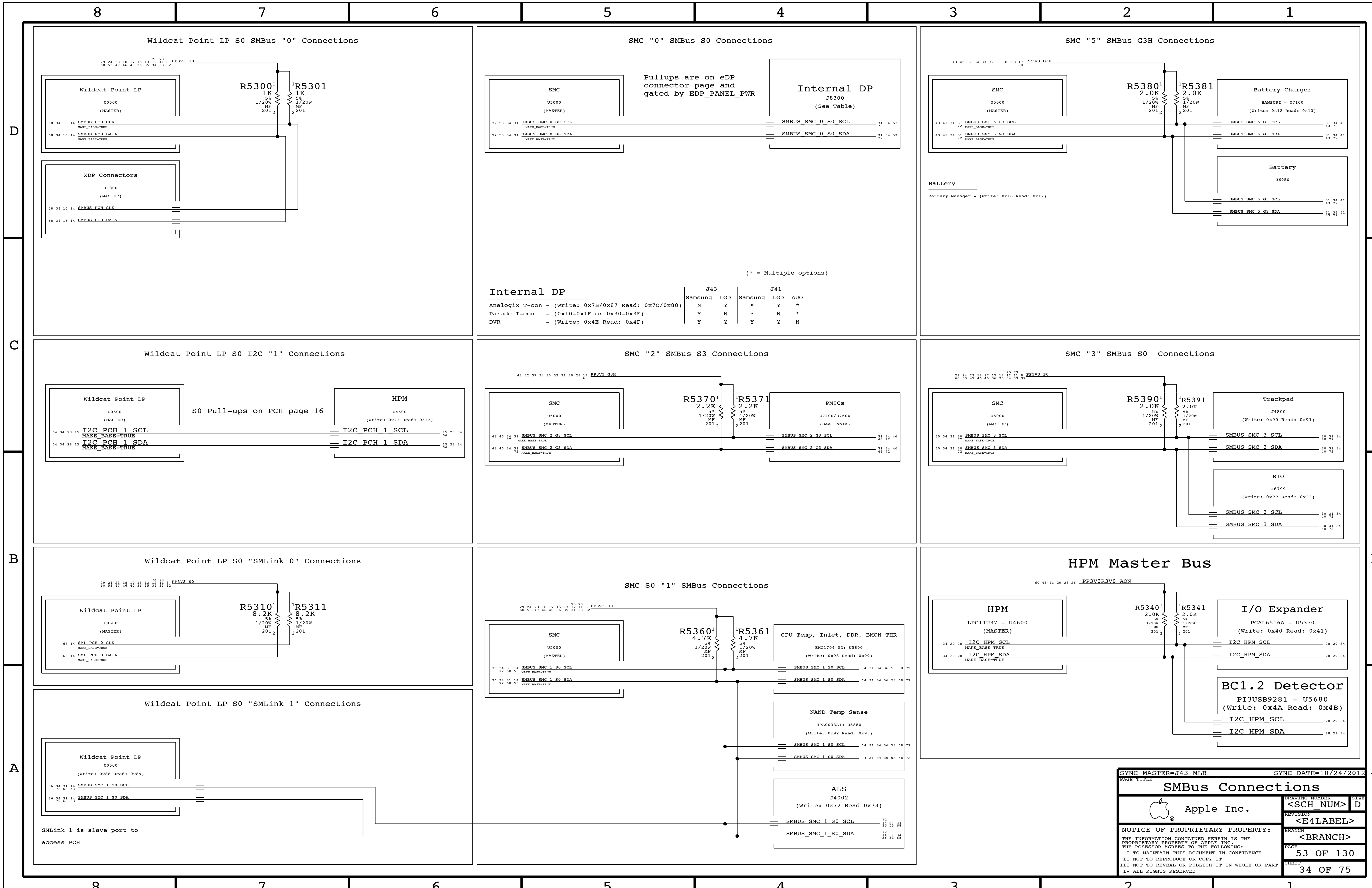
SMC Project Support

Apple Inc.

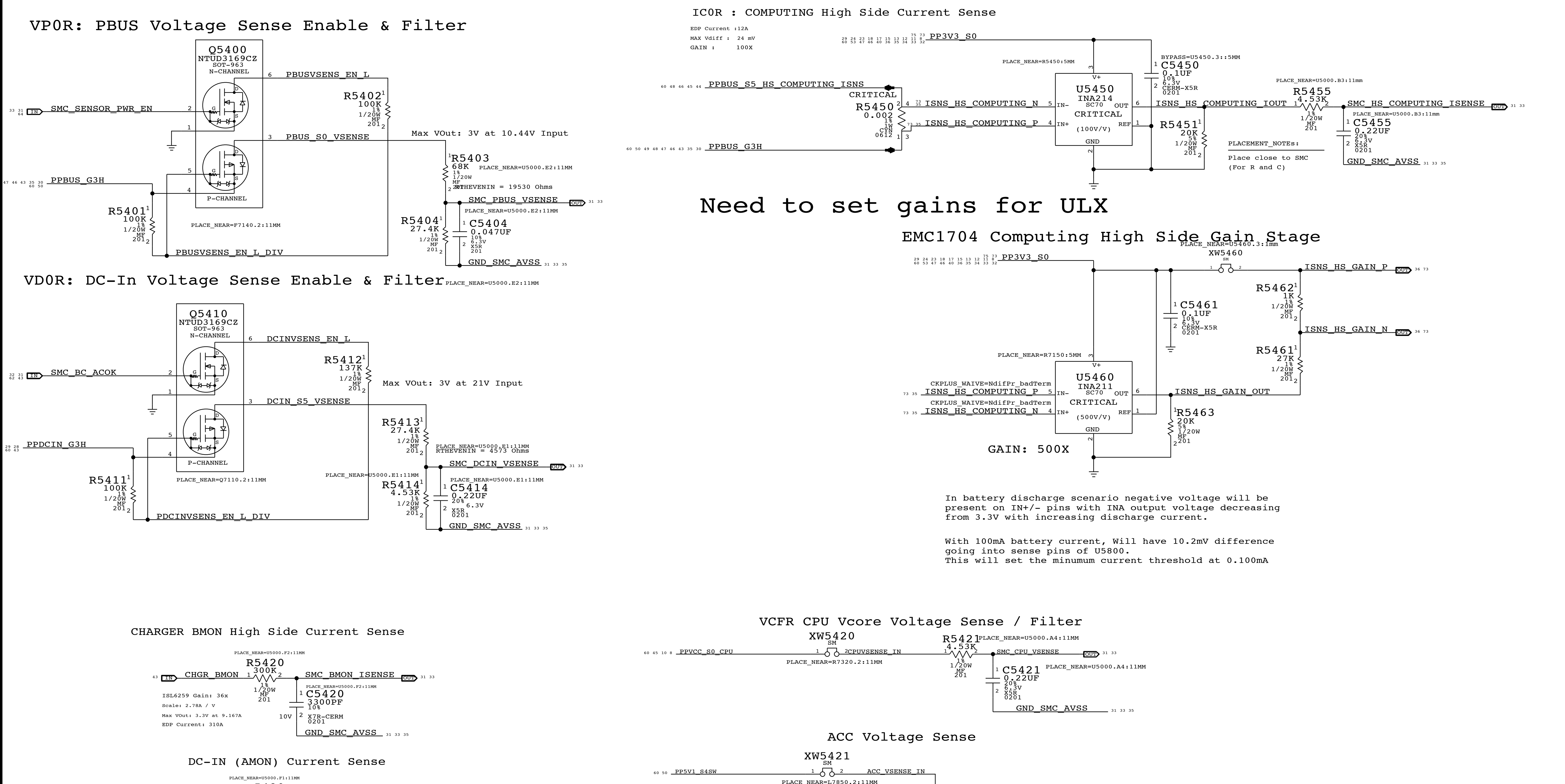
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>


Wildcat Point LP S0 SMBus "0" Connections
PP3V3_S0
Wildcat Point LP
U0500
(MASTER)
SMBUS_PCH_CLK
SMBUS_PCH_DATA
R5300 1K 5% 1/20W MF 201
R5301 1K 5% 1/20W MF 201
XDP Connectors
J1800
(MASTER)
SMC "0" SMBus S0 Connections
SMC
U5000
(MASTER)
SMBUS_SMC_0_S0_SCL
SMBUS_SMC_0_S0_SDA
Pullups are on eDP connector page and gated by EDP_PANEL_PWR
Internal DP
J8300
(See Table)
(* = Multiple options)
Internal DP
J43 Samsung LGD
J41 Samsung LGD AUO
Analogix T-con - (Write: 0x7B/0x87 Read: 0x7C/0x88) N Y * Y *
Parade T-con - (0x10-0x1F or 0x30-0x3F) Y N * N *
DVR - (Write: 0x4E Read: 0x4F) Y Y Y Y N
SMC "5" SMBus G3H Connections
PP3V3_G3H
R5380 2.0K 5% 1/20W MF 201
R5381 2.0K 5% 1/20W MF 201
Battery Charger
BANSURI - U7100
(Write: 0x12 Read: 0x13)
SMBUS_SMC_5_G3_SCL
SMBUS_SMC_5_G3_SDA
Battery
J6950
Battery
Battery Manager - (Write: 0x16 Read: 0x17)
Wildcat Point LP S0 I2C "1" Connections
S0 Pull-ups on PCH page 16
I2C_PCH_1_SCL
I2C_PCH_1_SDA
MAKE_BASE=TRUE
HPM
U4600
(Write: 0x77 Read: 0X77)
SMC "2" SMBus S3 Connections
R5370 2.2K 5% 1/20W MF 201
R5371 2.2K 5% 1/20W MF 201
PMICs
U7400/U7600
(See Table)
SMBUS_SMC_2_G3_SCL
SMBUS_SMC_2_G3_SDA
SMC "3" SMBus S0 Connections
R5390 2.0K 5% 1/20W MF 201
R5391 2.0K 5% 1/20W MF 201
Trackpad
J4800
(Write: 0x90 Read: 0x91)
SMBUS_SMC_3_SCL
SMBUS_SMC_3_SDA
RIO
J6799
(Write: 0x77 Read: 0x77)
Wildcat Point LP S0 "SMLink 0" Connections
R5310 8.2K 5% 1/20W MF 201
R5311 8.2K 5% 1/20W MF 201
SML_PCH_0_CLK
SML_PCH_0_DATA
SMC S0 "1" SMBus Connections
R5360 4.7K 5% 1/20W MF 201
R5361 4.7K 5% 1/20W MF 201
CPU Temp, Inlet, DDR, BMON THR
EMC1704-02: U5800
(Write: 0x98 Read: 0x99)
SMBUS_SMC_1_S0_SCL
SMBUS_SMC_1_S0_SDA
NAND Temp Sense
HPA0033AI: U5880
(Write: 0x92 Read: 0x93)
ALS
J4002
(Write: 0x72 Read 0x73)
HPM Master Bus
PP3V3R3V0_AON
HPM
LPC11U37 - U4600
(MASTER)
I2C_HPM_SCL
I2C_HPM_SDA
R5340 2.0K 5% 1/20W MF 201
R5341 2.0K 5% 1/20W MF 201
I/O Expander
PCAL6516A - U5350
(Write: 0x40 Read: 0x41)
BC1.2 Detector
PI3USB9281 - U5680
(Write: 0x4A Read: 0x4B)
Wildcat Point LP S0 "SMLink 1" Connections
Wildcat Point LP
U0500
(Write: 0x88 Read: 0x89)
SMLink 1 is slave port to access PCH
SYNC_MASTER=J43_MLB SYNC_DATE=10/24/2012
PAGE TITLE
SMBus Connections
Apple Inc.
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>
PAGE 53 OF 130
SHEET 34 OF 75
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

POR VOLTAGE / CURRENT SENSORS : TO BE USED IN PRODUCTION



SYNC_MASTER=J92_DEVMLB SYNC_DATE=02/07/2014

PAGE TITLE: Voltage & Current Sensing

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

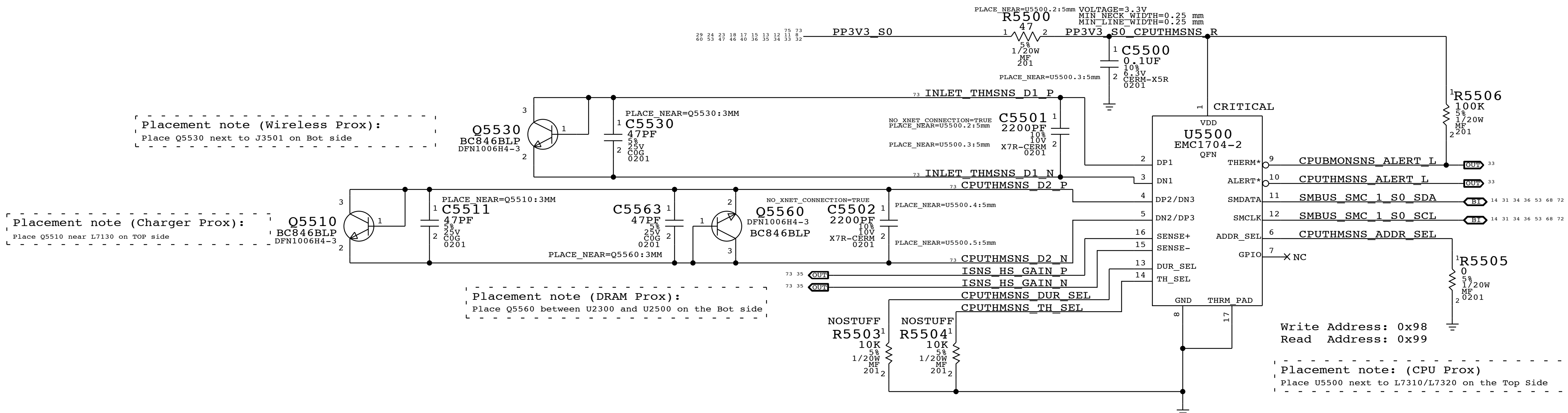
BRANCH: <BRANCH>

PAGE: 54 OF 130

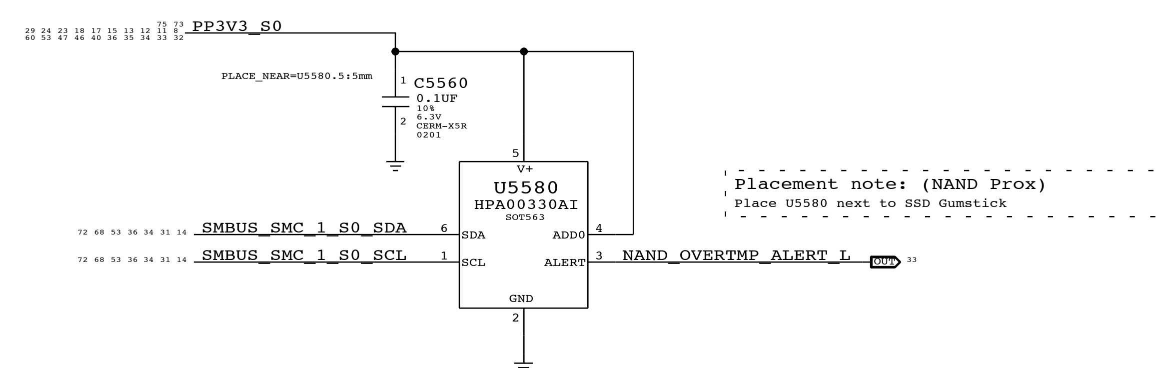
SHEET: 35 OF 75

POR THERMAL SENSORS : TO BE USED IN PRODUCTION

CPU Proximity, Inlet ,DDR and BMON THR Sensor



NAND Temp Sensor



SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/12/2013

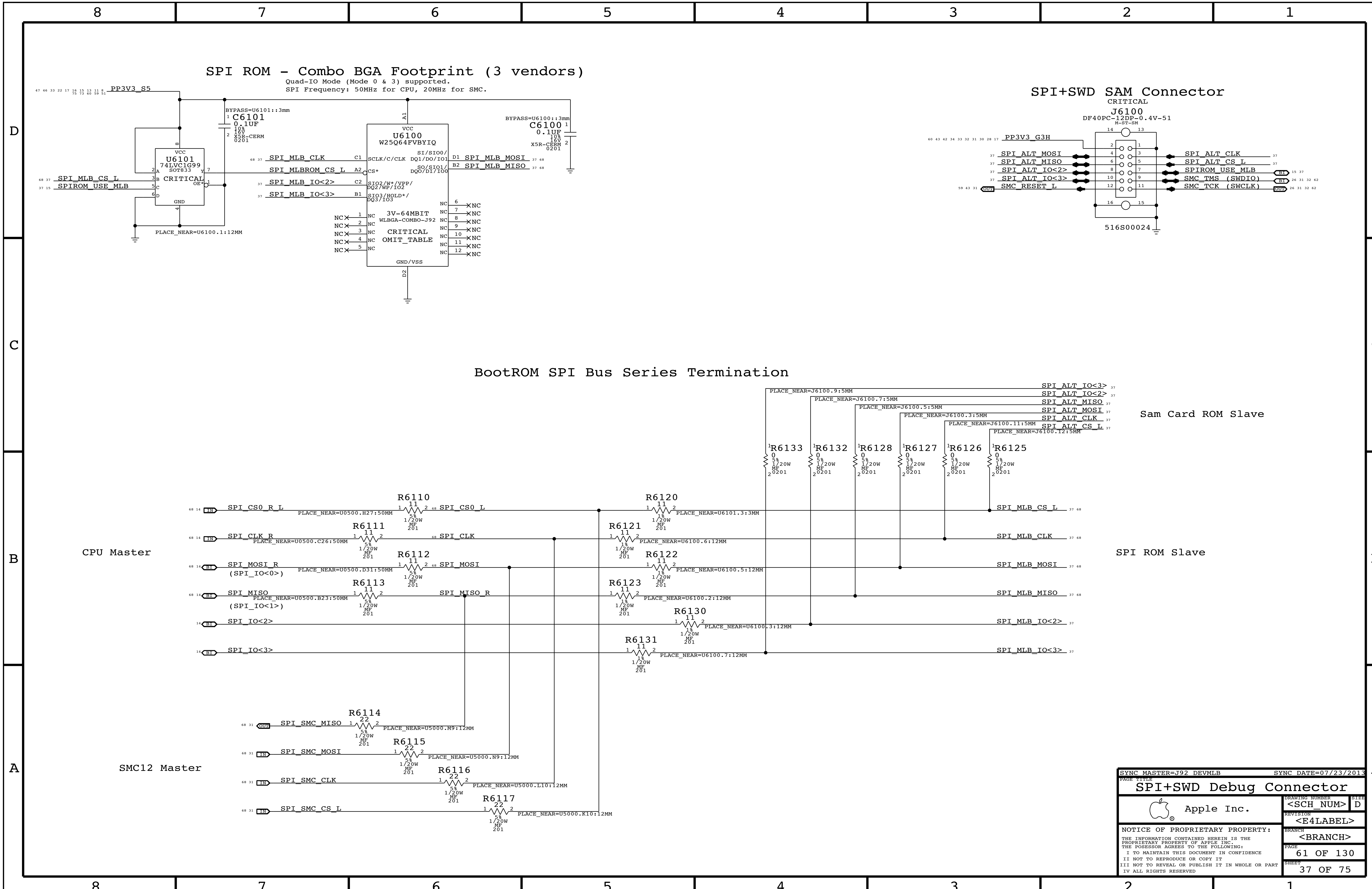
PAGE TITLE: Temperature Sensing

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

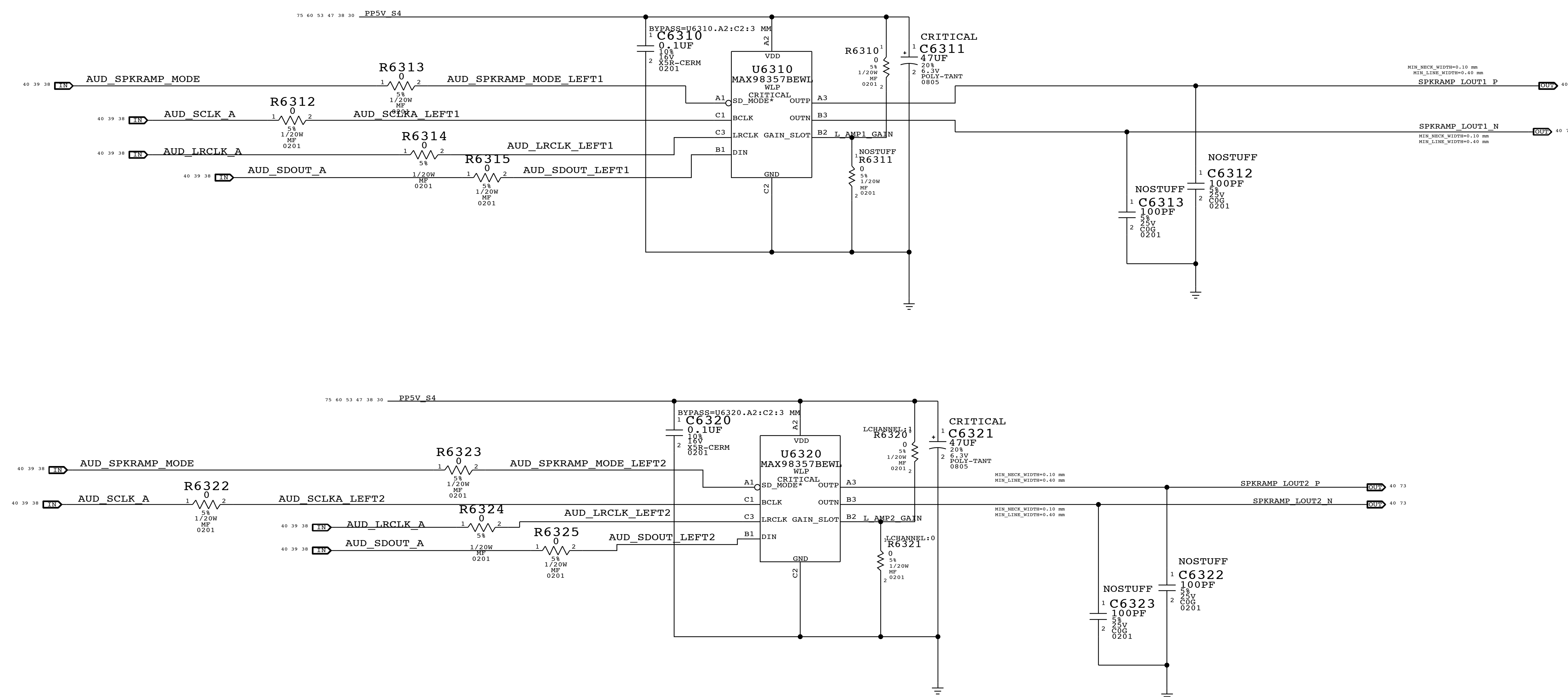
REVISION: <E4LABEL>

BRANCH: <BRANCH>


SPI ROM - Combo BGA Footprint (3 vendors)
Quad-IO Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 20MHz for SMC.
PP3V3_S5
BYPASS=U6101::3mm
C6101
0.1UF
10%
16V
X5R-CERM
0201
U6101
74LVC1G99
SOT833
CRITICAL
VCC
GND
OE*
SPI_MLB_CS_L
SPIROM_USE_MLB
PLACE_NEAR=U6100.1:12MM
U6100
W25Q64FVBYIQ
VCC
SPI_MLB_CLK
SPI_MLBROM_CS_L
SPI_MLB_IO<2>
SPI_MLB_IO<3>
SCLK/C/CLK
CS*
SIO2/W*/VPP/
DQ2/WP/IO2
SIO3/HOLD*/
DQ3/IO3
SI/SIO0/
DQ1/DO/IO1
SO/SIO1/
DQ0/DI/IO0
SPI_MLB_MOSI
SPI_MLB_MISO
3V-64MBIT
WLBGA-COMBO-J92
CRITICAL
OMIT_TABLE
GND/VSS
NC
BYPASS=U6100::3mm
C6100
0.1UF
10%
16V
X5R-CERM
0201
SPI+SWD SAM Connector
CRITICAL
J6100
DF40PC-12DP-0.4V-51
M-ST-SM
PP3V3_G3H
SPI_ALT_MOSI
SPI_ALT_MISO
SPI_ALT_IO<2>
SPI_ALT_IO<3>
SMC_RESET_L
SPI_ALT_CLK
SPI_ALT_CS_L
SPIROM_USE_MLB
SMC_TMS (SWDIO)
SMC_TCK (SWCLK)
516S00024
BootROM SPI Bus Series Termination
SPI_ALT_IO<3>
SPI_ALT_IO<2>
SPI_ALT_MISO
SPI_ALT_MOSI
SPI_ALT_CLK
SPI_ALT_CS_L
PLACE_NEAR=J6100.9:5MM
PLACE_NEAR=J6100.7:5MM
PLACE_NEAR=J6100.5:5MM
PLACE_NEAR=J6100.3:5MM
PLACE_NEAR=J6100.11:5MM
PLACE_NEAR=J6100.12:5MM
Sam Card ROM Slave
R6133
R6132
R6128
R6127
R6126
R6125
0
5%
1/20W
MF
0201
CPU Master
SPI_CS0_R_L
R6110
11
5%
1/20W
MF
201
PLACE_NEAR=U0500.H27:50MM
SPI_CS0_L
R6120
PLACE_NEAR=U6101.3:3MM
SPI_MLB_CS_L
SPI_CLK_R
R6111
PLACE_NEAR=U0500.C26:50MM
SPI_CLK
R6121
PLACE_NEAR=U6100.6:12MM
SPI_MLB_CLK
SPI_MOSI_R
(SPI_IO<0>)
R6112
PLACE_NEAR=U0500.D31:50MM
SPI_MOSI
R6122
PLACE_NEAR=U6100.5:12MM
SPI_MLB_MOSI
SPI_MISO
(SPI_IO<1>)
R6113
PLACE_NEAR=U0500.B23:50MM
SPI_MISO_R
R6123
PLACE_NEAR=U6100.2:12MM
SPI_MLB_MISO
SPI_IO<2>
R6130
PLACE_NEAR=U6100.3:12MM
SPI_MLB_IO<2>
SPI_IO<3>
R6131
PLACE_NEAR=U6100.7:12MM
SPI_MLB_IO<3>
SPI ROM Slave
SMC12 Master
SPI_SMC_MISO
R6114
22
PLACE_NEAR=U5000.M9:12MM
SPI_SMC_MOSI
R6115
PLACE_NEAR=U5000.N9:12MM
SPI_SMC_CLK
R6116
PLACE_NEAR=U5000.L10:12MM
SPI_SMC_CS_L
R6117
PLACE_NEAR=U5000.K10:12MM
SYNC_MASTER=J92_DEVMLB
SYNC_DATE=07/23/2013
PAGE TITLE
SPI+SWD Debug Connector
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
D
REVISION
<E4LABEL>
BRANCH
<BRANCH>
PAGE
61 OF 130
SHEET
37 OF 75
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

Left Speaker Amps

APPLE P/N 353S4265



SPEAKER CONFIGS

BOM GROUP	BOM OPTIONS
EQ:2CH	LCHANNEL:1,RCHANNEL:3
EQ:4CH	LCHANNEL:0,RCHANNEL:4

SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/19/2013

PAGE TITLE

Audio:Left Speaker Amps

Apple Inc.

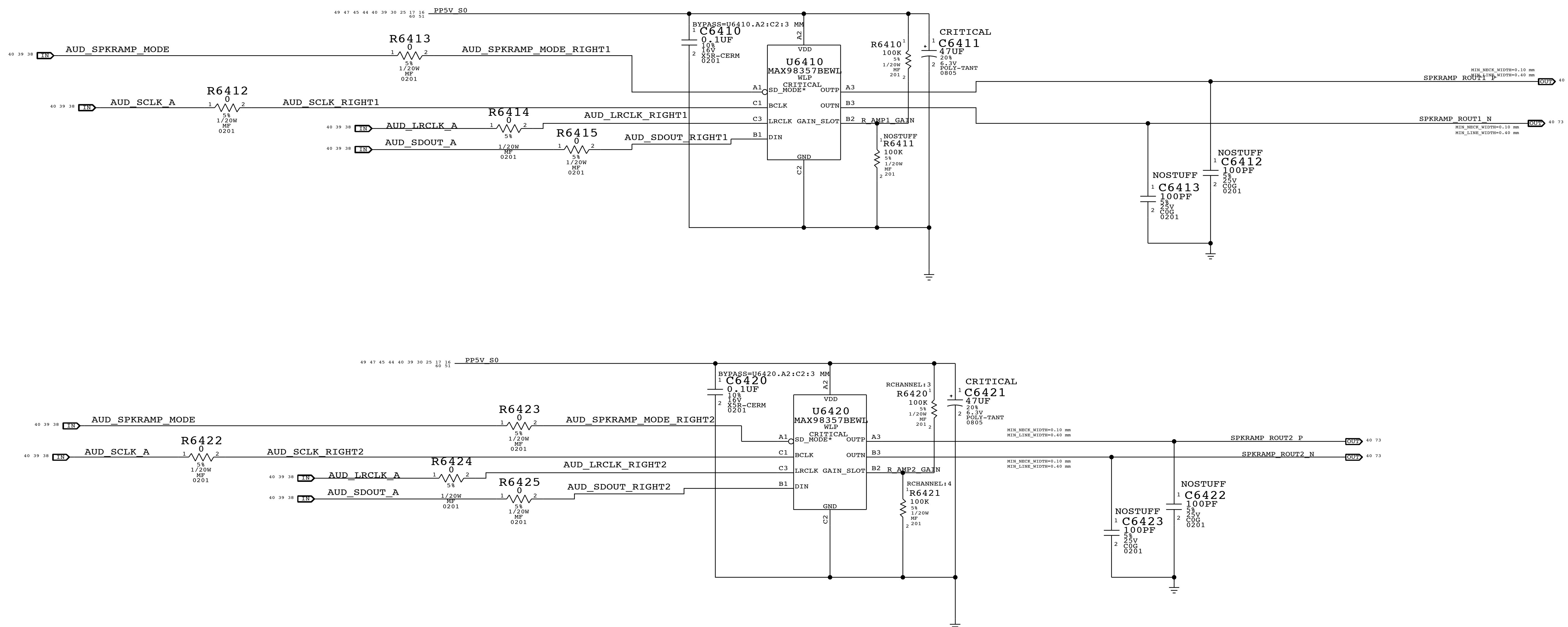
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Right Speaker Amps

APPLE P/N 353S4265



SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/19/2013

PAGE TITLE

Audio:Right Speaker Amps

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9,A)		0X09 (A)
SPEAKERS	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A

CODEC INPUT SIGNAL PATHS

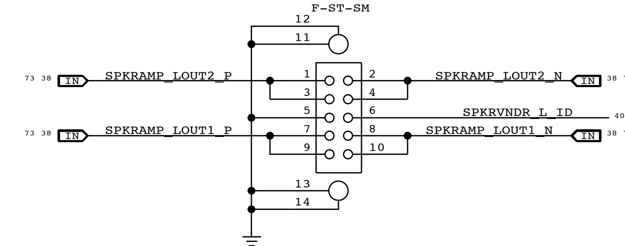
FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
BUILT-IN MIC	0X06 (6)	0X0D (13,B,RIGHT)	MIC_BIAS (80%)	N/A
HEADSET MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	MIKEY	MIKEY

SOUTHBRIDGE RESOURCE/PIN ALLOCATIONS

FUNCTION	NET NAME	SB GPIO/INT
PERIPHERAL/EXTRACTION DETECT	AUD_IP_PERIPHERAL_DET	GPIO 3
MIKEY INTERRUPT	AUD_I2C_INT_L	GPIO 5
MIKEY ENABLE	AUD_IPHS_SWITCH_EN	GPIO 16
MIKEY I2C BUS	I2C_MIKEY_SDA/SCL	PCH SMBUS 0

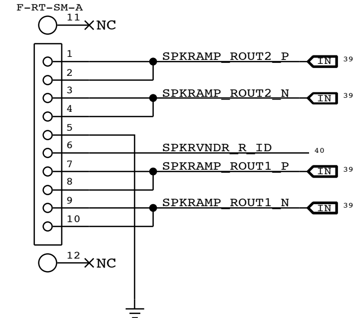
LEFT SPEAKER CONN

516S0899
CRITICAL
J6750
AA07A-S010-VA1
F-ST-SM

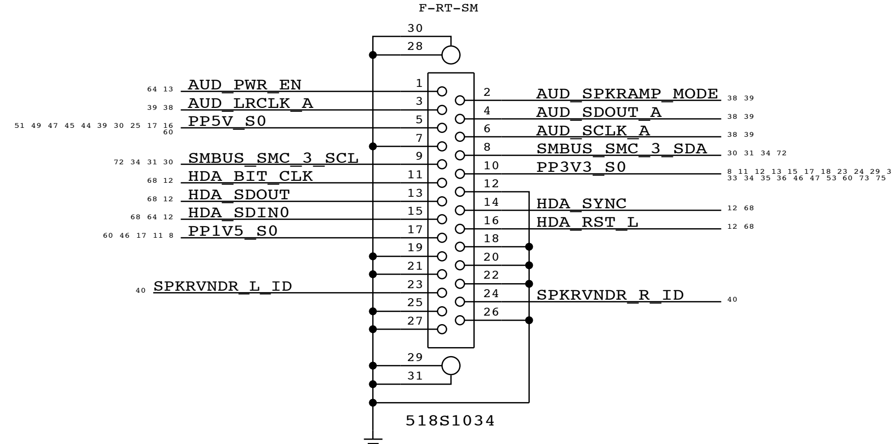


RIGHT SPEAKER CONN

518S00014
CRITICAL
J6751
FF18-10A-R11AD-B-3H
F-RT-SM-A



CRITICAL
J6799
502250-8027
F-RT-SM



518S1034

RIO FLEX CONN

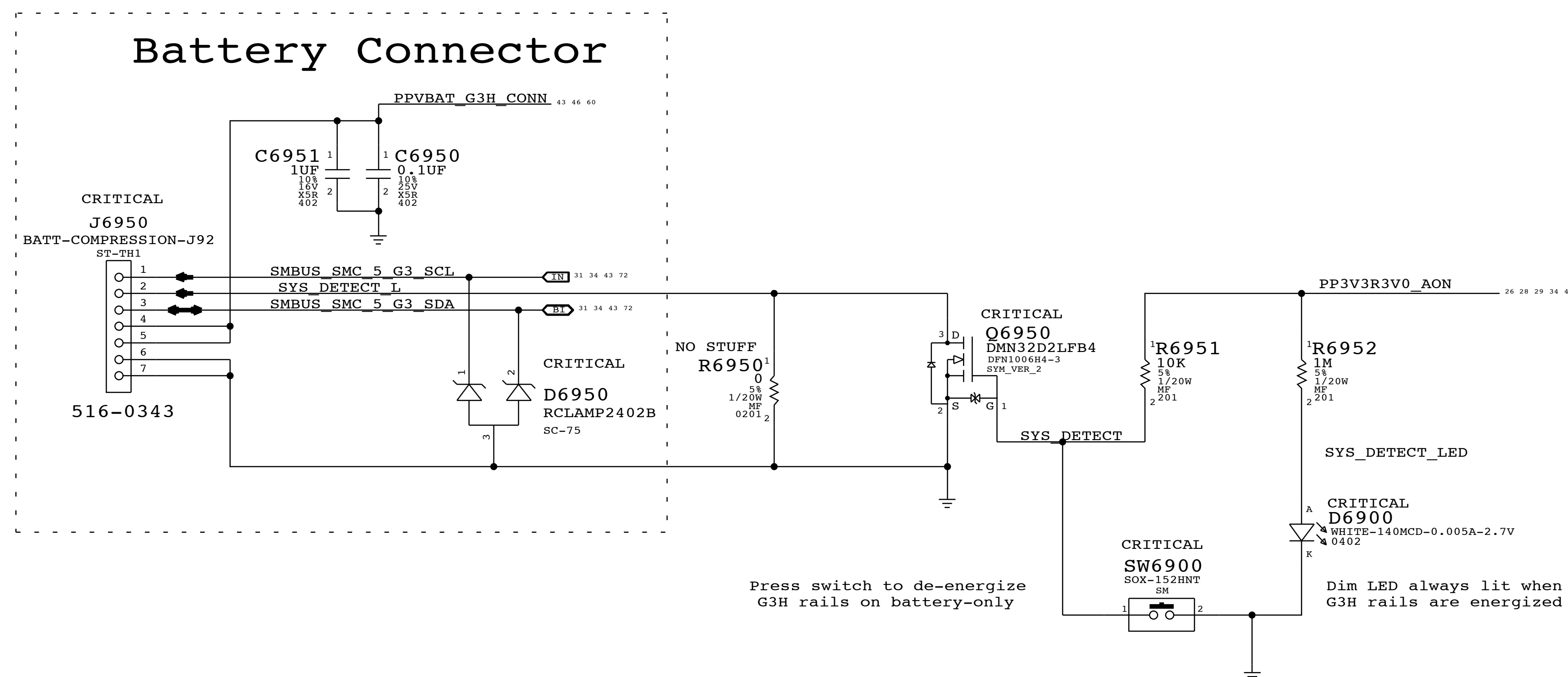
SYNC_MASTER=CARA_J92 SYNC_DATE=04/17/2014

PAGE TITLE
AUDIO: CONNECTORS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>
PAGE 67 OF 130
SHEET 40 OF 75

Battery Connector



SYNC_MASTER=J43_MLB SYNC_DATE=10/24/2012

PAGE TITLE

Battery Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

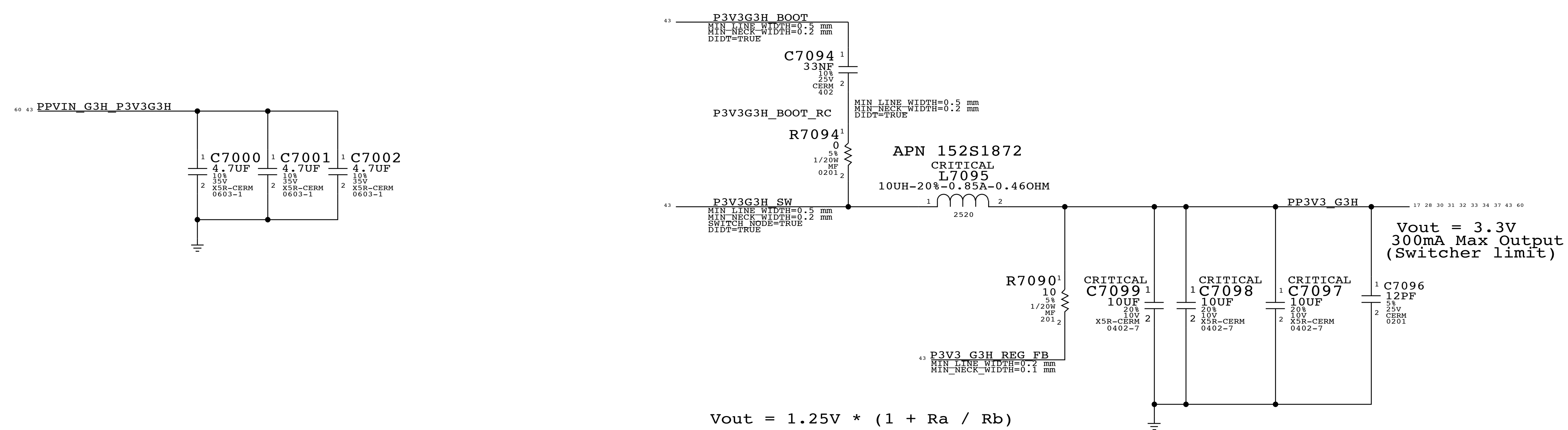
REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 69 OF 130

SHEET 41 OF 75

3.3V G3H VR - Bansuri



Vout = 1.25V * (1 + Ra / Rb)

SYNC_MASTER=J92_WILL SYNC_DATE=02/04/2013

PAGE TITLE

3.3V G3Hot Regulator

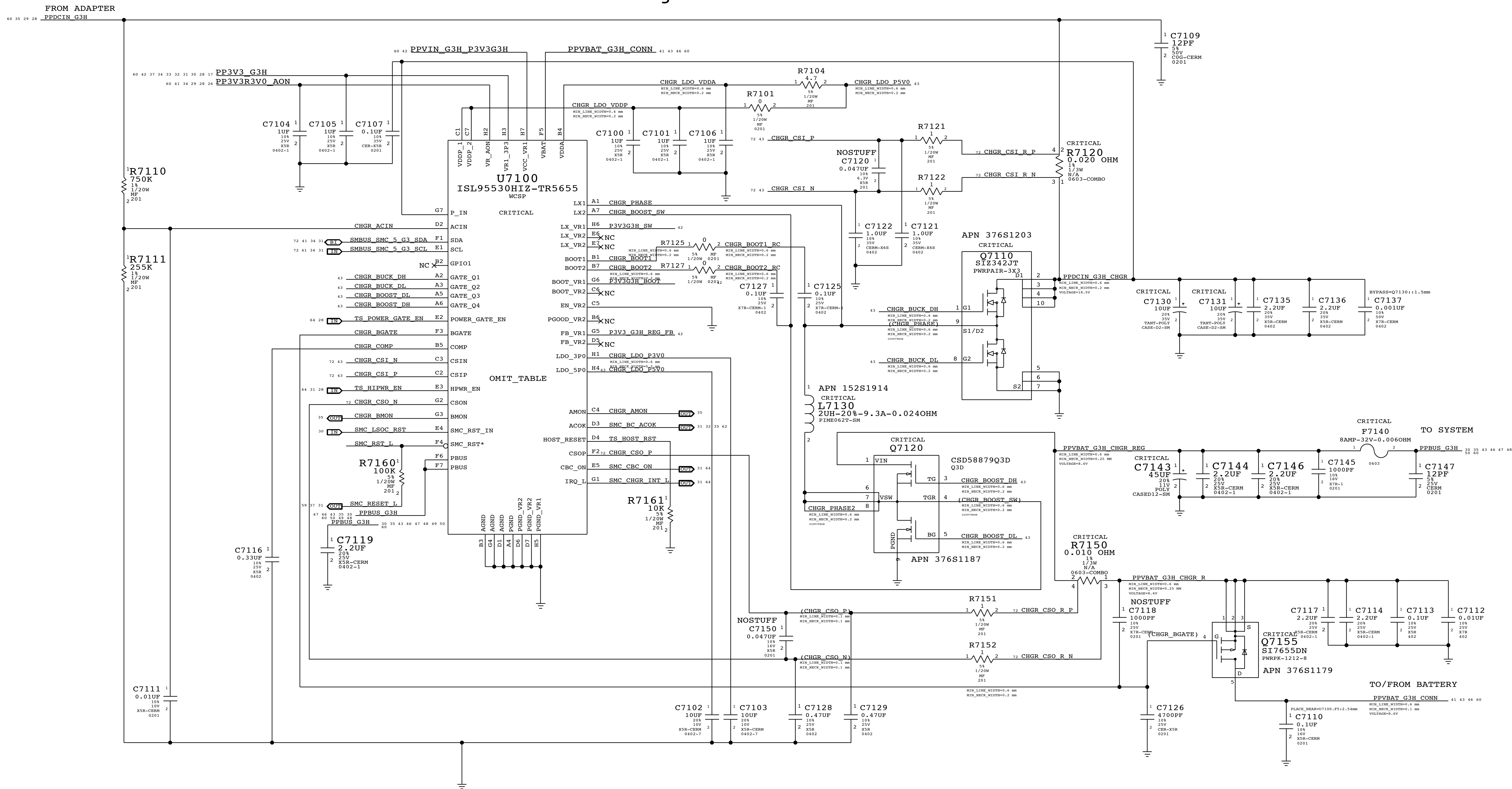
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Charger PMIC - Bansuri



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00347	1	IC,ISL95530B1T11,CHGR PMU,BANSURI,WCSP56	U7100	CRITICAL	

SYNC_MASTER=J92_DEVMLB SYNC_DATE=04/04/2014

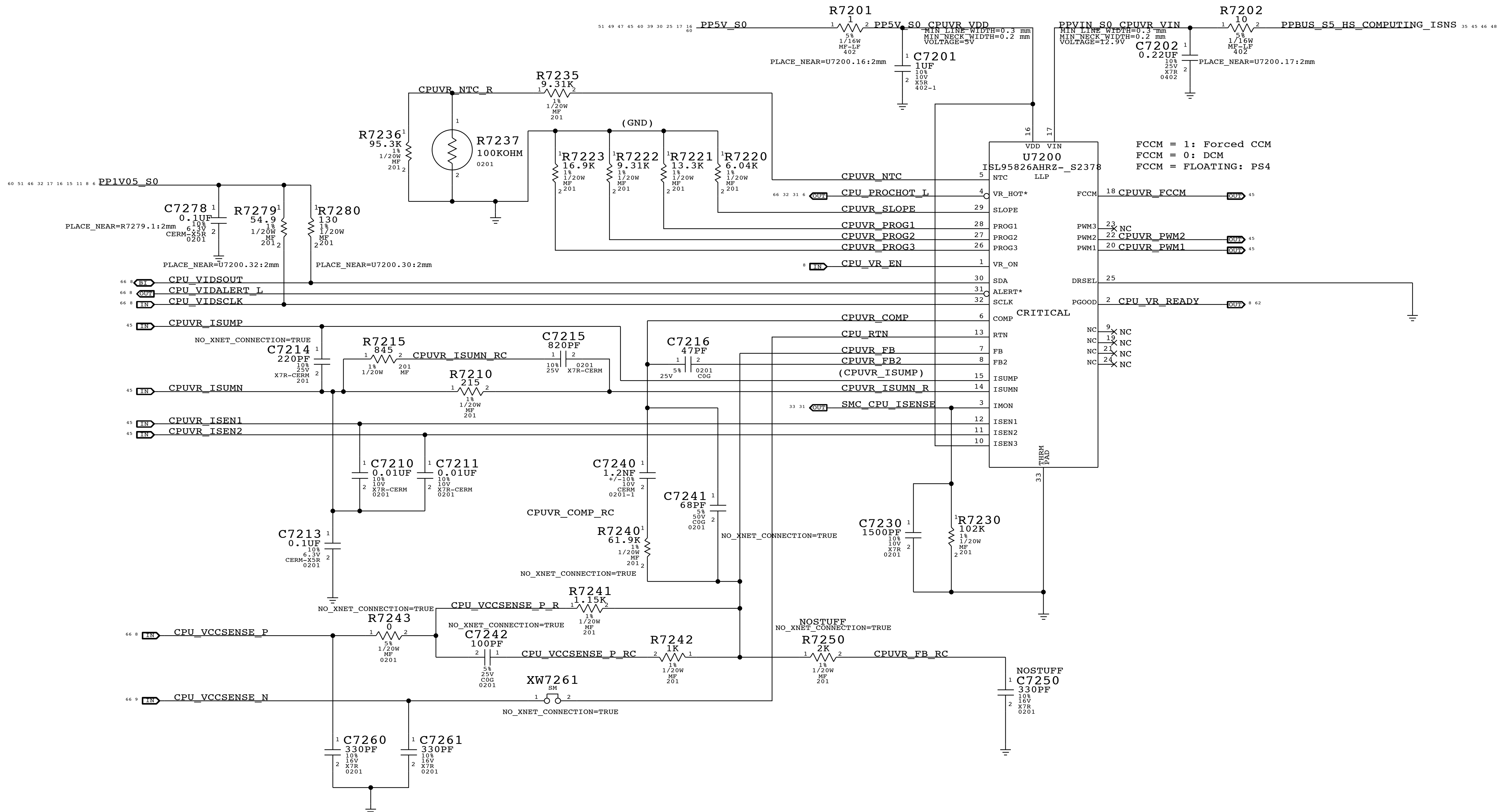
PAGE TITLE
PBus Supply & Battery Charger

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J43_MLB SYNC_DATE=10/09/2012

PAGE TITLE

CPU VR12.6 VCC Regulator IC

Apple Inc.

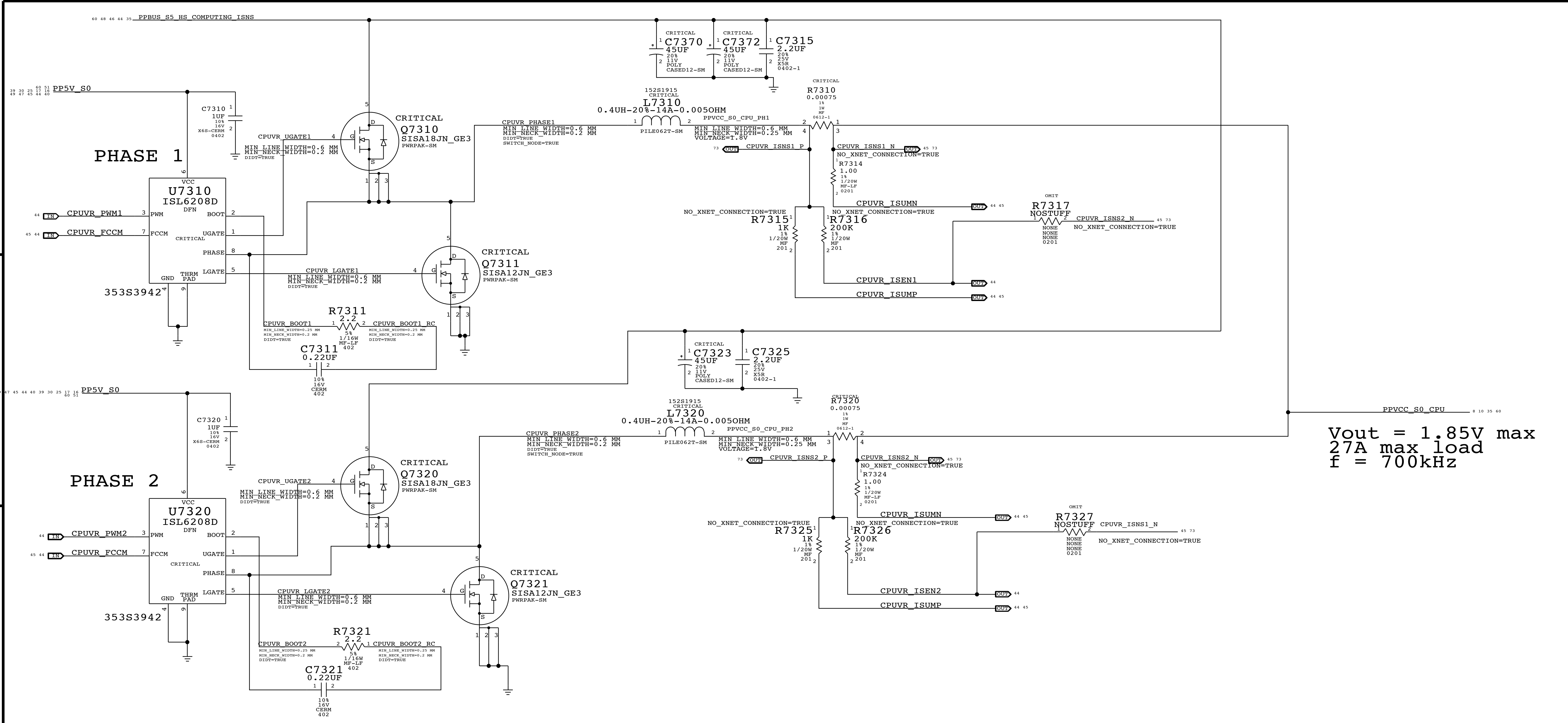
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 72 OF 130

SHEET 44 OF 75



SYNC_MASTER=DEV_MLB SYNC_DATE=04/17/2014

PAGE TITLE

CPU VR12.5 VCC Power Stage

Apple Inc.

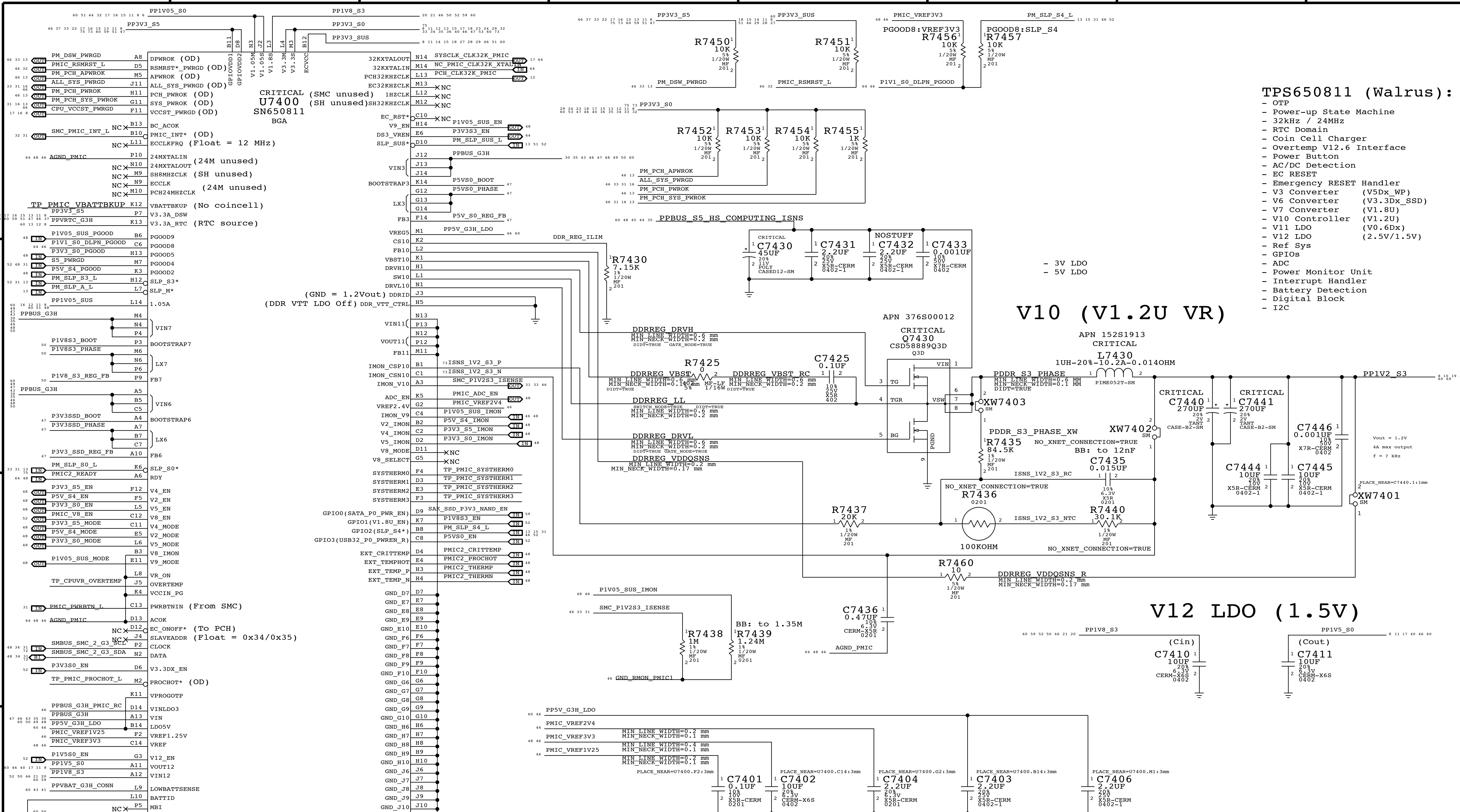
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 73 OF 130

SHEET 45 OF 75



SYNC_MASTER=J92_DEVMLB SYNC_DATE=04/04/2014

PAGE TITLE: LPDDR3 Supply

Apple Inc.

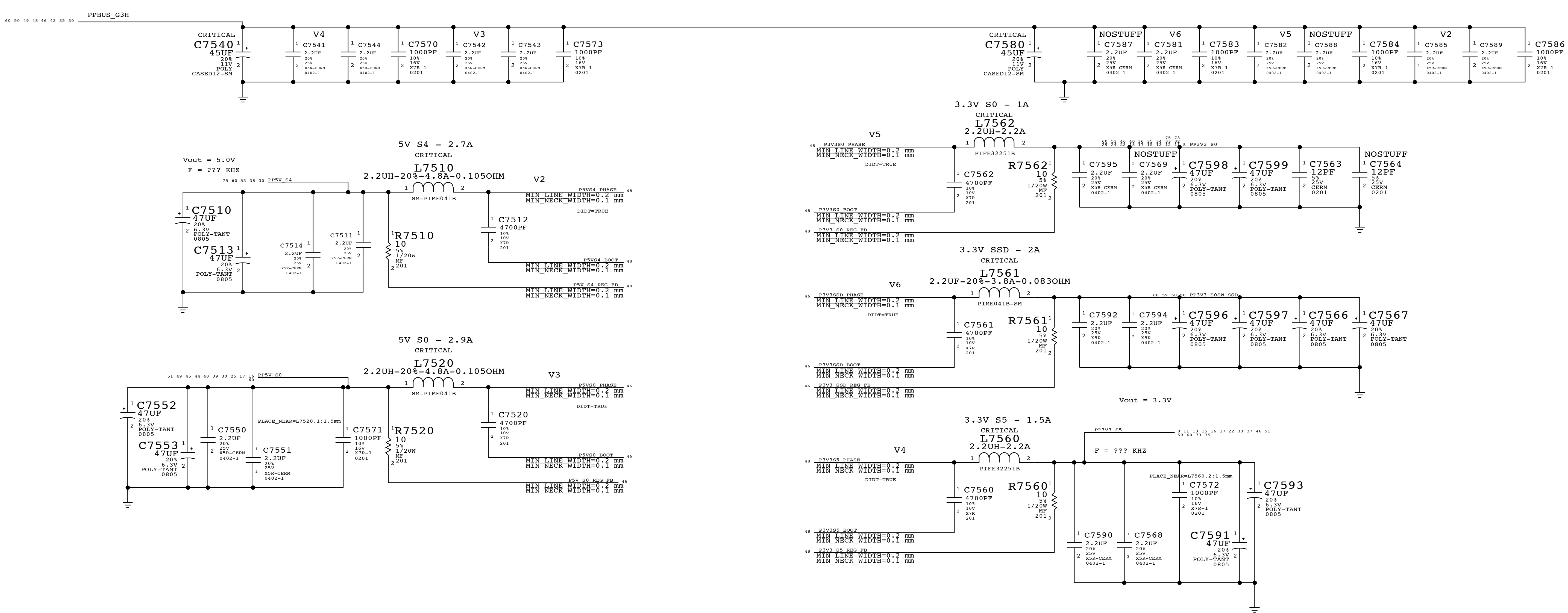
DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

PAGE: 74 OF 130

SHEET: 46 OF 75



SYNC_MASTER=DEV_MLB SYNC_DATE=04/17/2014

PAGE TITLE

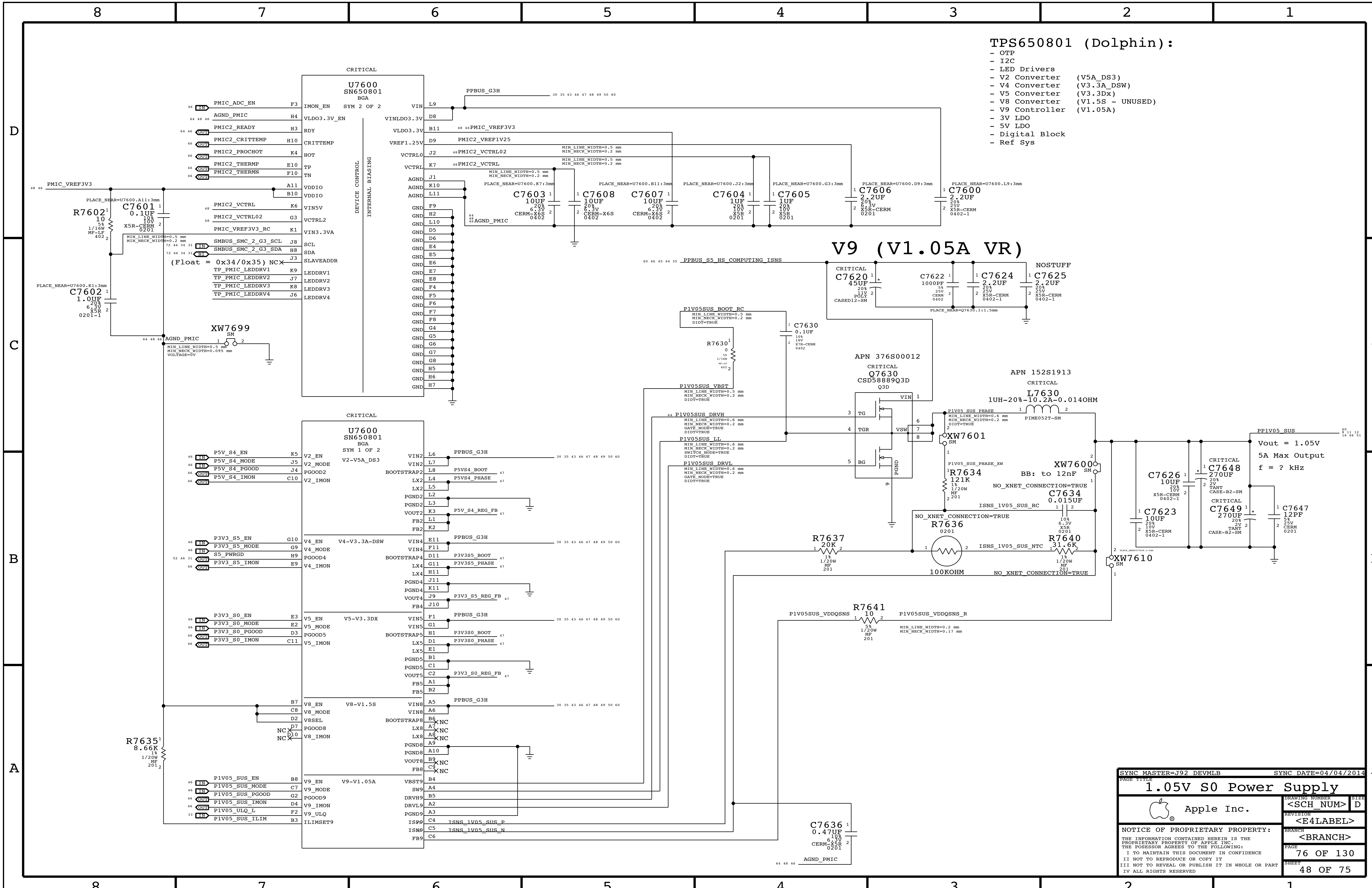
5V & 3.3V Power Supplies

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

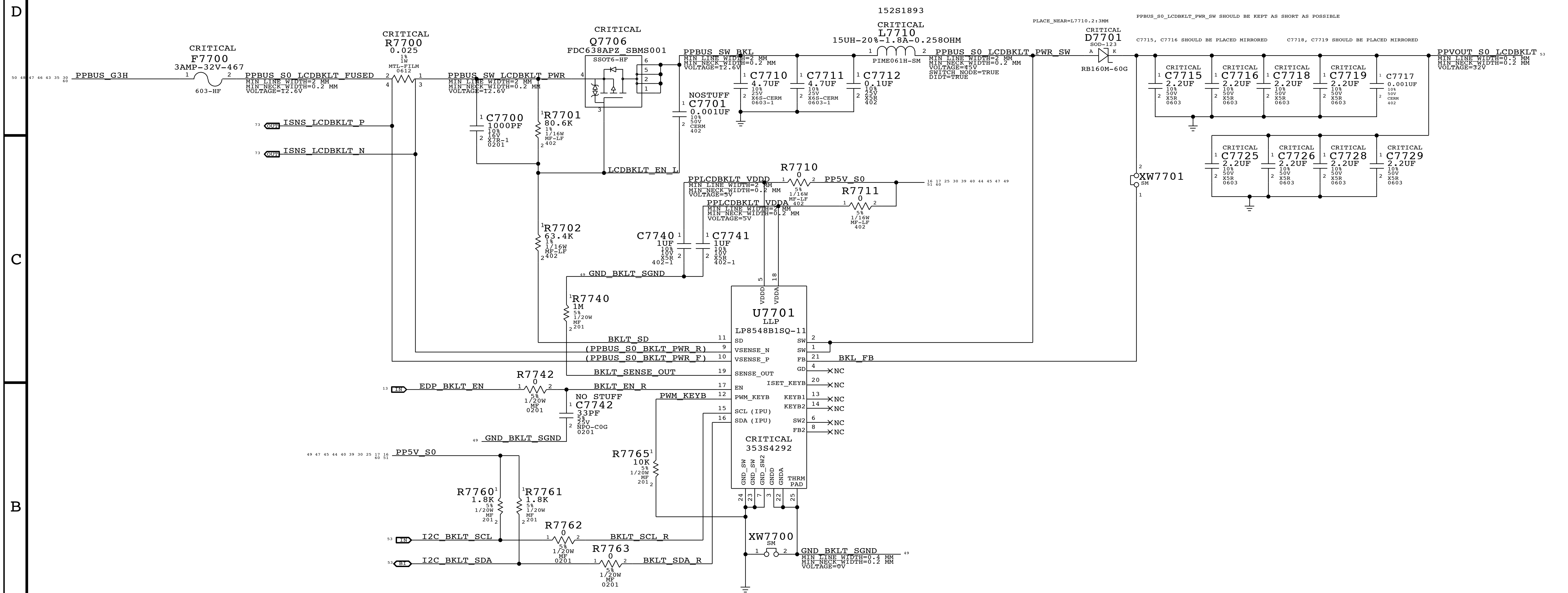
BRANCH <BRANCH>


TPS650801 (Dolphin):
- OTP
- I2C
- LED Drivers
- V2 Converter (V5A_DS3)
- V4 Converter (V3.3A_DSW)
- V5 Converter (V3.3Dx)
- V8 Converter (V1.5S - UNUSED)
- V9 Controller (V1.05A)
- 3V LDO
- 5V LDO
- Digital Block
- Ref Sys
CRITICAL
U7600
SN650801
BGA
SYM 2 OF 2
DEVICE CONTROL
INTERNAL BIASING
SYM 1 OF 2
V2-V5A_DS3
V4-V3.3A-DSW
V5-V3.3DX
V8-V1.5S
V9-V1.05A
(Float = 0x34/0x35) NC
XW7699
V9 (V1.05A VR)
PPBUS_S5_HS_COMPUTING_ISNS
CRITICAL
C7620
45UF
C7622
1000PF
C7624
2.2UF
NOSTUFF
C7625
2.2UF
APN 376S00012
CRITICAL
Q7630
CSD58889Q3D
APN 152S1913
CRITICAL
L7630
1UH-20%-10.2A-0.014OHM
PP1V05_SUS
Vout = 1.05V
5A Max Output
f = ? kHz
XW7600
BB: to 12nF
R7634
121K
C7634
0.015UF
R7636
100KOHM
R7637
20K
R7640
31.6K
XW7610
R7641
10
C7636
0.47UF
R7635
8.66K
C7626
10UF
C7648
270UF
C7649
270UF
C7623
10UF
C7647
12PF
C7603
C7608
C7607
C7604
C7605
C7606
C7600
C7601
C7602
R7602
C7630
R7630
XW7601
SYNC_MASTER=J92_DEVMLB
SYNC_DATE=04/04/2014
PAGE TITLE
1.05V S0 Power Supply
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
D
REVISION
<E4LABEL>
BRANCH
<BRANCH>
PAGE
76 OF 130
SHEET
48 OF 75
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

Page Notes

Power aliases required by this page:
- =PPVIN_S0_LCDBKLT (6-8.6V LCD Backlight Input)
- =PP5V_S0_BKLTCTRL (5V Backlight Driver Input)
- =PP5V_S0_KBDLED (5V Keyboard Backlight Input)

BOM options provided by this page:
BKLT:ENG - Stuffs 10.2 ohm series R for engineering builds
BKLT:PROD - Stuffs 0 ohm series R for production



SYNC_MASTER=J92_DEVMLB SYNC_DATE=10/01/2013

PAGE TITLE

LCD Backlight Driver

Apple Inc.

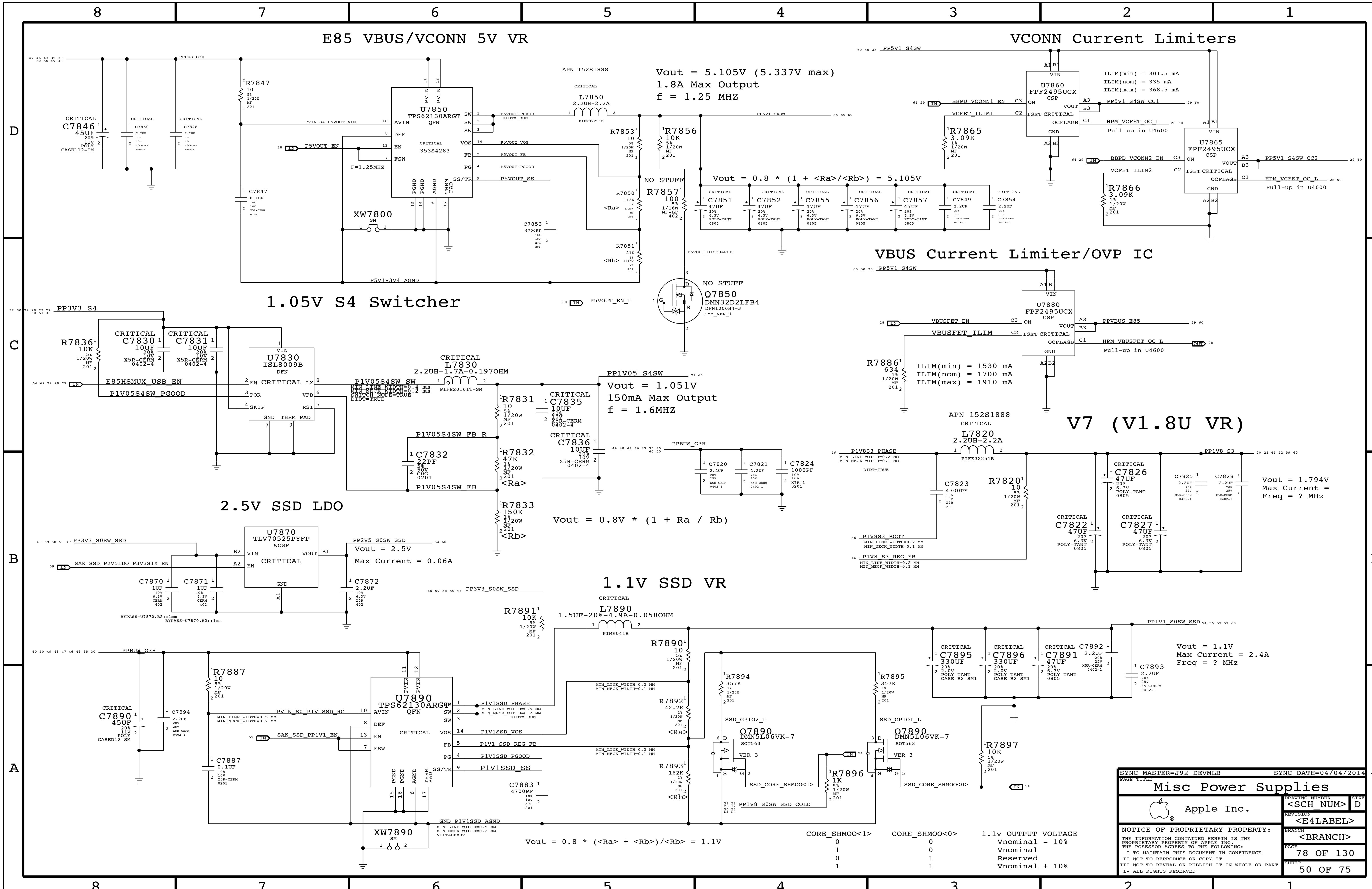
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

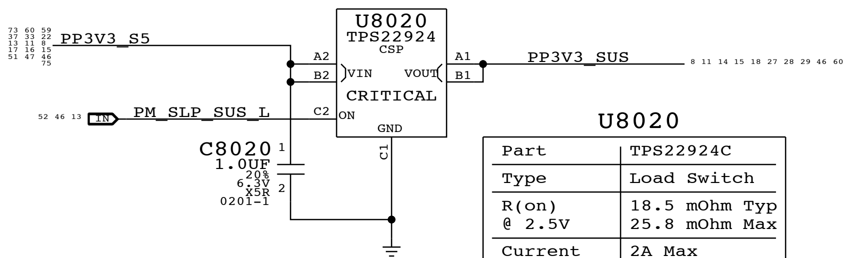
BRANCH <BRANCH>

PAGE 77 OF 130

SHEET 49 OF 75


E85 VBUS/VCONN 5V VR
Vout = 5.105V (5.337V max)
1.8A Max Output
f = 1.25 MHZ
APN 152S1888
Vout = 0.8 * (1 + <Ra>/<Rb>) = 5.105V
VCONN Current Limiters
ILIM(min) = 301.5 mA
ILIM(nom) = 335 mA
ILIM(max) = 368.5 mA
Pull-up in U4600
VBUS Current Limiter/OVP IC
ILIM(min) = 1530 mA
ILIM(nom) = 1700 mA
ILIM(max) = 1910 mA
1.05V S4 Switcher
Vout = 1.051V
150mA Max Output
f = 1.6MHZ
Vout = 0.8V * (1 + Ra / Rb)
2.5V SSD LDO
Vout = 2.5V
Max Current = 0.06A
V7 (V1.8U VR)
Vout = 1.794V
Max Current =
Freq = ? MHz
1.1V SSD VR
Vout = 1.1V
Max Current = 2.4A
Freq = ? MHz
Vout = 0.8 * (<Ra> + <Rb>)/<Rb> = 1.1V
CORE_SHMOO<1> CORE_SHMOO<0> 1.1v OUTPUT VOLTAGE
0 0 Vnominal - 10%
1 0 Vnominal
0 1 Reserved
1 1 Vnominal + 10%
SYNC_MASTER=J92_DEVMLB
SYNC_DATE=04/04/2014
PAGE TITLE
Misc Power Supplies
Apple Inc.
DRAWING NUMBER <SCH_NUM>
SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>
PAGE 78 OF 130
SHEET 50 OF 75
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

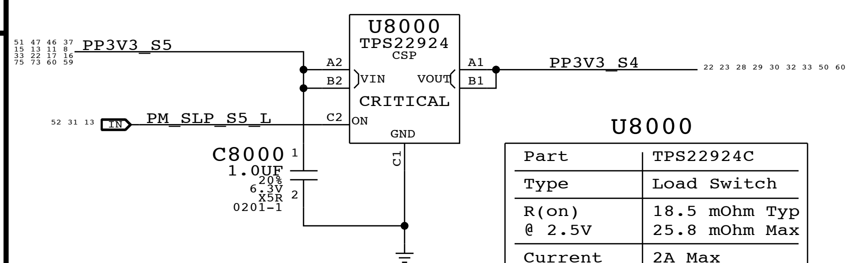
3.3V SUS Switch



U8020

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

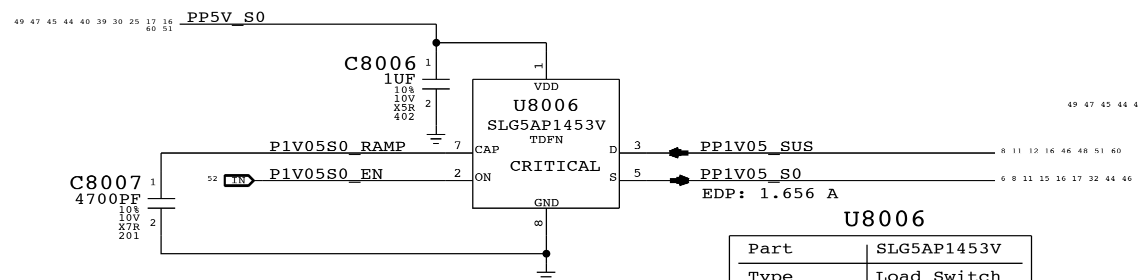
3.3V S4 Switch



U8000

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

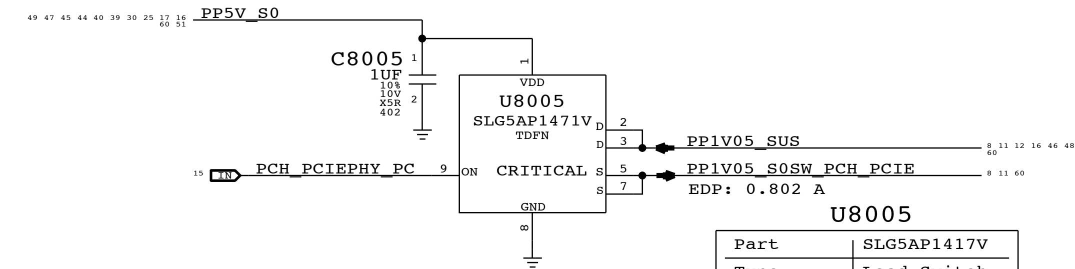
1.05V S0 Switch



U8006

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

1.05V PCH PCIe Switch

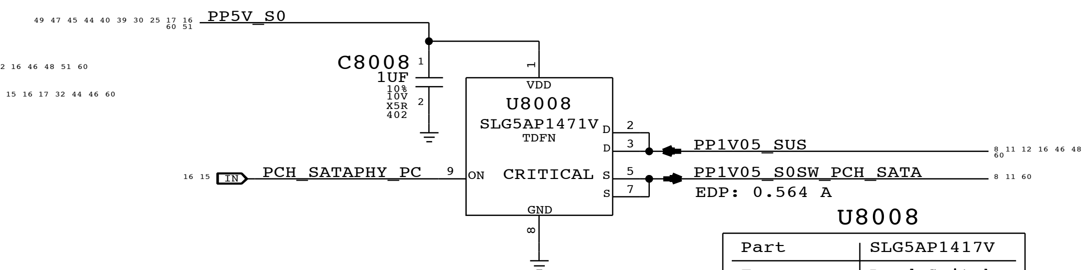


HSIO has turn-on requirement of
<0.1V/uS ramp rate and
<65uS from EN to 95% (1.05V)

U8005

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

1.05V PCH SATA Switch

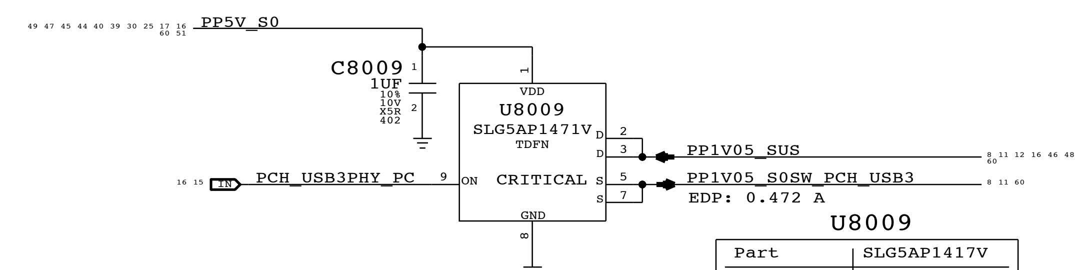


HSIO has turn-on requirement of
<0.1V/uS ramp rate and
<65uS from EN to 95% (1.05V)

U8008

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

1.05V PCH USB3 Switch



HSIO has turn-on requirement of
<0.1V/uS ramp rate and
<65uS from EN to 95% (1.05V)

U8009

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

SYNC_MASTER=J92_DEVMLB SYNC_DATE=07/24/2013

PAGE TITLE

Power FETs

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

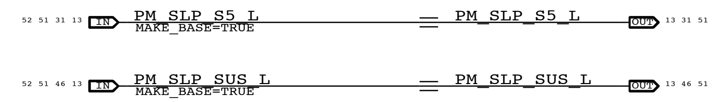
PAGE 80 OF 130

SHEET 51 OF 75

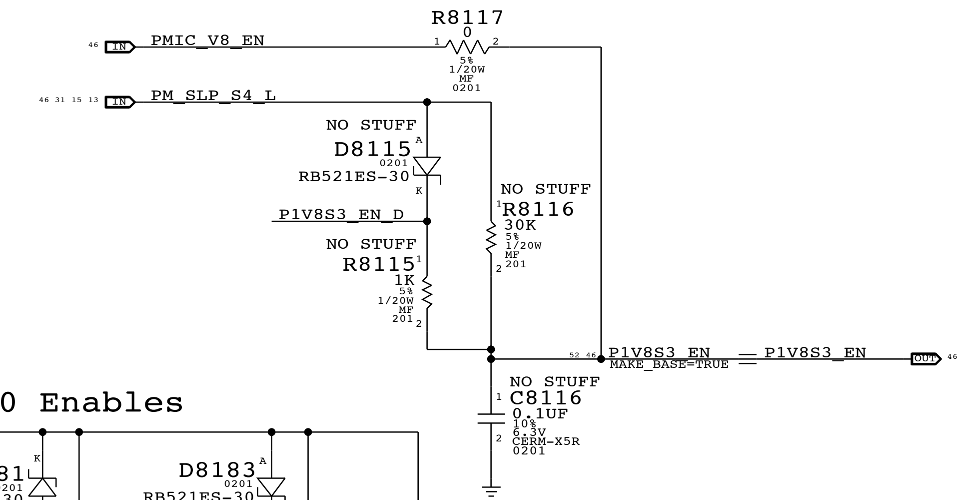
Mobile System Power State Table

State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SUS_EN	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (S4AC)	1	1	1	0	0	0	0
Deep Sleep (S4)	0	1	1	0	0	0	0
Deep Sleep (S5AC)	1	1	0	0	0	0	0
Deep Sleep (S5)	0	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

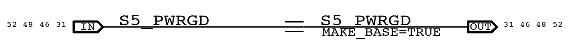
SUS & S4 Enables



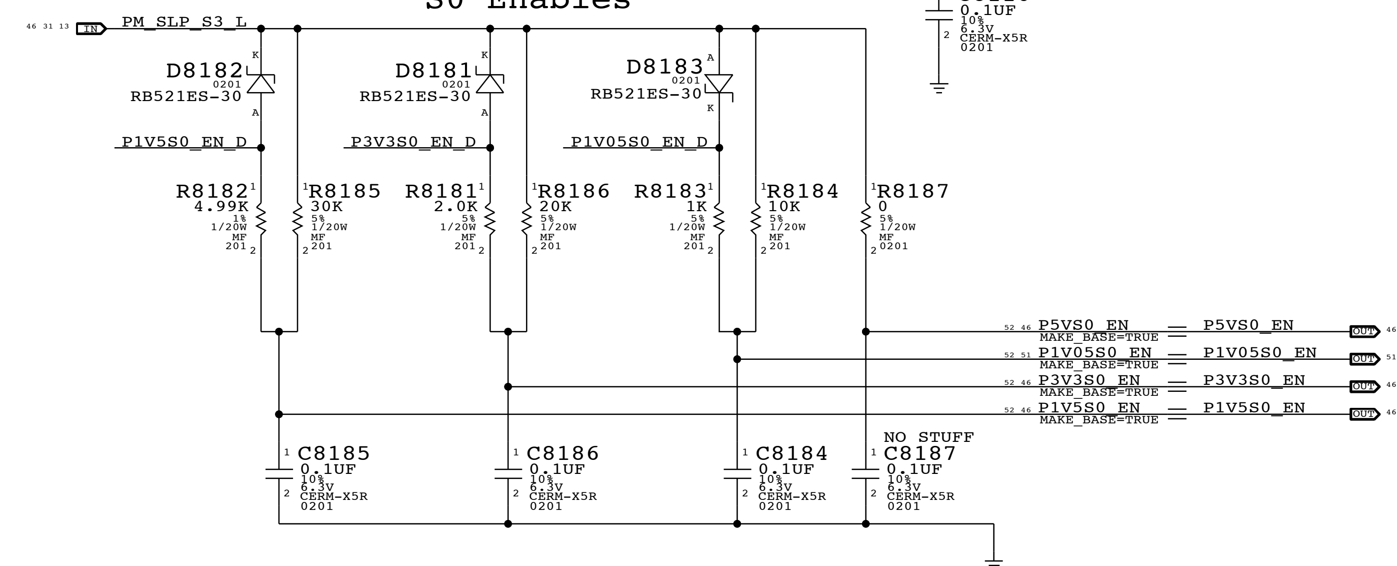
S3 Enables



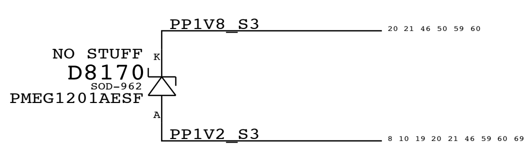
S5 Power Good



S0 Enables



LPDDR power down sequencing support



SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/20/2013

PAGE TITLE

Power Control

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

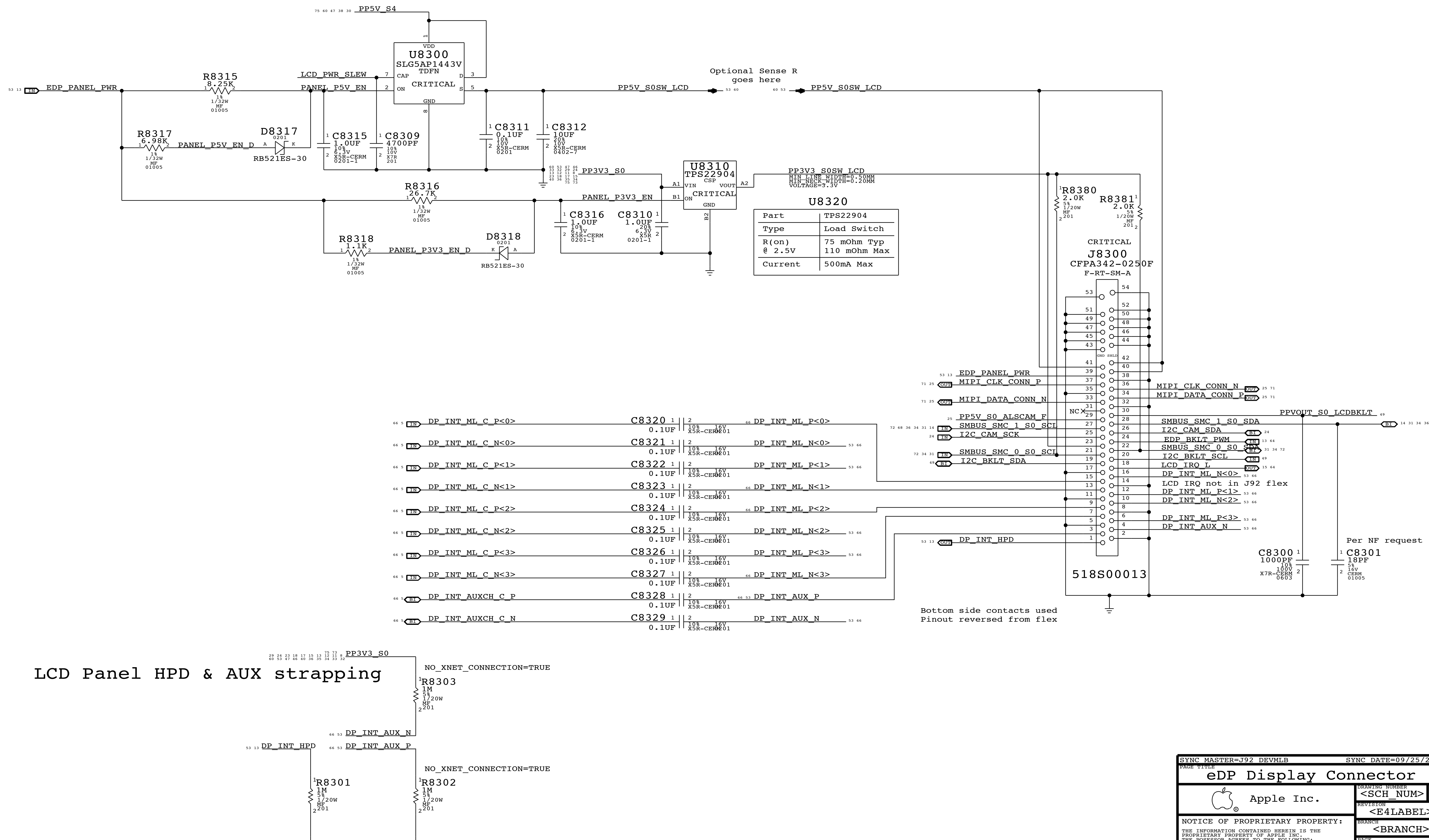
REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 81 OF 130

SHEET 52 OF 75

LCD PANEL INTERFACE (eDP) + Camera (MIPI)



U8320

Part	TPS22904
Type	Load Switch
R(on) @ 2.5V	75 mOhm Typ 110 mOhm Max
Current	500mA Max

SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/25/2013

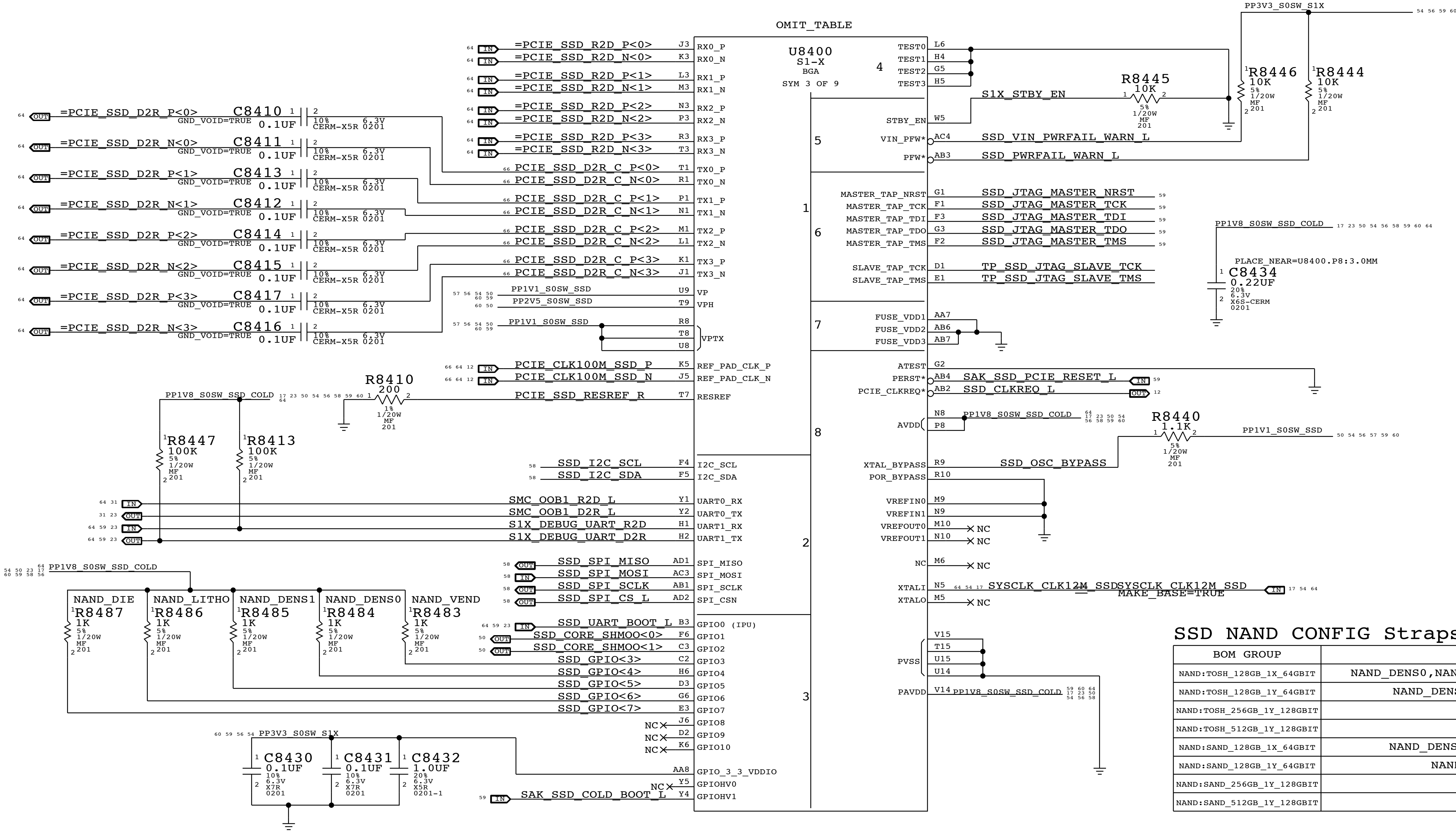
PAGE TITLE
eDP Display Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SSD NAND CONFIG Straps

BOM GROUP	BOM OPTIONS
NAND:TOSH_128GB_1X_64GBIT	NAND_DENS0,NAND_DENS1,NAND_VEND,NAND_LITHO,NAND_DIE,NAND_TYPE:TOSH_128GB_1X_64GBIT
NAND:TOSH_128GB_1Y_64GBIT	NAND_DENS0,NAND_DENS1,NAND_VEND,NAND_DIE,NAND_TYPE:TOSH_128GB_1Y_64GBIT
NAND:TOSH_256GB_1Y_128GBIT	NAND_VEND,NAND_DENS1,NAND_TYPE:TOSH_256GB_1Y_128GBIT
NAND:TOSH_512GB_1Y_128GBIT	NAND_VEND,NAND_DENS0,NAND_TYPE:TOSH_512GB_1Y_128GBIT
NAND:SAND_128GB_1X_64GBIT	NAND_DENS0,NAND_DENS1,NAND_LITHO,NAND_DIE,NAND_TYPE:SAND_128GB_1X_64GBIT
NAND:SAND_128GB_1Y_64GBIT	NAND_DENS0,NAND_DENS1,NAND_DIE,NAND_TYPE:SAND_128GB_1Y_64GBIT
NAND:SAND_256GB_1Y_128GBIT	NAND_DENS1,NAND_TYPE:SAND_256GB_1Y_128GBIT
NAND:SAND_512GB_1Y_128GBIT	NAND_DENS0,NAND_TYPE:SAND_512GB_1Y_128GBIT

SSD NAND Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
998-7360	2	NAND,19NM,64GX8,MLC,RAW,64G,2-CH,LGA60	U8801,U8810	CRITICAL	NAND_TYPE:TOSH_128GB_1X_64GBIT
335S00004	2	NAND,TOGGLE DDR2,2CH,64GB,1YNM,3.3V,LGA60	U8801,U8810	CRITICAL	NAND_TYPE:TOSH_128GB_1Y_64GBIT
335S1031	2	NAND,TOGGLE DDR2,2CH,128GB,1YNM,3.3V,LGA60	U8801,U8810	CRITICAL	NAND_TYPE:TOSH_256GB_1Y_128GBIT
335S1032	2	NAND,TOGGLE DDR2,2CH,256GB,1YNM,3.3V,LGA60	U8801,U8810	CRITICAL	NAND_TYPE:TOSH_512GB_1Y_128GBIT
998-7361	2	NAND,19NM,64GX8,MLC,RAW,64G,2-CH,LGA60	U8801,U8810	CRITICAL	NAND_TYPE:SAND_128GB_1X_64GBIT
335S1044	2	NAND,TOGGLE DDR2,2CH,64GB,1YNM,3.3V,LGA60	U8801,U8810	CRITICAL	NAND_TYPE:SAND_128GB_1Y_64GBIT
335S1041	2	NAND,TOGGLE DDR2,2CH,128GB,1YNM,3.3V,LGA60	U8801,U8810	CRITICAL	NAND_TYPE:SAND_256GB_1Y_128GBIT
335S1045	2	NAND,TOGGLE DDR2,2CH,256GB,1YNM,3.3V,LGA60	U8801,U8810	CRITICAL	NAND_TYPE:SAND_512GB_1Y_128GBIT

S1-X GPIO[10:1] IPD

VENDOR	NAND_VEND(GPIO3)
TOSHIBA	1
SANDISK	0

DENSITY	NAND_DENS0(GPIO4)	NAND_DENS1(GPIO5)
128GB	1	1
256GB	0	1
512GB	1	0

LITHO	NAND_LITHO(GPIO6)
1X	1
1Y	0

DIE SIZE	NAND_DIE(GPIO7)
64Gb	1
128Gb	0

# NAND_CH	(GPIO8)
8 (X56)	1
4 (J92)	0

SYNC_MASTER=J92_SSD SYNC_DATE=02/11/2014

PAGE TITLE

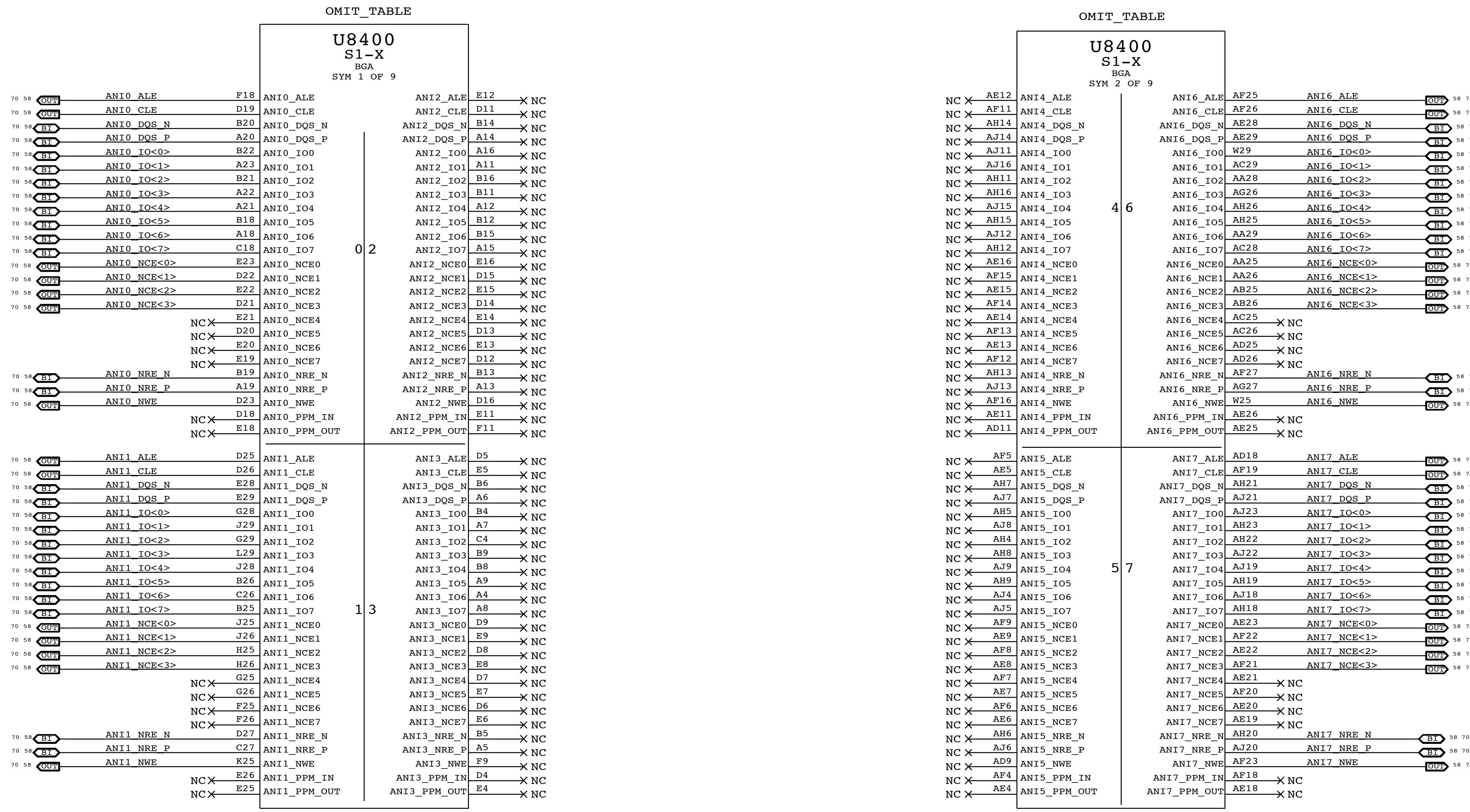
SSD Controller (1 of 4)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_DEVMLB SYNC_DATE=10/10/2013

PAGE TITLE

SSD Controller (2 of 4)

Apple Inc.

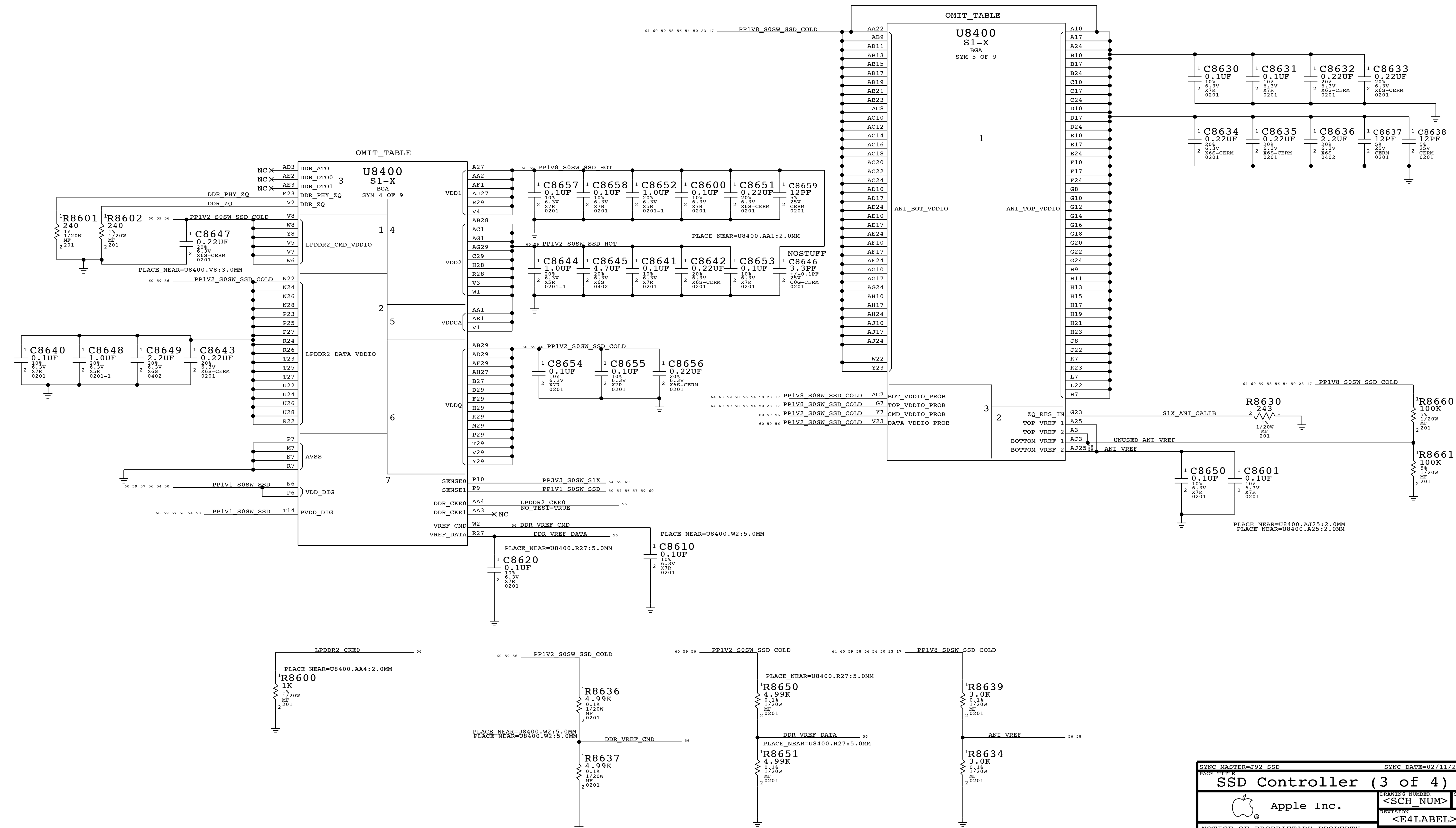
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 85 OF 130

SHEET 55 OF 75



SYNC_MASTER=J92_SSD SYNC_DATE=02/11/2014

PAGE TITLE

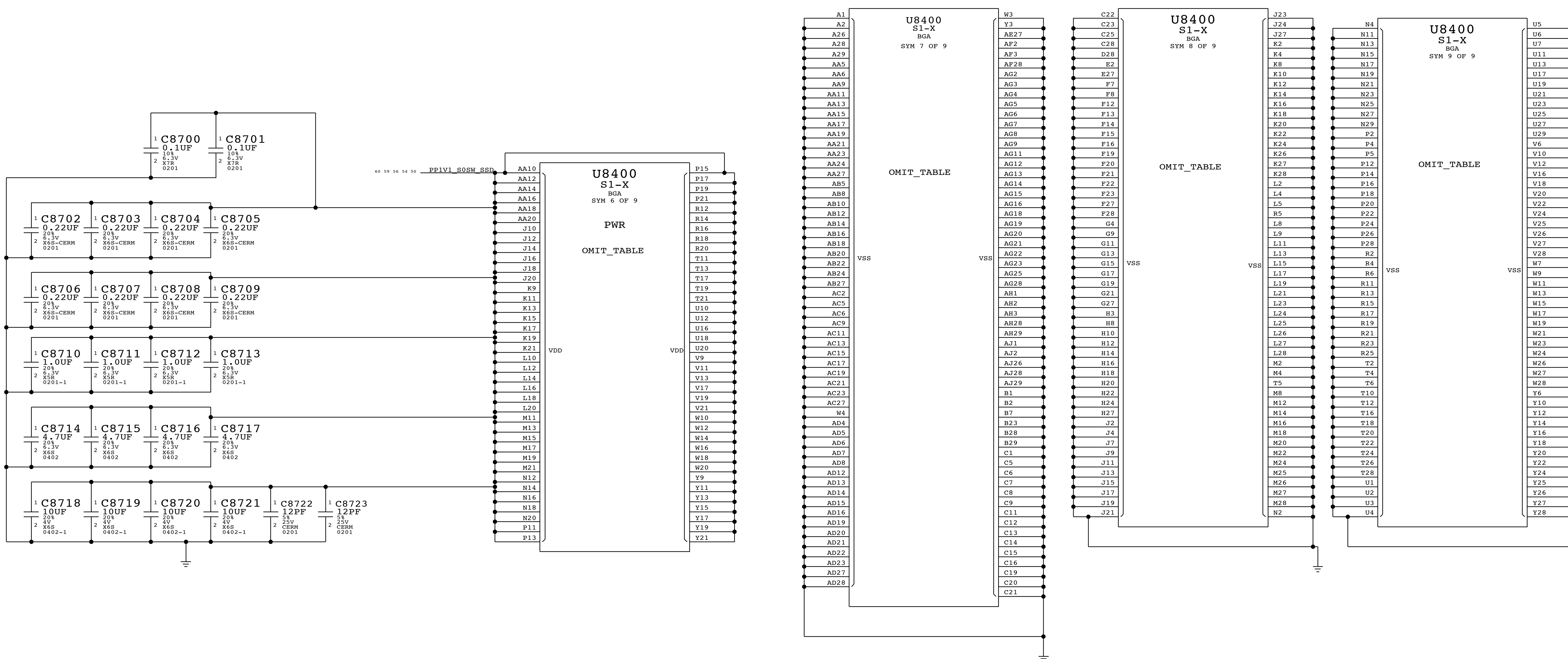
SSD Controller (3 of 4)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_DEVMLB SYNC_DATE=10/10/2013

PAGE TITLE

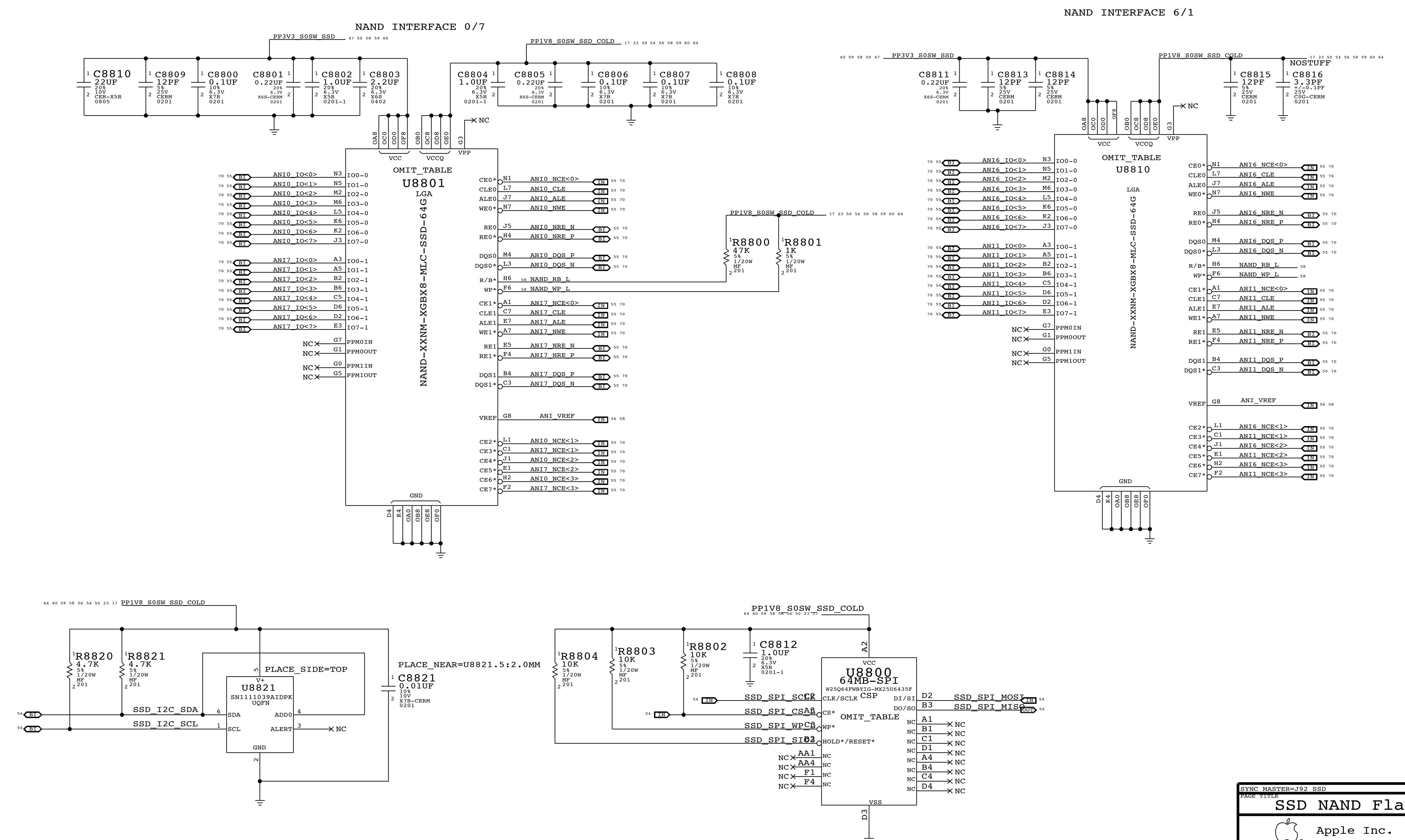
SSD Controller (4 of 4)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



SYNC_MASTER=J92_SSD SYNC_DATE=10/07/2013

PAGE TITLE

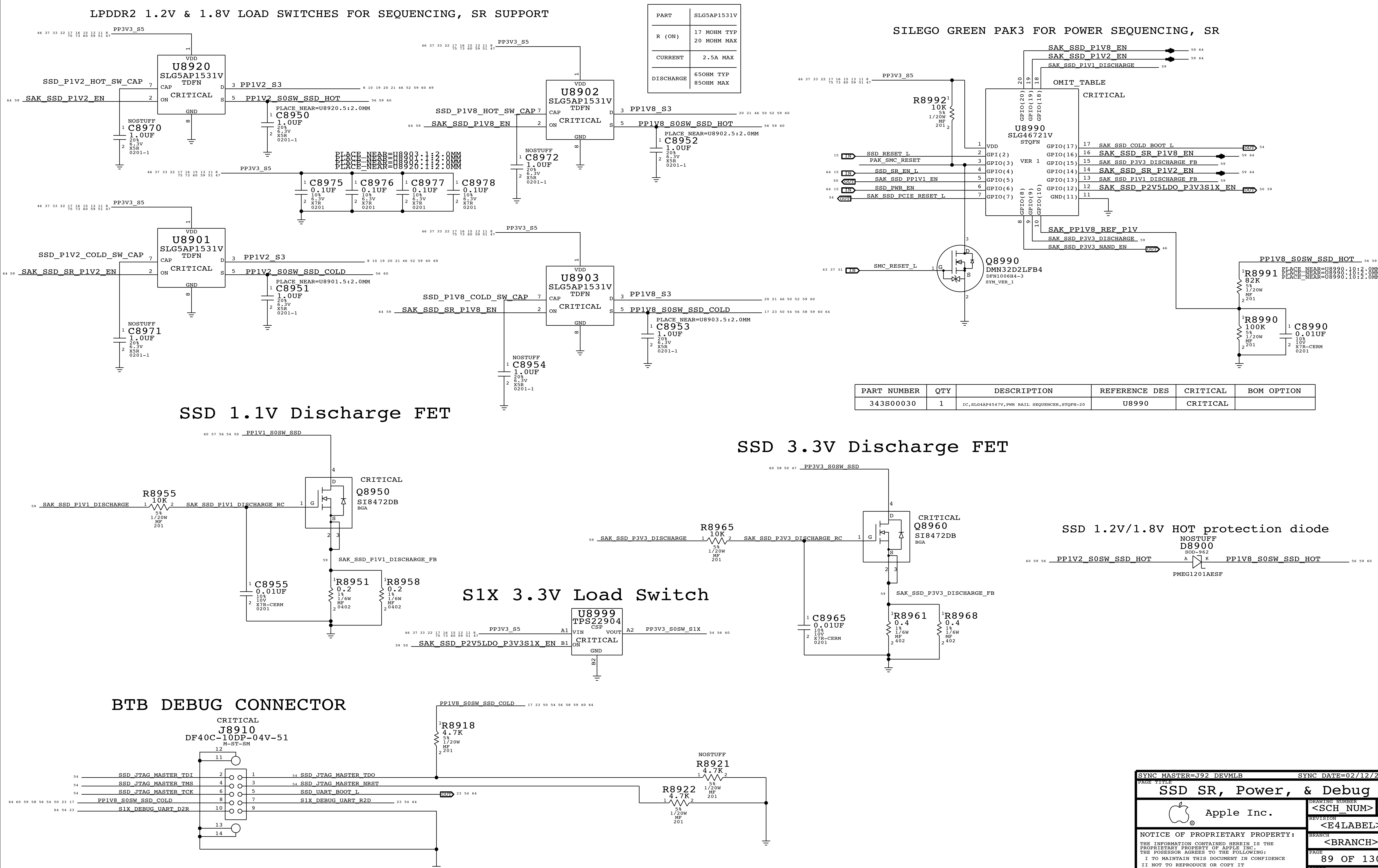
SSD NAND Flash & ROM

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

REVISION <E4LABEL>

BRANCH <BRANCH>



PART	SLG5AP1531V
R (ON)	17 MOHM TYP 20 MOHM MAX
CURRENT	2.5A MAX
DISCHARGE	65OHM TYP 85OHM MAX

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
343S00030	1	IC,SLG4AP4547V,PWR RAIL SEQUENCER,STQFN-20	U8990	CRITICAL	

SYNC_MASTER=J92_DEVMLB SYNC_DATE=02/12/2014

PAGE TITLE

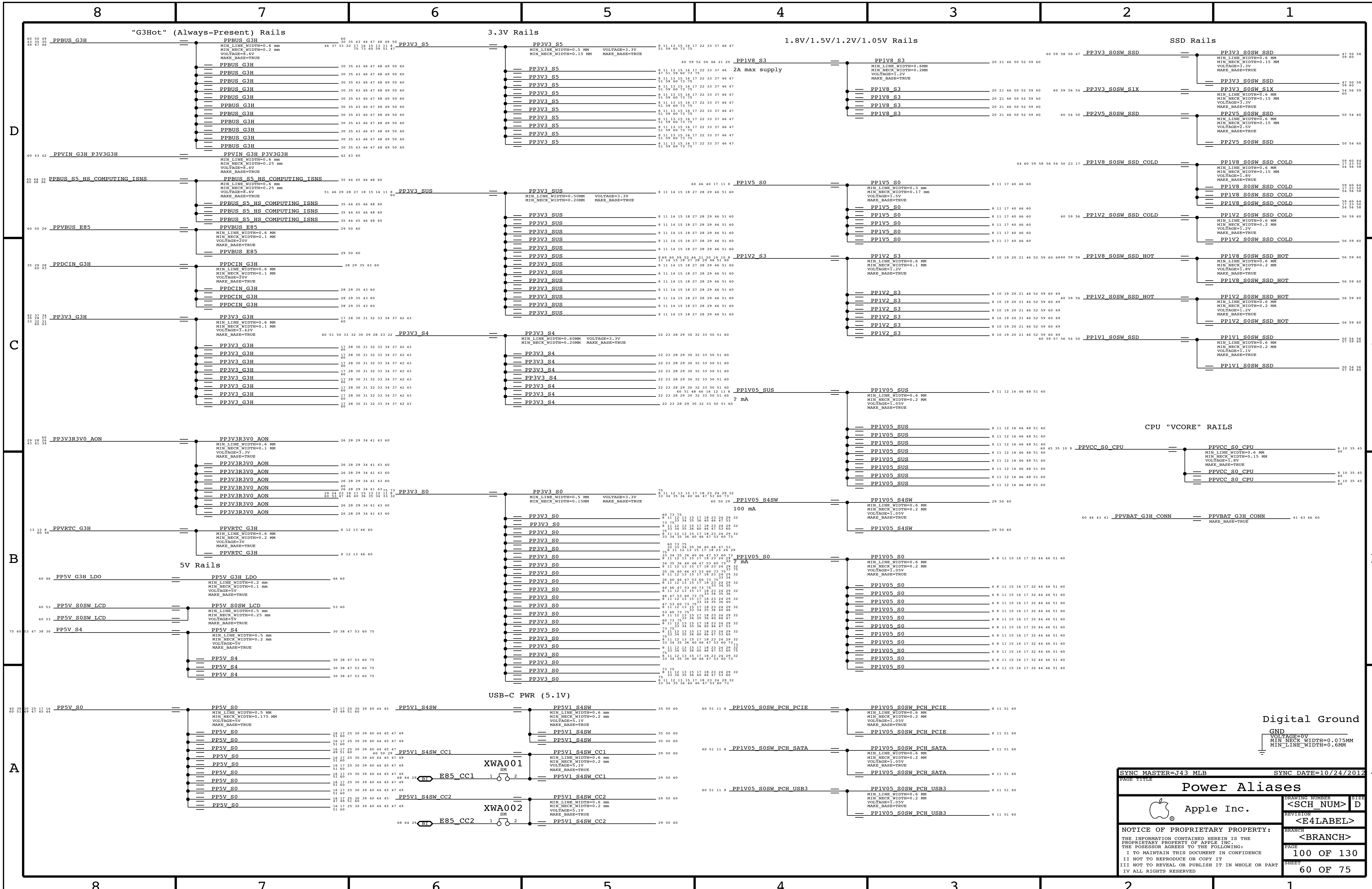
SSD SR, Power, & Debug

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>



"G3Hot" (Always-Present) Rails
PPBUS_G3H
MIN_LINE_WIDTH=0.6 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=8.6V
MAKE_BASE=TRUE
PPVIN_G3H_P3V3G3H
PPBUS_S5_HS_COMPUTING_ISNS
PPVBUS_E85
PPDCIN_G3H
PP3V3_G3H
PP3V3R3V0_AON
PPVRTC_G3H
5V Rails
PP5V_G3H_LDO
PP5V_S0SW_LCD
PP5V_S4
PP5V_S0
3.3V Rails
PP3V3_S5
PP3V3_SUS
PP3V3_S4
PP3V3_S0
USB-C PWR (5.1V)
PP5V1_S4SW
PP5V1_S4SW_CC1
PP5V1_S4SW_CC2
XWA001
XWA002
E85_CC1
E85_CC2
1.8V/1.5V/1.2V/1.05V Rails
PP1V8_S3
2A max supply
PP1V5_S0
PP1V2_S3
PP1V05_SUS
? mA
PP1V05_S4SW
100 mA
PP1V05_S0
PP1V05_S0SW_PCH_PCIE
PP1V05_S0SW_PCH_SATA
PP1V05_S0SW_PCH_USB3
SSD Rails
PP3V3_S0SW_SSD
PP3V3_S0SW_S1X
PP2V5_S0SW_SSD
PP1V8_S0SW_SSD_COLD
PP1V2_S0SW_SSD_COLD
PP1V8_S0SW_SSD_HOT
PP1V2_S0SW_SSD_HOT
PP1V1_S0SW_SSD
CPU "VCORE" RAILS
PPVCC_S0_CPU
PPVBAT_G3H_CONN
Digital Ground
GND
VOLTAGE=0V
MIN_NECK_WIDTH=0.075MM
MIN_LINE_WIDTH=0.6MM
SYNC_MASTER=J43_MLB
SYNC_DATE=10/24/2012
PAGE TITLE
Power Aliases
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
D
REVISION
<E4LABEL>
BRANCH
<BRANCH>
PAGE
100 OF 130
SHEET
60 OF 75
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

Memory Bit/Byte Swizzle

LPDDR3 Command/Address

MAKE_BASE

Ref	IN		OUT	Ref
7	=MEM_A_A<5>	TRUE	MEM_A_CAA<0>	20 69
7	=MEM_A_A<9>	TRUE	MEM_A_CAA<1>	20 69
7	=MEM_A_A<6>	TRUE	MEM_A_CAA<2>	20 69
7	=MEM_A_A<8>	TRUE	MEM_A_CAA<3>	20 69
7	=MEM_A_A<7>	TRUE	MEM_A_CAA<4>	20 69
7	=MEM_A_BA<2>	TRUE	MEM_A_CAA<5>	20 69
69 61 20 7	MEM_A_CAA<6>	TRUE	MEM_A_CAA<6>	7 20 61 69
7	=MEM_A_A<11>	TRUE	MEM_A_CAA<7>	20 69
7	=MEM_A_A<15>	TRUE	MEM_A_CAA<8>	20 69
7	=MEM_A_A<14>	TRUE	MEM_A_CAA<9>	20 69
7	=MEM_A_A<13>	TRUE	MEM_A_CAB<0>	20 69
7	=MEM_A_CAS_L	TRUE	MEM_A_CAB<1>	20 69
7	=MEM_A_WE_L	TRUE	MEM_A_CAB<2>	20 69
7	=MEM_A_RAS_L	TRUE	MEM_A_CAB<3>	20 69
7	=MEM_A_BA<0>	TRUE	MEM_A_CAB<4>	20 69
7	=MEM_A_A<2>	TRUE	MEM_A_CAB<5>	20 69
69 61 20 7	MEM_A_CAB<6>	TRUE	MEM_A_CAB<6>	7 20 61 69
7	=MEM_A_A<10>	TRUE	MEM_A_CAB<7>	20 69
7	=MEM_A_A<1>	TRUE	MEM_A_CAB<8>	20 69
7	=MEM_A_A<0>	TRUE	MEM_A_CAB<9>	20 69
69 61 20 7	MEM_A_ODT<0>	TRUE	MEM_A_ODT<0>	7 20 61 69
61 7	TP_LPDDR3_RSVD1	TRUE	TP_LPDDR3_RSVD1	7 61
61 7	TP_LPDDR3_RSVD2	TRUE	TP_LPDDR3_RSVD2	7 61
7	=MEM_B_A<5>	TRUE	MEM_B_CAA<0>	21 64 69
7	=MEM_B_A<9>	TRUE	MEM_B_CAA<1>	21 69
7	=MEM_B_A<6>	TRUE	MEM_B_CAA<2>	21 64 69
7	=MEM_B_A<8>	TRUE	MEM_B_CAA<3>	21 64 69
7	=MEM_B_A<7>	TRUE	MEM_B_CAA<4>	21 69
7	=MEM_B_BA<2>	TRUE	MEM_B_CAA<5>	21 69
69 61 21 7	MEM_B_CAA<6>	TRUE	MEM_B_CAA<6>	7 21 61 69
7	=MEM_B_A<11>	TRUE	MEM_B_CAA<7>	21 69
7	=MEM_B_A<15>	TRUE	MEM_B_CAA<8>	21 69
7	=MEM_B_A<14>	TRUE	MEM_B_CAA<9>	21 69
7	=MEM_B_A<13>	TRUE	MEM_B_CAB<0>	21 69
7	=MEM_B_CAS_L	TRUE	MEM_B_CAB<1>	21 64 69
7	=MEM_B_WE_L	TRUE	MEM_B_CAB<2>	21 69
7	=MEM_B_RAS_L	TRUE	MEM_B_CAB<3>	21 69
7	=MEM_B_BA<0>	TRUE	MEM_B_CAB<4>	21 69
7	=MEM_B_A<2>	TRUE	MEM_B_CAB<5>	21 69
69 61 21 7	MEM_B_CAB<6>	TRUE	MEM_B_CAB<6>	7 21 61 69
7	=MEM_B_A<10>	TRUE	MEM_B_CAB<7>	21 69
7	=MEM_B_A<1>	TRUE	MEM_B_CAB<8>	21 69
7	=MEM_B_A<0>	TRUE	MEM_B_CAB<9>	21 69
69 61 21 7	MEM_B_ODT<0>	TRUE	MEM_B_ODT<0>	7 21 61 69
61 7	TP_LPDDR3_RSVD3	TRUE	TP_LPDDR3_RSVD3	7 61
61 7	TP_LPDDR3_RSVD4	TRUE	TP_LPDDR3_RSVD4	7 61

MAKE_BASE

Ref	BI		BI	Ref
20	=MEM_A_DQ<0>	TRUE	MEM_A_DQ<34>	7 69
20	=MEM_A_DQ<1>	TRUE	MEM_A_DQ<38>	7 69
69 61 20 7	MEM_A_DQ<33>	TRUE	MEM_A_DQ<33>	7 20 61 69
20	=MEM_A_DQ<3>	TRUE	MEM_A_DQ<36>	7 69
20	=MEM_A_DQ<4>	TRUE	MEM_A_DQ<35>	7 69
20	=MEM_A_DQ<5>	TRUE	MEM_A_DQ<37>	7 69
20	=MEM_A_DQ<6>	TRUE	MEM_A_DQ<32>	7 69
20	=MEM_A_DQ<7>	TRUE	MEM_A_DQ<39>	7 69
20	=MEM_A_DQ<8>	TRUE	MEM_A_DQ<44>	7 69
20	=MEM_A_DQ<9>	TRUE	MEM_A_DQ<43>	7 69
20	=MEM_A_DQ<10>	TRUE	MEM_A_DQ<46>	7 69
20	=MEM_A_DQ<11>	TRUE	MEM_A_DQ<45>	7 69
20	=MEM_A_DQ<12>	TRUE	MEM_A_DQ<42>	7 69
20	=MEM_A_DQ<13>	TRUE	MEM_A_DQ<40>	7 69
20	=MEM_A_DQ<14>	TRUE	MEM_A_DQ<47>	7 69
20	=MEM_A_DQ<15>	TRUE	MEM_A_DQ<41>	7 69
20	=MEM_A_DQ<16>	TRUE	MEM_A_DQ<55>	7 69
20	=MEM_A_DQ<17>	TRUE	MEM_A_DQ<48>	7 69
20	=MEM_A_DQ<18>	TRUE	MEM_A_DQ<54>	7 69
20	=MEM_A_DQ<19>	TRUE	MEM_A_DQ<53>	7 69
20	=MEM_A_DQ<20>	TRUE	MEM_A_DQ<49>	7 69
20	=MEM_A_DQ<21>	TRUE	MEM_A_DQ<51>	7 69
20	=MEM_A_DQ<22>	TRUE	MEM_A_DQ<50>	7 69
20	=MEM_A_DQ<23>	TRUE	MEM_A_DQ<52>	7 69
20	=MEM_A_DQ<24>	TRUE	MEM_A_DQ<61>	7 69
20	=MEM_A_DQ<25>	TRUE	MEM_A_DQ<60>	7 69
20	=MEM_A_DQ<26>	TRUE	MEM_A_DQ<63>	7 69
20	=MEM_A_DQ<27>	TRUE	MEM_A_DQ<62>	7 69
20	=MEM_A_DQ<28>	TRUE	MEM_A_DQ<57>	7 69
20	=MEM_A_DQ<29>	TRUE	MEM_A_DQ<58>	7 69
20	=MEM_A_DQ<30>	TRUE	MEM_A_DQ<56>	7 69
20	=MEM_A_DQ<31>	TRUE	MEM_A_DQ<59>	7 69
20	=MEM_A_DQ<32>	TRUE	MEM_A_DQ<2>	7 69
20	=MEM_A_DQ<33>	TRUE	MEM_A_DQ<5>	7 69
20	=MEM_A_DQ<34>	TRUE	MEM_A_DQ<7>	7 69
20	=MEM_A_DQ<35>	TRUE	MEM_A_DQ<3>	7 69
20	=MEM_A_DQ<36>	TRUE	MEM_A_DQ<6>	7 69
20	=MEM_A_DQ<37>	TRUE	MEM_A_DQ<0>	7 69
20	=MEM_A_DQ<38>	TRUE	MEM_A_DQ<1>	7 69
20	=MEM_A_DQ<39>	TRUE	MEM_A_DQ<4>	7 69
20	=MEM_A_DQ<40>	TRUE	MEM_A_DQ<14>	7 69
20	=MEM_A_DQ<41>	TRUE	MEM_A_DQ<15>	7 69
20	=MEM_A_DQ<42>	TRUE	MEM_A_DQ<8>	7 69
20	=MEM_A_DQ<43>	TRUE	MEM_A_DQ<11>	7 69
20	=MEM_A_DQ<44>	TRUE	MEM_A_DQ<10>	7 69
20	=MEM_A_DQ<45>	TRUE	MEM_A_DQ<12>	7 69
20	=MEM_A_DQ<46>	TRUE	MEM_A_DQ<13>	7 69
20	=MEM_A_DQ<47>	TRUE	MEM_A_DQ<9>	7 69
20	=MEM_A_DQ<48>	TRUE	MEM_A_DQ<22>	7 69
20	=MEM_A_DQ<49>	TRUE	MEM_A_DQ<23>	7 69
20	=MEM_A_DQ<50>	TRUE	MEM_A_DQ<19>	7 69
20	=MEM_A_DQ<51>	TRUE	MEM_A_DQ<17>	7 69
20	=MEM_A_DQ<52>	TRUE	MEM_A_DQ<21>	7 69
20	=MEM_A_DQ<53>	TRUE	MEM_A_DQ<16>	7 69
20	=MEM_A_DQ<54>	TRUE	MEM_A_DQ<18>	7 69
20	=MEM_A_DQ<55>	TRUE	MEM_A_DQ<20>	7 69
20	=MEM_A_DQ<56>	TRUE	MEM_A_DQ<24>	7 69
20	=MEM_A_DQ<57>	TRUE	MEM_A_DQ<29>	7 69
20	=MEM_A_DQ<58>	TRUE	MEM_A_DQ<26>	7 69
20	=MEM_A_DQ<59>	TRUE	MEM_A_DQ<28>	7 69
20	=MEM_A_DQ<60>	TRUE	MEM_A_DQ<25>	7 69
20	=MEM_A_DQ<61>	TRUE	MEM_A_DQ<31>	7 69
20	=MEM_A_DQ<62>	TRUE	MEM_A_DQ<30>	7 69
20	=MEM_A_DQ<63>	TRUE	MEM_A_DQ<27>	7 69
20	=MEM_A_DQS_P<0>	TRUE	MEM_A_DQS_P<4>	7 69
20	=MEM_A_DQS_N<0>	TRUE	MEM_A_DQS_N<4>	7 69
20	=MEM_A_DQS_P<1>	TRUE	MEM_A_DQS_P<5>	7 69
20	=MEM_A_DQS_N<1>	TRUE	MEM_A_DQS_N<5>	7 69
69 61 20	MEM_A_DQS_P<6>	TRUE	MEM_A_DQS_P<6>	7 20 61 69
69 61 20	MEM_A_DQS_N<6>	TRUE	MEM_A_DQS_N<6>	7 20 61 69
20	=MEM_A_DQS_P<3>	TRUE	MEM_A_DQS_P<7>	7 69
20	=MEM_A_DQS_N<3>	TRUE	MEM_A_DQS_N<7>	7 69
20	=MEM_A_DQS_P<4>	TRUE	MEM_A_DQS_P<0>	7 69
20	=MEM_A_DQS_N<4>	TRUE	MEM_A_DQS_N<0>	7 69
20	=MEM_A_DQS_P<5>	TRUE	MEM_A_DQS_P<1>	7 69
20	=MEM_A_DQS_N<5>	TRUE	MEM_A_DQS_N<1>	7 69
20	=MEM_A_DQS_P<6>	TRUE	MEM_A_DQS_P<2>	7 69
20	=MEM_A_DQS_N<6>	TRUE	MEM_A_DQS_N<2>	7 69
20	=MEM_A_DQS_P<7>	TRUE	MEM_A_DQS_P<3>	7 69
20	=MEM_A_DQS_N<7>	TRUE	MEM_A_DQS_N<3>	7 69

MAKE_BASE

Ref	BI		BI	Ref
21	=MEM_B_DQ<0>	TRUE	MEM_B_DQ<35>	7 69
21	=MEM_B_DQ<1>	TRUE	MEM_B_DQ<38>	7 69
61 21 7 69	MEM_B_DQ<32>	TRUE	MEM_B_DQ<32>	7 21 61 69
21	=MEM_B_DQ<3>	TRUE	MEM_B_DQ<36>	7 69
21	=MEM_B_DQ<4>	TRUE	MEM_B_DQ<37>	7 69
21	=MEM_B_DQ<5>	TRUE	MEM_B_DQ<39>	7 69
21	=MEM_B_DQ<6>	TRUE	MEM_B_DQ<34>	7 69
21	=MEM_B_DQ<7>	TRUE	MEM_B_DQ<33>	7 69
21	=MEM_B_DQ<8>	TRUE	MEM_B_DQ<41>	7 69
21	=MEM_B_DQ<9>	TRUE	MEM_B_DQ<43>	7 69
21	=MEM_B_DQ<10>	TRUE	MEM_B_DQ<46>	7 69
21	=MEM_B_DQ<11>	TRUE	MEM_B_DQ<45>	7 69
21	=MEM_B_DQ<12>	TRUE	MEM_B_DQ<47>	7 69
21	=MEM_B_DQ<13>	TRUE	MEM_B_DQ<44>	7 69
21	=MEM_B_DQ<14>	TRUE	MEM_B_DQ<42>	7 69
21	=MEM_B_DQ<15>	TRUE	MEM_B_DQ<40>	7 69
21	=MEM_B_DQ<16>	TRUE	MEM_B_DQ<55>	7 69
21	=MEM_B_DQ<17>	TRUE	MEM_B_DQ<50>	7 69
21	=MEM_B_DQ<18>	TRUE	MEM_B_DQ<48>	7 69
21	=MEM_B_DQ<19>	TRUE	MEM_B_DQ<53>	7 69
21	=MEM_B_DQ<20>	TRUE	MEM_B_DQ<51>	7 69
21	=MEM_B_DQ<21>	TRUE	MEM_B_DQ<52>	7 69
21	=MEM_B_DQ<22>	TRUE	MEM_B_DQ<54>	7 69
21	=MEM_B_DQ<23>	TRUE	MEM_B_DQ<49>	7 69
21	=MEM_B_DQ<24>	TRUE	MEM_B_DQ<60>	7 69
21	=MEM_B_DQ<25>	TRUE	MEM_B_DQ<57>	7 69
21	=MEM_B_DQ<26>	TRUE	MEM_B_DQ<59>	7 69
21	=MEM_B_DQ<27>	TRUE	MEM_B_DQ<63>	7 69
21	=MEM_B_DQ<28>	TRUE	MEM_B_DQ<61>	7 69
21	=MEM_B_DQ<29>	TRUE	MEM_B_DQ<56>	7 69
21	=MEM_B_DQ<30>	TRUE	MEM_B_DQ<58>	7 69
21	=MEM_B_DQ<31>	TRUE	MEM_B_DQ<62>	7 69
21	=MEM_B_DQ<32>	TRUE	MEM_B_DQ<6>	7 69
21	=MEM_B_DQ<33>	TRUE	MEM_B_DQ<0>	7 64 69
21	=MEM_B_DQ<34>	TRUE	MEM_B_DQ<3>	7 69
21	=MEM_B_DQ<35>	TRUE	MEM_B_DQ<5>	7 69
21	=MEM_B_DQ<36>	TRUE	MEM_B_DQ<7>	7 69
21	=MEM_B_DQ<37>	TRUE	MEM_B_DQ<1>	7 69
21	=MEM_B_DQ<38>	TRUE	MEM_B_DQ<4>	7 69
21	=MEM_B_DQ<39>	TRUE	MEM_B_DQ<2>	7 69
21	=MEM_B_DQ<40>	TRUE	MEM_B_DQ<8>	7 69
21	=MEM_B_DQ<41>	TRUE	MEM_B_DQ<11>	7 69
21	=MEM_B_DQ<42>	TRUE	MEM_B_DQ<9>	7 69
21	=MEM_B_DQ<43>	TRUE	MEM_B_DQ<12>	7 69
21	=MEM_B_DQ<44>	TRUE	MEM_B_DQ<15>	7 69
21	=MEM_B_DQ<45>	TRUE	MEM_B_DQ<14>	7 69
21	=MEM_B_DQ<46>	TRUE	MEM_B_DQ<13>	7 69
21	=MEM_B_DQ<47>	TRUE	MEM_B_DQ<10>	7 69
21	=MEM_B_DQ<48>	TRUE	MEM_B_DQ<23>	7 69
21	=MEM_B_DQ<49>	TRUE	MEM_B_DQ<18>	7 69
21	=MEM_B_DQ<50>	TRUE	MEM_B_DQ<19>	7 69
21	=MEM_B_DQ<51>	TRUE	MEM_B_DQ<17>	7 69
21	=MEM_B_DQ<52>	TRUE	MEM_B_DQ<20>	7 69
21	=MEM_B_DQ<53>	TRUE	MEM_B_DQ<16>	7 69
21	=MEM_B_DQ<54>	TRUE	MEM_B_DQ<22>	7 69
21	=MEM_B_DQ<55>	TRUE	MEM_B_DQ<21>	7 69
21	=MEM_B_DQ<56>	TRUE	MEM_B_DQ<29>	7 69
21	=MEM_B_DQ<57>	TRUE	MEM_B_DQ<25>	7 69
21	=MEM_B_DQ<58>	TRUE	MEM_B_DQ<24>	7 69
21	=MEM_B_DQ<59>	TRUE	MEM_B_DQ<26>	7 69
21	=MEM_B_DQ<60>	TRUE	MEM_B_DQ<27>	7 69
21	=MEM_B_DQ<61>	TRUE	MEM_B_DQ<28>	7 69
21	=MEM_B_DQ<62>	TRUE	MEM_B_DQ<30>	7 69
21	=MEM_B_DQ<63>	TRUE	MEM_B_DQ<31>	7 69
21	=MEM_B_DQS_P<0>	TRUE	MEM_B_DQS_P<4>	7 69
21	=MEM_B_DQS_N<0>	TRUE	MEM_B_DQS_N<4>	7 69
21	=MEM_B_DQS_P<1>	TRUE	MEM_B_DQS_P<5>	7 69
21	=MEM_B_DQS_N<1>	TRUE	MEM_B_DQS_N<5>	7 69
61 21 7 69	MEM_B_DQS_P<6>	TRUE	MEM_B_DQS_P<6>	7 21 61 69
61 21 7 69	MEM_B_DQS_N<6>	TRUE	MEM_B_DQS_N<6>	7 21 61 69
21	=MEM_B_DQS_P<3>	TRUE	MEM_B_DQS_P<7>	7 69
21	=MEM_B_DQS_N<3>	TRUE	MEM_B_DQS_N<7>	7 69
21	=MEM_B_DQS_P<4>	TRUE	MEM_B_DQS_P<0>	7 64 69
21	=MEM_B_DQS_N<4>	TRUE	MEM_B_DQS_N<0>	7 64 69
21	=MEM_B_DQS_P<5>	TRUE	MEM_B_DQS_P<1>	7 69
21	=MEM_B_DQS_N<5>	TRUE	MEM_B_DQS_N<1>	7 69
21	=MEM_B_DQS_P<6>	TRUE	MEM_B_DQS_P<2>	7 69
21	=MEM_B_DQS_N<6>	TRUE	MEM_B_DQS_N<2>	7 69
21	=MEM_B_DQS_P<7>	TRUE	MEM_B_DQS_P<3>	7 69
21	=MEM_B_DQS_N<7>	TRUE	MEM_B_DQS_N<3>	7 69

61 6 TP_CPU_MEMVTT_PWR_EN_LSVDDQ — TP_CPU_MEMVTT_PWR_EN_LSVDDQ 6 61
MAKE_BASE=TRUE

D C B A

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE
Memory Signal Swaps

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>
PAGE 102 OF 130
SHEET 61 OF 75

MAKE_BASE

Left refs	Net	Alias	Net	Right refs
64 62 28 15	QR_SWITCH_EN	TRUE	QR_SWITCH_EN	15 28 62 64
62 44 8	CPU_VR_READY	TRUE	CPU_VR_READY	8 44 62
62 31 13	SMC_PCH_SUSWARN_L	TRUE	SMC_PCH_SUSWARN_L	13 31 62
62 31 13	SMC_PCH_SUSACK_L	TRUE	SMC_PCH_SUSACK_L	13 31 62
62 43 35 32 31	SMC_BC_ACOK	TRUE	SMC_BC_ACOK	31 32 35 43 62
62 5	TP_ULX_SPARE1	TRUE	TP_ULX_SPARE1	5 62
62 5	TP_ULX_SSP_SPARE	TRUE	TP_ULX_SSP_SPARE	5 62
68 62 12	TP_LPC_CLK24M_LPCPLUS_R	TRUE	TP_LPC_CLK24M_LPCPLUS_R	12 62 68
64 62 28 13	HPM_I2C_INT_L	TRUE	HPM_I2C_INT_L	13 28 62 64
64 62 28 13	HPM_I2C_INT_L			
63 62 28 16 14 64	XDP_USB_EXTA_OC_L	TRUE	XDP_USB_EXTA_OC_L	14 16 28 62 63 64
62 37 32 31 26	SMC_TCK	TRUE	SMC_TCK	26 31 32 37 62
62 37 32 31 26	SMC_TMS	TRUE	SMC_TMS	26 31 32 37 62
62 28	TP_HPM_XTALOUT	TRUE	TP_HPM_XTALOUT	28 62
	GND	TRUE	GND	
68 64 62 29 12	PCIE_CLK100M_TBT_P	TRUE	PCIE_CLK100M_TBT_P	12 29 62 64 68
68 64 62 29 12	PCIE_CLK100M_TBT_N	TRUE	PCIE_CLK100M_TBT_N	12 29 62 64 68
29	=PCIE_E85_D2R_P	TRUE	PCIE_TBT_D2R_P<0>	14 68
29	=PCIE_E85_D2R_N	TRUE	PCIE_TBT_D2R_N<0>	14 68
29	=PCIE_E85_R2D_P	TRUE	PCIE_TBT_R2D_C_P<0>	14 68
29	=PCIE_E85_R2D_N	TRUE	PCIE_TBT_R2D_C_N<0>	14 68
62 29 12	TBT_CLKREQ_L	TRUE	TBT_CLKREQ_L	12 29 62
62 29 15	PCH_TBT_PCIE_RESET_L	TRUE	PCH_TBT_PCIE_RESET_L	15 29 62
64 62 29 28	E85_TEST_MODE_L	TRUE	E85_TEST_MODE_L	28 29 62 64

64 62 50 29 28 27 IN E85HSMUX_USB_EN MAKE_BASE=TRUE — E85HSMUX_USB_EN OUT 27 28 29 50 62 64

SYNC_MASTER=J92_DEVMLB SYNC_DATE=07/08/2014

PAGE TITLE

J92 Signal Aliases

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

NO_TEST Nets

NO_TEST MAKE_BASE

Pages	Net	NO_TEST	MAKE_BASE	Net	Pages	Group
63 12	NC_PCIE_CLK100M_SDP	TRUE	TRUE	NC_PCIE_CLK100M_SDP	12 63	CPU/PCH
63 12	NC_PCIE_CLK100M_SDN	TRUE	TRUE	NC_PCIE_CLK100M_SDN	12 63	
63	NC_PCIE_TBT_D2RP<1>	TRUE	TRUE	NC_PCIE_TBT_D2RP<1>	63	
63	NC_PCIE_TBT_D2RN<1>	TRUE	TRUE	NC_PCIE_TBT_D2RN<1>	63	
63	NC_PCIE_TBT_R2D_CP<1>	TRUE	TRUE	NC_PCIE_TBT_R2D_CP<1>	63	
63	NC_PCIE_TBT_R2D_CN<1>	TRUE	TRUE	NC_PCIE_TBT_R2D_CN<1>	63	
63 14	NC_USB_CAMERAP	TRUE	TRUE	NC_USB_CAMERAP	14 63	
63 14	NC_USB_CAMERAN	TRUE	TRUE	NC_USB_CAMERAN	14 63	
63 14	NC_USB_SDP	TRUE	TRUE	NC_USB_SDP	14 63	
63 14	NC_USB_SDN	TRUE	TRUE	NC_USB_SDN	14 63	
63 12	NC_HDA_SDIN1	TRUE	TRUE	NC_HDA_SDIN1	12 63	
63 13	NC_PCI_PME_L	TRUE	TRUE	NC_PCI_PME_L	13 63	
63 14	NC_CLINK_CLK	TRUE	TRUE	NC_CLINK_CLK	14 63	
63 14	NC_CLINK_DATA	TRUE	TRUE	NC_CLINK_DATA	14 63	
63 14	NC_CLINK_RESET_L	TRUE	TRUE	NC_CLINK_RESET_L	14 63	
	=DP_TBTSNK1_ML_C_P<3..0>	TRUE	TRUE	NC_DP_TBTSNK1_ML_CP<3..0>	5	
	=DP_TBTSNK1_ML_C_N<3..0>	TRUE	TRUE	NC_DP_TBTSNK1_ML_CN<3..0>	5	
63 13	NC_DP_TBTSNK1_DDC_CLK	TRUE	TRUE	NC_DP_TBTSNK1_DDC_CLK	13 63	
63 13	NC_DP_TBTSNK1_DDC_DATA	TRUE	TRUE	NC_DP_TBTSNK1_DDC_DATA	13 63	
63 13	NC_DP_TBTSNK1_AUXCH_CP	TRUE	TRUE	NC_DP_TBTSNK1_AUXCH_CP	13 63	
63 13	NC_DP_TBTSNK1_AUXCH_CN	TRUE	TRUE	NC_DP_TBTSNK1_AUXCH_CN	13 63	
63 13	NC_DP_TBTSNK1_HPD	TRUE	TRUE	NC_DP_TBTSNK1_HPD	13 63	
63 31	NC_SMC_SYS_LED	TRUE	TRUE	NC_SMC_SYS_LED	31 63	SMC
63	NC_SMC_DEBUGPRT_EN_L	TRUE	TRUE	NC_SMC_DEBUGPRT_EN_L	63	
63 31	NC_SMC_SYS_KBDLED	TRUE	TRUE	NC_SMC_SYS_KBDLED	31 63	
63	NC_SMC_T25_EN_L	TRUE	TRUE	NC_SMC_T25_EN_L	63	
63 31	NC_SMC_GFX_THROTTLE_L	TRUE	TRUE	NC_SMC_GFX_THROTTLE_L	31 63	
63 31	NC_SMC_FAN_0_TACH	TRUE	TRUE	NC_SMC_FAN_0_TACH	31 63	
63 31	NC_SMC_FAN_1_CTL	TRUE	TRUE	NC_SMC_FAN_1_CTL	31 63	
63 31	NC_SMC_FAN_1_TACH	TRUE	TRUE	NC_SMC_FAN_1_TACH	31 63	
63 31	NC_SMBUS_SMC_4_ASF_SCL	TRUE	TRUE	NC_SMBUS_SMC_4_ASF_SCL	31 63	
63 31	NC_SMBUS_SMC_4_ASF_SDA	TRUE	TRUE	NC_SMBUS_SMC_4_ASF_SDA	31 63	
64 63 31	NC_SYS_ONEWIRE	TRUE	TRUE	NC_SYS_ONEWIRE	31 63 64	

Unused nets with offpage

(Nets with offpages not used on this project)

Net	Pages
SD_RESET_L	15
SD_PWR_EN	15
ENET_MEDIA_SENSE	15
BT_PWRRST_L	15
TPAD_USB_IF_EN	15
PCH_GPIO12	15
TBT_PWR_EN	15
TBT_CIO_PLUG_EVENT_L	15
JTAG_TBT_TMS_PCH	15
ODD_PWR_EN_L	13
HDMITBTMUX_FLAG	13
HDMITBTMUX_LATCH	13
CAM_SENSOR_WAKE_L	24
LCD_PSR_EN	15
XDP_USB_EXTA_OC_L	14 16 28 62 64
AUD_WAKE_L	15
MIKEY_SPI_CS_L	15
MIKEY_SPI_CLK	15
MIKEY_SPI_MISO	15
MIKEY_SPI_MOSI	15

SYNC_MASTER=J92_DEVMLB SYNC_DATE=07/08/2014

PAGE TITLE

Func Test / No Test

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

REVISION <E4LABEL>

BRANCH <BRANCH>

EE Chaz Probe Points

- PPA501 P3V3S3_EN MAKE_BASE=TRUE = P3V3S3_EN
- PPA502 PMIC2_READY
- PPA503 P1V05SUS_DRVH
- PPA504 SMC_CBC_ON
- PPA505 SMC_CHGR_INT_L
- PPA506 MEM_B_CAA<0>
- PPA507 MEM_B_CLK_P<0>
- PPA508 MEM_B_CLK_N<0>
- PPA509 MEM_B_CS_L<0>
- PPA510 MEM_B_DQ<0>
- PPA511 MEM_B_DQS_P<0>
- PPA512 MEM_B_DQS_N<0>
- PPA513 AP_PCIE_DEV_WAKE
- PPA514 AP_S0IX_WAKE_L
- PPA515 AP_S0IX_WAKE_SEL
- PPA516 E85HSMUX_USB_EN

- PPA519 PCH_BT_UART_D2R
- PPA520 BT_UART_CTS_L
- PPA521 PCH_UART1_R2D
- PPA522 PCIE_CAMERA_D2R_P
- PPA523 PCIE_CLK100M_SSD_P
- PPA524 PCIE_CLK100M_TBT_P

- AGND_PMIC = AGND_PMIC MAKE_BASE=TRUE
- AGND_PMIC
- AGND_PMIC
- AGND_PMIC
- TS_HIPWR_EN = TS_HIPWR_EN MAKE_BASE=TRUE
- TS_POWER_GATE_EN = TS_POWER_GATE_EN MAKE_BASE=TRUE
- SSD_PWR_EN = SSD_PWR_EN MAKE_BASE=TRUE
- SSD_UART_BOOT_L = SSD_UART_BOOT_L MAKE_BASE=TRUE
- S1X_DEBUG_UART_D2R = S1X_DEBUG_UART_D2R MAKE_BASE=TRUE
- S1X_DEBUG_UART_R2D = S1X_DEBUG_UART_R2D MAKE_BASE=TRUE
- PP1V8_S0SW_SSD_COLD = PP1V8_S0SW_SSD_COLD MAKE_BASE=TRUE
- PCIE_CLK100M_SSD_P = PCIE_CLK100M_SSD_P MAKE_BASE=TRUE
- PCIE_CLK100M_SSD_N = PCIE_CLK100M_SSD_N MAKE_BASE=TRUE
- =PCIE_SSD_D2R_P<0..3> = PCIE_SSD_D2R_P<0..3> MAKE_BASE=TRUE
- =PCIE_SSD_D2R_N<0..3> = PCIE_SSD_D2R_N<0..3> MAKE_BASE=TRUE
- =PCIE_SSD_R2D_P<0..3> = PCIE_SSD_R2D_P<0..3> MAKE_BASE=TRUE
- =PCIE_SSD_R2D_N<0..3> = PCIE_SSD_R2D_N<0..3> MAKE_BASE=TRUE
- P1V1_S0_DLPN_PGOOD = P1V1_S0_DLPN_PGOOD MAKE_BASE=TRUE
- SYSCLK_CLK24M_CAMERA = SYSCLK_CLK24M_CAMERA MAKE_BASE=TRUE
- SYSCLK_CLK32K_PMIC = SYSCLK_CLK32K_PMIC MAKE_BASE=TRUE
- TP_SMC_PWRFAIL_WARN_L = TP_SMC_PWRFAIL_WARN_L MAKE_BASE=TRUE

- PPA525 E85_RFU<1>
- PPA526 E85_RFU<2>
- PPA527 E85_CC1
- PPA528 E85_CC2
- PPA529 AUD_PWR_EN
- PPA530 BT_UART_RTS_L
- PPA531 CAMERA_PWR_EN
- PPA532 HPM_I2C_INT_L
- PPA533 EDP_BKLT_PWM
- PPA534 HDA_SDIN0
- PPA535 I2C_PCH_1_SCL
- PPA536 I2C_PCH_1_SDA
- PPA537 LCD_IRQ_L
- PPA538 MEM_CAM_A<8>
- PPA539 MEM_CAM_CS_L

- PPA545 PCH_BT_UART_CTS_L
- PPA546 PCH_BT_UART_R2D
- PPA547 PCH_UART1_D2R
- PPA548 PCIE_CAMERA_D2R_N
- PPA549 PCIE_CLK100M_SSD_N
- PPA550 PCIE_CLK100M_TBT_N
- PPA551 PCIE_SSD_D2R_N<0>
- PPA552 PCIE_SSD_D2R_P<0>

- PPA555 QR_SWITCH_EN
- PPA556 SAK_SSD_P1V2_EN
- PPA557 SAK_SSD_P1V8_EN
- PPA558 SAK_SSD_SR_P1V2_EN
- PPA559 SAK_SSD_SR_P1V8_EN
- PPA560 SMC_BT_PWR_EN
- PPA561 SMC_CLK32K_PMIC
- PPA562 SMC_OOB1_R2D_L
- PPA563 SMC_PROCHOT
- PPA564 SMC_SENSOR_PWR_EN
- PPA565 SMC_WAKE_SCI_L
- PPA566 SMC_WIFI_PWR_EN
- PPA567 AP_PCIE_WAKE_L
- PPA568 SSD_BOOT
- PPA569 SSD_SR_EN_L
- PPA570 SYSCLK_CLK12M_SMC
- PPA571 SYSCLK_CLK12M_SSD
- PPA572 SYSCLK_CLK24M_CAMERA
- PPA573 SYSCLK_CLK32K_PMIC
- PPA574 BT_DEV_WAKE
- PPA575 TPAD_SPI_CS_L
- PPA576 TPAD_SPI_INT_L

- PPA577 USB3_EXTA_D2R_N
- PPA578 USB3_EXTA_D2R_P
- PPA579 CPU_CATERR_L
- PPA580 E85_TEST_MODE_L
- PPA581 MEM_B_CS_L<1>
- PPA582 MEM_B_CKE<1>
- PPA583 MEM_B_CAB<1>
- PPA584 MEM_B_CAA<2>
- PPA585 MCP_RSVD_CK13 MAKE_BASE=TRUE = MCP_RSVD_CK13
- PPA586 MCP_DC_CV45 MAKE_BASE=TRUE = MCP_DC_CV45
- PPA587 MEM_B_CAA<3>
- PPA588 E85_LS_P<1>
- PPA589 E85_LS_N<1>
- PPA590 E85_LS_P<2>
- PPA591 E85_LS_N<2>
- PPA592 XDP_PCH_UART_SSD_L_BT_H
- PPA593 DP_E85SNK_HPD
- PPA594 XDP_USB_EXTA_OC_L
- PPA595 XDP_USB_EXTB_OC_L
- PPA596 USB_EXTA_P
- PPA597 USB_EXTA_N

- PPA5A0 BBPD_IPU_EN
- PPA5A1 BBPD_MISO_LS1V1
- PPA5A2 BBPD_PD_EN
- PPA5A3 BBPD_PD_SEL_CC2
- PPA5A4 BBPD_RPD_EN
- PPA5A5 BBPD_RX_L_TX_H
- PPA5A6 BBPD_SPI_MOSI
- PPA5A7 BBPD_VCONN1_EN
- PPA5A8 BBPD_VCONN2_EN
- PPA5A9 BBPD_VCONN_EN

- TP_ULX_DDR_VCCDDQG = TRUE TP_ULX_DDR_VCCDDQG
- SMC_DEBUGPRT_TX_L = TRUE SMC_DEBUGPRT_TX_L
- SMC_DEBUGPRT_RX_L = TRUE SMC_DEBUGPRT_RX_L

NO_TEST Nets

CPU/PCH

NO_TEST MAKE_BASE

- NC_USB3_EXTB_D2RP = TRUE TRUE NC_USB3_EXTB_D2RP
- NC_USB3_EXTB_D2RN = TRUE TRUE NC_USB3_EXTB_D2RN
- NC_USB3_EXTB_R2D_CP = TRUE TRUE NC_USB3_EXTB_R2D_CP
- NC_USB3_EXTB_R2D_CN = TRUE TRUE NC_USB3_EXTB_R2D_CN

NO_TEST MAKE_BASE

- NC_USB_IRP = TRUE TRUE NC_USB_IRP
- NC_USB_IRN = TRUE TRUE NC_USB_IRN
- NC_USB_EXTBP = TRUE TRUE NC_USB_EXTBP
- NC_USB_EXTBN = TRUE TRUE NC_USB_EXTBN

NO_TEST MAKE_BASE

- NC_USB_TPADP = TRUE TRUE NC_USB_TPADP
- NC_USB_TPADN = TRUE TRUE NC_USB_TPADN
- NC_PCIE_CLK100M_FWP = TRUE TRUE NC_PCIE_CLK100M_FWP
- NC_PCIE_CLK100M_FWN = TRUE TRUE NC_PCIE_CLK100M_FWN

- NC_SYS_ONEWIRE = TRUE TRUE NC_SYS_ONEWIRE
- NC_PMIC_CLK32K_XTALIN = TRUE TRUE NC_PMIC_CLK32K_XTALIN
- NC_SMC_GFX_OVERTEMP = TRUE TRUE NC_SMC_GFX_OVERTEMP
- NC_SMC_FAN_0_CTL = TRUE TRUE NC_SMC_FAN_0_CTL
- NC_SMC_FAN_4_CTL = TRUE TRUE NC_SMC_FAN_4_CTL

Unused nets with offpage

(Nets with offpages not used on this project)

- XDP_USB_EXTB_OC_L

SYNC_MASTER=J41_MLB	SYNC_DATE=10/24/2012
PAGE TITLE: Project FCT/NC/Aliases	
Apple Inc.	DRAWING NUMBER <SCH_NUM> SIZE D
	REVISION <E4LABEL>
NOTICE OF PROPRIETARY PROPERTY:	BRANCH <BRANCH>
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	PAGE 105 OF 130
	SHEET 64 OF 75

J92 Board-Specific Spacing & Physical Constraints

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,MEM_TERM	MM	16.5

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	TOP,BOTTOM	Y	=50_OHM_SE	=50_OHM_SE			
DEFAULT	ISL6,ISL8,ISL10	Y	=45_OHM_SE	=45_OHM_SE			
DEFAULT	ISL3,ISL4	Y	=45_OHM_SE	=45_OHM_SE			
DEFAULT	*	Y	100 MM	100 MM	10 MM	0 MM	0 MM
STANDARD	*	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT

Single-ended Physical Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.310 MM	0.260 MM			
27P4_OHM_SE	ISL6,ISL8,ISL10	Y	0.134 MM	0.134 MM			
27P4_OHM_SE	ISL3,ISL4	Y	0.190 MM	0.190 MM			
27P4_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.144 MM	0.144 MM			
40_OHM_SE	ISL6	Y	0.070 MM	0.070 MM			
40_OHM_SE	ISL8	Y	0.073 MM	0.073 MM			
40_OHM_SE	ISL10	Y	0.071 MM	0.071 MM			
40_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.120 MM	0.095 MM			
45_OHM_SE	ISL6	Y	0.057 MM	0.057 MM			
45_OHM_SE	ISL8	Y	0.058 MM	0.058 MM			
45_OHM_SE	ISL10	Y	0.057 MM	0.057 MM			
45_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.095 MM	0.095 MM			
50_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE_RF	TOP,BOTTOM	Y	0.225 MM	0.225 MM			
50_OHM_SE_RF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

Differential Pair Physical Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	TOP,BOTTOM	Y	0.147 MM	0.060 MM		0.100 MM	0.100 MM
70_OHM_DIFF	ISL6	Y	0.082 MM	0.082 MM	=STANDARD	0.080 MM	0.080 MM
70_OHM_DIFF	ISL8	Y	0.085 MM	0.085 MM	=STANDARD	0.080 MM	0.080 MM
70_OHM_DIFF	ISL10	Y	0.081 MM	0.081 MM	=STANDARD	0.080 MM	0.080 MM
70_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	TOP,BOTTOM	Y	0.099 MM	0.060 MM		0.100 MM	0.100 MM
85_OHM_DIFF	ISL6,ISL9,ISL11	Y	0.058 MM	0.058 MM	=STANDARD	0.100 MM	0.100 MM
85_OHM_DIFF	ISL8	Y	0.060 MM	0.060 MM	=STANDARD	0.100 MM	0.100 MM
85_OHM_DIFF	ISL10	Y	0.057 MM	0.057 MM	=STANDARD	0.100 MM	0.100 MM
85_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

Note: 80ohm copied from 85ohm (pending stack-up calcs)

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	TOP,BOTTOM	Y	0.099 MM	0.060 MM		0.100 MM	0.100 MM
80_OHM_DIFF	ISL6,ISL9,ISL11	Y	0.058 MM	0.058 MM	=STANDARD	0.100 MM	0.100 MM
80_OHM_DIFF	ISL8	Y	0.060 MM	0.060 MM	=STANDARD	0.100 MM	0.100 MM
80_OHM_DIFF	ISL10	Y	0.056 MM	0.056 MM	=STANDARD	0.101 MM	0.101 MM
80_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

Spacing Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.100 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1x_DIELECTRIC	TOP,BOTTOM	0.057 MM	?
1x_DIELECTRIC	ISL2,ISL11	0.057 MM	?
1x_DIELECTRIC	ISL3,ISL10	0.057 MM	?
1x_DIELECTRIC	ISL4,ISL9	0.054 MM	?
1x_DIELECTRIC	ISL5,ISL8	0.058 MM	?
1x_DIELECTRIC	ISL6,ISL7	0.051 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P075MM	*	0.075 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P075MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	BGA	P070MM_BGA

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P070MM_BGA	*			0.070 MM	5 MM		0.075 MM

SYNC_MASTER=J43_MLB SYNC_DATE=10/24/2012

PAGE TITLE
PCB Rule Definitions

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	0.100 MM	0.100 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?
CPU_AGTL	*	=STANDARD	?

Note: CPU_8MIL and CPU_ITP can be converted back to TABLE_SPACING_RULE once rdar://10308147 is resolved

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_8MIL	*	*	CPU_8MIL_2ANY

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_8MIL_2ANY	*	8 MIL	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_ITP	*	*	CPU_ITP_2ANY

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ITP_2ANY	*	=4x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	CPU_COMP	*	CPU_COMP_2SELF
CPU_COMP	*	*	CPU_COMP_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_COMP_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CPU_COMP_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_COMP_2SELF	*	=4x_DIELECTRIC	?
CPU_COMP_2OTHER	*	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_VCCSENSE	CPU_VCCSENSE	*	CPU_VCCSENSE_2SELF
CPU_VCCSENSE	*	*	CPU_VCCSENSE_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CPU_VCCSENSE_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE_2SELF	*	=4x_DIELECTRIC	?
CPU_VCCSENSE_2OTHER	*	=6x_DIELECTRIC	?

PCI-Express Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
CLK_PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

PCIE Clock Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	CLK_PCIE	*	CLK_PCIE_2SELF
CLK_PCIE	*	*	CLK_PCIE_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CLK_PCIE_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_2SELF	*	=4x_DIELECTRIC	?
CLK_PCIE_2OTHER	*	=6x_DIELECTRIC	?

CPU PCIE Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_CPU_TX	PCIE_CPU_TX	*	PCIE_TX2TX
PCIE_CPU_RX	PCIE_CPU_RX	*	PCIE_RX2RX
PCIE_CPU_TX	*_CPU_TX	*	PCIE_TX2OTHERTX
PCIE_CPU_RX	*_CPU_RX	*	PCIE_RX2OTHERRX
PCIE_CPU_TX	*_CPU_RX	*	PCIE_TX2RX
PCIE_CPU_RX	*_CPU_TX	*	PCIE_RX2TX
PCIE_CPU_TX	*_TX	*	PCIE_2OTHERHS
PCIE_CPU_RX	*_TX	*	PCIE_2OTHERHS
PCIE_CPU_TX	*_RX	*	PCIE_2OTHERHS
PCIE_CPU_RX	*_RX	*	PCIE_2OTHERHS
PCIE_CPU_TX	*	*	PCIE_2OTHER
PCIE_CPU_RX	*	*	PCIE_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_TX2TX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_RX2RX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_TX2OTHERTX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_RX2OTHERRX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_TX2RX	TOP,BOTTOM	=7x_DIELECTRIC	?
PCIE_RX2TX	TOP,BOTTOM	=7x_DIELECTRIC	?
PCIE_2OTHERHS	TOP,BOTTOM	=6x_DIELECTRIC	?
PCIE_2OTHER	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_TX2TX	*	=2.5x_DIELECTRIC	?
PCIE_RX2RX	*	=2.5x_DIELECTRIC	?
PCIE_TX2OTHERTX	*	=4x_DIELECTRIC	?
PCIE_RX2OTHERRX	*	=4x_DIELECTRIC	?
PCIE_TX2RX	*	=6x_DIELECTRIC	?
PCIE_RX2TX	*	=6x_DIELECTRIC	?
PCIE_2OTHERHS	*	=4x_DIELECTRIC	?
PCIE_2OTHER	*	=3x_DIELECTRIC	?

PCH PCIE Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_PCH_TX	PCIE_PCH_TX	*	PCIE_TX2TX
PCIE_PCH_RX	PCIE_PCH_RX	*	PCIE_RX2RX
PCIE_PCH_TX	*_PCH_TX	*	PCIE_TX2OTHERTX
PCIE_PCH_RX	*_PCH_RX	*	PCIE_RX2OTHERRX
PCIE_PCH_TX	*_PCH_RX	*	PCIE_TX2RX
PCIE_PCH_RX	*_PCH_TX	*	PCIE_RX2TX
PCIE_PCH_TX	*_TX	*	PCIE_2OTHERHS
PCIE_PCH_RX	*_TX	*	PCIE_2OTHERHS
PCIE_PCH_TX	*_RX	*	PCIE_2OTHERHS
PCIE_PCH_RX	*_RX	*	PCIE_2OTHERHS
PCIE_PCH_TX	*	*	PCIE_2OTHER
PCIE_PCH_RX	*	*	PCIE_2OTHER

SOURCE: 471984_Chief_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

Note: DisplayPort tables are on Page 113

CPU Net Properties

Note: 80ohm constraints are actually 85ohm

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
CPU_PECI	CPU_45S	CPU_COMP	CPU_PECI	6 32
PM_SYNC	CPU_45S	CPU_AGTL	PM_SYNC	
PM_MEM_PWRGD	CPU_45S	CPU_AGTL	PM_MEM_PWRGD	
	CPU_45S	CPU_ITP	XDP_DBRESET_L	16 17
	CPU_45S	CPU_ITP	XDP_CPU_PRDY_L	6 16
	CPU_45S	CPU_ITP	XDP_CPU_PREQ_L	6 16
	CPU_27P4S	CPU_COMP	EDP_COMP	
	CPU_27P4S	CPU_COMP	CPU_PEG_COMP	
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<0>	6
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<1>	6
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<2>	6
CPU_CFG	CPU_45S	CPU_ITP	CPU_CFG<2..0>	6 16
CPU_CFG3	CPU_45S	CPU_ITP	CPU_CFG<3>	6 16
CPU_CFG	CPU_45S	CPU_ITP	CPU_CFG<19..4>	6 16
CPU_CATERR_L	CPU_45S	CPU_AGTL	CPU_CATERR_L	6 31 64
	CPU_45S	CPU_AGTL	CPU_VCCIO_SEL	
CPU_PROCHOT_L	CPU_45S	CPU_AGTL	CPU_PROCHOT_L	6 31 32 44
CPU_PWRGD	CPU_45S	CPU_AGTL	CPU_PWRGD	6
PM_THRMTRIP_L	CPU_45S	CPU_8MIL	PM_THRMTRIP_L	15 31 32
DMI_CLK100M	CLK_PCIE_80D	CLK_PCIE	DMI_CLK100M_CPU_P	
DMI_CLK100M	CLK_PCIE_80D	CLK_PCIE	DMI_CLK100M_CPU_N	
DPLL_REF_CLK120M	CLK_PCIE_80D	CLK_PCIE	DPLL_REF_CLKP	
DPLL_REF_CLK120M	CLK_PCIE_80D	CLK_PCIE	DPLL_REF_CLKN	
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPCPU_CLK100M_P	
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPCPU_CLK100M_N	
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPXDP_CLK100M_P	
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPXDP_CLK100M_N	
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	XDP_CPU_CLK100M_P	
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	XDP_CPU_CLK100M_N	
XDP_TDI	CPU_45S	CPU_ITP	XDP_CPU_TDI	6 16
XDP_TDO	CPU_45S	CPU_ITP	XDP_CPU_TDO	6 16
XDP_TMS	CPU_45S	CPU_ITP	XDP_CPU_TMS	6 16
XDP_TCK	CPU_45S	CPU_ITP	XDP_CPU_TCK	6 16
XDP_TRST_L	CPU_45S	CPU_ITP	XDP_CPUPCH_TRST_L	6 12 16
XDP_BPM_L	CPU_45S	CPU_ITP	XDP_BPM_L<1..0>	6 16
	CPU_45S	CPU_ITP	XDP_BPM_L<7..2>	6 16
	CPU_45S	CPU_ITP	XDP_OBSDATA_B<3..0>	
(FSB_CPURST_L)	CPU_45S	CPU_ITP	XDP_CPURST_L	16
CPU_VCCSENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_VCCSENSE_P	8 44
CPU_VCCSENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_VCCSENSE_N	9 44
CPU_VCCIOSENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_VCCIOSENSE_P	
CPU_VCCIOSENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_VCCIOSENSE_N	
CPU_AXG_SENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_AXG_SENSE_P	
CPU_AXG_SENSE	SENSE_1TO1_P2MM	CPU_VCCSENSE	CPU_AXG_SENSE_N	
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VDDQ_SENSE_P	
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VDDQ_SENSE_N	
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_P	
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_N	
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_P	
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_N	
CPU_SVIDALERT_L	CPU_45S	CPU_COMP	CPU_VIDALERT_L	8 44
CPU_SVIDSCLK	CPU_45S	CPU_COMP	CPU_VIDSCLK	8 44
CPU_SVIDSOUT	CPU_45S	CPU_COMP	CPU_VIDSOUT	8 44
PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_C_P<3..0>	14 23
PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_C_N<3..0>	14 23
	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_P<3..0>	23 64
	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_N<3..0>	23 64
	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_C_P<3..0>	54
	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_C_N<3..0>	54
PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_P<3..0>	14 64
PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_N<3..0>	14 64
PCIE_CLK100M_SSD	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_SSD_P	12 54 64
PCIE_CLK100M_SSD	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_SSD_N	12 54 64

PCIe SSD

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I208 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_E85SNK_AUXCH_P	26 29
I206 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_E85SNK_AUXCH_N	26 29
I207	DP_80D	DP_AUX	DP_TBTSNK0_AUXCH_C_P	13 29
I205	DP_80D	DP_AUX	DP_TBTSNK0_AUXCH_C_N	13 29
I204 DP_TBT_ML	DP_80D	DP_TX	DP_TBTSNK1_ML_P<3..0>	
I203 DP_TBT_ML	DP_80D	DP_TX	DP_TBTSNK1_ML_N<3..0>	
I202	DP_80D	DP_TX	DP_TBTSNK1_ML_C_P<3..0>	
I201	DP_80D	DP_TX	DP_TBTSNK1_ML_C_N<3..0>	
I200 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_TBTSNK1_AUXCH_P	
I199 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_TBTSNK1_AUXCH_N	
I198	DP_80D	DP_AUX	DP_TBTSNK1_AUXCH_C_P	
I197	DP_80D	DP_AUX	DP_TBTSNK1_AUXCH_C_N	
DP_INT_ML	DP_80D	DP_TX	DP_INT_ML_P<3..0>	53
DP_INT_ML	DP_80D	DP_TX	DP_INT_ML_N<3..0>	53
	DP_80D	DP_TX	DP_INT_ML_C_P<3..0>	5 53
	DP_80D	DP_TX	DP_INT_ML_C_N<3..0>	5 53
I215 DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUX_CH_C_P	
I214 DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUX_CH_C_N	
DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUXCH_C_P	5 53
DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUXCH_C_N	5 53
	DP_80D	DP_AUX	DP_INT_AUX_P	53
	DP_80D	DP_AUX	DP_INT_AUX_N	53

DP

SYNC_MASTER=J92_DEVMLB SYNC_DATE=07/08/2014

PAGE TITLE

CPU Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PAGE 111 OF 130

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ICOMP	*	=4x_DIELECTRIC	?

E85 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
E85_HS_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
E85_LS_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
E85_HS	*	=6x_DIELECTRIC	?
E85_LS	*	=4x_DIELECTRIC	?
E85_CC	*	=4x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
E85_HS	TOP,BOTTOM	=10x_DIELECTRIC	?
E85_LS	TOP,BOTTOM	=6x_DIELECTRIC	?
E85_CC	TOP,BOTTOM	=6x_DIELECTRIC	?

SOURCE: 471984_Chief_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

UART Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
UART_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
UART	*	=2x_DIELECTRIC	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.8

USB 3.0 Interface Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_PCH_TX	USB3_PCH_TX	*	USB3_TX2TX
USB3_PCH_RX	USB3_PCH_RX	*	USB3_RX2RX
USB3_PCH_TX	*_PCH_TX	*	USB3_TX2OTHERTX
USB3_PCH_RX	*_PCH_RX	*	USB3_RX2OTHERRX
USB3_PCH_TX	*_PCH_RX	*	USB3_TX2RX
USB3_PCH_RX	*_PCH_TX	*	USB3_RX2TX
USB3_PCH_TX	*_TX	*	USB3_2OTHERHS
USB3_PCH_RX	*_TX	*	USB3_2OTHERHS
USB3_PCH_TX	*_RX	*	USB3_2OTHERHS
USB3_PCH_RX	*_RX	*	USB3_2OTHERHS
USB3_PCH_TX	*	*	USB3_2OTHER
USB3_PCH_RX	*	*	USB3_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_TX2TX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_RX2RX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_TX2OTHERTX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_RX2OTHERRX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_TX2RX	TOP,BOTTOM	=7x_DIELECTRIC	?
USB3_RX2TX	TOP,BOTTOM	=7x_DIELECTRIC	?
USB3_2OTHERHS	TOP,BOTTOM	=6x_DIELECTRIC	?
USB3_2OTHER	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_TX2TX	*	=2.5x_DIELECTRIC	?
USB3_RX2RX	*	=2.5x_DIELECTRIC	?
USB3_TX2OTHERTX	*	=4x_DIELECTRIC	?
USB3_RX2OTHERRX	*	=4x_DIELECTRIC	?
USB3_TX2RX	*	=6x_DIELECTRIC	?
USB3_RX2TX	*	=6x_DIELECTRIC	?
USB3_2OTHERHS	*	=4x_DIELECTRIC	?
USB3_2OTHER	*	=3x_DIELECTRIC	?

SOURCE: 471984_Cheif_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

PCH Net Properties

Note: 80ohm constraints are actually 85ohm

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES	GROUP
	SPI_45S	SPI	TPAD_SPI_MOSI	15 30	TP SPI
	SPI_45S	SPI	TPAD_SPI_MISO	15 30	
	SPI_45S	SPI	TPAD_SPI_CLK	15 30	
USB_BT	USB_80D	USB	USB_BT_P	14 22	Internal USB
USB_BT	USB_80D	USB	USB_BT_N	14 22	
	USB_80D	USB	USB_BT_R_P	22	
	USB_80D	USB	USB_BT_R_N	22	
USB_BT	USB_80D	USB	USB_BT_CONN_P		
USB_BT	USB_80D	USB	USB_BT_CONN_N		
USB_TPAD	USB_80D	USB	NC_USB_TPADP	14 64	
USB_TPAD	USB_80D	USB	NC_USB_TPADN	14 64	
USB_HPM	USB_85D	USB	USB_HPM_P	28 29	I/O Port Device USB
USB_HPM	USB_85D	USB	USB_HPM_N	28 29	
	USB_85D	USB	USB_HPM_R_P	26	
	USB_85D	USB	USB_HPM_R_N	26	
	USB_80D	USB	USB_EXTA_P	14 26 64	I/O Port Host USB
	USB_80D	USB	USB_EXTA_N	14 26 64	
	USB_85D	USB	USB_EXTA_F_P	26	
	USB_85D	USB	USB_EXTA_F_N	26	
USB3_EXTA_RX	USB_80D	USB3_PCH_RX	USB3_EXTA_D2R_P	14 29 64	
USB3_EXTA_RX	USB_80D	USB3_PCH_RX	USB3_EXTA_D2R_N	14 29 64	
	USB_80D	USB3_PCH_RX	USB3_EXTA_D2R_C_P		
	USB_80D	USB3_PCH_RX	USB3_EXTA_D2R_C_N		
USB3_EXTA_TX	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_P	29	
USB3_EXTA_TX	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_N	29	
	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_C_P	14 29	
	USB_80D	USB3_PCH_TX	USB3_EXTA_R2D_C_N	14 29	
	USB_80D	USB3_PCH_RX	USB3_EXTD_D2R_P	14 27	I/O Port DCI
	USB_80D	USB3_PCH_RX	USB3_EXTD_D2R_N	14 27	
	USB_80D	USB3_PCH_TX	USB3_EXTD_R2D_P	27 29	
	USB_80D	USB3_PCH_TX	USB3_EXTD_R2D_N	27 29	
	USB_80D	USB3_PCH_TX	USB3_EXTD_R2D_C_P	14 29	
	USB_80D	USB3_PCH_TX	USB3_EXTD_R2D_C_N	14 29	
	UART_45S	UART	SMC_DEBUGPRT_TX_L	26 31 32 64	
	UART_45S	UART	SMC_DEBUGPRT_RX_L	26 31 32 64	
USB_EXTB	USB_80D	USB	NC_USB_EXTBP	14 64	USB EXTB nets (Left USB port)
USB_EXTB	USB_80D	USB	NC_USB_EXTBN	14 64	
USB_EXTB	USB_80D	USB	USB2_EXTB_F_P		
USB_EXTB	USB_80D	USB	USB2_EXTB_F_N		
USB3_EXTB_RX	USB_80D	USB3_PCH_RX	NC_USB3_EXTB_D2RP	14 64	
USB3_EXTB_RX	USB_80D	USB3_PCH_RX	NC_USB3_EXTB_D2RN	14 64	
USB3_EXTB_TX	USB_80D	USB3_PCH_TX	USB3_EXTB_R2D_P		
USB3_EXTB_TX	USB_80D	USB3_PCH_TX	USB3_EXTB_R2D_N		
	USB_80D	USB3_PCH_TX	NC_USB3_EXTB_R2D_CP	14 64	
	USB_80D	USB3_PCH_TX	NC_USB3_EXTB_R2D_CN	14 64	
USB_EXTC	USB_80D	USB	USB2_EXTC_P		USB EXTC nets (Left USB port)
USB_EXTC	USB_80D	USB	USB2_EXTC_N		
USB_EXTC	USB_80D	USB	USB2_EXTC_F_P		
USB_EXTC	USB_80D	USB	USB2_EXTC_F_N		
USB3_EXTC_RX	USB_80D	USB3_PCH_RX	USB3_EXTC_D2R_P		
USB3_EXTC_RX	USB_80D	USB3_PCH_RX	USB3_EXTC_D2R_N		
USB3_EXTC_TX	USB_80D	USB3_PCH_TX	USB3_EXTC_R2D_P		
USB3_EXTC_TX	USB_80D	USB3_PCH_TX	USB3_EXTC_R2D_N		
	USB_80D	USB3_PCH_TX	USB3_EXTC_R2D_C_P		
	USB_80D	USB3_PCH_TX	USB3_EXTC_R2D_C_N		
USB3_SD_RX	USB_80D	USB3_PCH_RX	USB3RPCIE_SD_D2R_P		
USB3_SD_RX	USB_80D	USB3_PCH_RX	USB3RPCIE_SD_D2R_N		
USB3_SD_TX	USB_80D	USB3_PCH_TX	USB3RPCIE_SD_R2D_C_P		
USB3_SD_TX	USB_80D	USB3_PCH_TX	USB3RPCIE_SD_R2D_C_N		
	USB_80D	USB3_PCH_RX	USB3_SD_D2R_C_P		
	USB_80D	USB3_PCH_RX	USB3_SD_D2R_C_N		
	USB_80D	USB3_PCH_TX	USB3_SD_R2D_P		
	USB_80D	USB3_PCH_TX	USB3_SD_R2D_N		
PCH_USB_RBIAS	PCH_USB_RBIAS		PCH_USB_RBIAS	14	
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_PCH_P		
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_PCH_N		
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCH_CLK96M_DOT_P		
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCH_CLK96M_DOT_N		
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCH_CLK100M_SATA_P		
PCH_DIFFCLK_UNUSED_	CLK_PCIE_80D	CLK_PCIE	PCH_CLK100M_SATA_N		
	CPU_45S	CLK_PCIE	PCH_CLK14P3M_REFCLK		

SYNC_MASTER=DEV_MLB	SYNC_DATE=04/17/2014
PAGE TITLE	PCH Constraints 1
DRAWING NUMBER	<SCH_NUM>
SIZE	D
Apple Inc.	
REVISION	<E4LABEL>
BRANCH	<BRANCH>
PAGE	112 OF 130
SHEET	67 OF 75

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=3x_DIELECTRIC	?
CLK_LPC	*	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S_R_50S	TOP,BOTTOM	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE		
SMB_45S_R_50S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=4x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=4x_DIELECTRIC	?

XDP Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_ITP	*	=2:1_SPACING	?

DisplayPort

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2DP	*	=3x_DIELECTRIC	?
DP_2OTHERHS	*	=4x_DIELECTRIC	?
DP_2OTHER	*	=3x_DIELECTRIC	?
DP_AUX	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2DP	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_2OTHERHS	TOP,BOTTOM	=6x_DIELECTRIC	?
DP_2OTHER	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_AUX	TOP,BOTTOM	=4x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DP_TX	DP_TX	*	DP_2DP
DP_TX	*_TX	*	DP_2OTHERHS
DP_TX	*_RX	*	DP_2OTHERHS
DP_TX	*	*	DP_2OTHER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	14 31
LPC_FRAME_L	LPC_45S	LPC	LPC_FRAME_L	14 31
	LPC_45S	LPC	LPCPLUS_RESET_L	
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC	17 31
	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC_R	12 17
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS	
	CLK_LPC_45S	CLK_LPC	TP_LPC_CLK24M_LPCPLUS_R	12 62
SMBUS_PCH_CLK	SMB_45S_R_50S	SMB	SMBUS_PCH_CLK	14 16 34
SMBUS_PCH_DATA	SMB_45S_R_50S	SMB	SMBUS_PCH_DATA	14 16 34
SMBUS_PCH_0_CLK	SMB_45S_R_50S	SMB	SML_PCH_0_CLK	14 34
SMBUS_PCH_0_DATA	SMB_45S_R_50S	SMB	SML_PCH_0_DATA	14 34
SMBUS_SMC_1_S0_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SCL	14 31 34 36 53 72
SMBUS_SMC_1_S0_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SDA	14 31 34 36 53 72
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	12 40
	HDA_45S	HDA	HDA_BIT_CLK_R	12
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	12 40
	HDA_45S	HDA	HDA_SYNC_R	12
HDA_RST_L	HDA_45S	HDA	HDA_RST_R_L	12
	HDA_45S	HDA	HDA_RST_L	12 40
HDA_SDIN0	HDA_45S	HDA	HDA_SDIN0	12 40 64
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	12 40
	HDA_45S	HDA	HDA_SDOUT_R	12 17
PM_SUS_CLK	CLK_SLOW_45S	CLK_SLOW	PM_CLK32K_SUSCLK_R	13 32
	CLK_SLOW_45S	CLK_SLOW	SMC_CLK32K	
SPI_CLK	SPI_45S	SPI	SPI_CLK_R	14 37
	SPI_45S	SPI	SPI_CLK	37
SPI_MOSI	SPI_45S	SPI	SPI_MOSI_R	14 37
	SPI_45S	SPI	SPI_MOSI	37
SPI_MISO	SPI_45S	SPI	SPI_MISO	14 37
SPI_CS0	SPI_45S	SPI	SPI_CS0_R_L	14 37
	SPI_45S	SPI	SPI_CS0_L	37
	SPI_45S	SPI	SPI_SMC_CLK	31 37
	SPI_45S	SPI	SPI_SMC_MOSI	31 37
	SPI_45S	SPI	SPI_SMC_MISO	31 37
	SPI_45S	SPI	SPI_SMC_CS_L	31 37
	SPI_45S	SPI	SPI_MLB_CLK	37
	SPI_45S	SPI	SPI_MLB_MOSI	37
	SPI_45S	SPI	SPI_MLB_MISO	37
	SPI_45S	SPI	SPI_MLB_CS_L	37
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_P	22
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_N	22
	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_C_P	14 22
	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_C_N	14 22
PCIE_AP_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_P	14 22
PCIE_AP_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_N	14 22
I401	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_C_P	22
I402	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_C_N	22
PCIE_CLK100M_AP	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_AP_P	12 22
PCIE_CLK100M_AP	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_AP_N	12 22
I399	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_AP_C_P	22
I400	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_AP_C_N	22
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_P<3..0>	
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_N<3..0>	
	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_C_P<3..0>	14 62
	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_C_N<3..0>	14 62
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_P<3..0>	14 62
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_N<3..0>	14 62
	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_C_P<3..0>	
	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_C_N<3..0>	
PCIE_CLK100M_TBT	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_TBT_P	12 29 62 64
PCIE_CLK100M_TBT	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_TBT_N	12 29 62 64
XDP_TDI	PCH_45S	PCH_ITP	XDP_PCH_TDI	12 16
XDP_TDO	PCH_45S	PCH_ITP	XDP_PCH_TDO	12 16
XDP_TMS	PCH_45S	PCH_ITP	XDP_PCH_TMS	12 16
XDP_TCK	PCH_45S	PCH_ITP	XDP_PCH_TCK	12 16
PCIE_CAM	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_P	24 25
PCIE_CAM	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_N	24 25
	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_C_P	14 25
	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_C_N	14 25
PCIE_CAM	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_P	14 25 64
PCIE_CAM	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_N	14 25 64
	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_C_P	24 25
	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_C_N	24 25
PCIE_CLK100M_CAM	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_P	12 25
PCIE_CLK100M_CAM	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_N	12 25
	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P	24 25
	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N	24 25
I385 PCIE_CLK100M_DBG	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_DEBUG_P	
I386 PCIE_CLK100M_DBG	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_DEBUG_N	
I387 DP_TBT_ML	DP_80D	DP_TX	DP_ML_P<3..0>	
I388 DP_TBT_ML	DP_80D	DP_TX	DP_ML_N<3..0>	
I389 DP_TBT_ML	DP_80D	DP_TX	DP_ML_C_P<3..0>	
I390 DP_TBT_ML	DP_80D	DP_TX	DP_ML_C_N<3..0>	
I391 DP_TBT_ML	DP_80D	DP_TX	DP_ML_CONN_P<3..0>	
I392 DP_TBT_ML	DP_80D	DP_TX	DP_ML_CONN_N<3..0>	
I393 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_AUX_CH_P	
I394 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_AUX_CH_N	
I395 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_AUX_CH_C_P	
I396 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_AUX_CH_C_N	
I397 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_AUX_CH_CONN_P	
I398 DP_TBT_AUXCH	DP_80D	DP_AUX	DP_AUX_CH_CONN_N	

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SYSCLK_CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK_CLK32K_RTCX1	
SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	
	CLK_25M_45S	CLK_25M	SYSCLK_CLK24M_CAMERA	17 24 64
	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R	
	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP	
	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN	
	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN	24
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	
	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	
SYSCLK_CLK25M_XTAL	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1	
	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2	
	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R	
	CLK_25M_45S	CLK_25M	SDCLK_CLK25M_X2	
	CLK_25M_45S	CLK_25M	SDCLK_CLK25M_X2_R	
	CLK_25M_45S	CLK_25M	SDSCLK_CLK25M_X1	

E85 Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
E85_HS_XBAR_D2R	DP_85D	DP_TX	DP_TBTSNK0_ML_C_P<0>	5 29
E85_HS_XBAR_D2R	DP_85D	DP_TX	DP_TBTSNK0_ML_C_N<0>	5 29
E85_HS_XBAR_R2D	DP_85D	DP_TX	DP_TBTSNK0_ML_C_P<1>	5 29
E85_HS_XBAR_R2D	DP_85D	DP_TX	DP_TBTSNK0_ML_C_N<1>	5 29
E85_HS	DP_85D	DP_TX	DP_TBTSNK0_ML_C_P<3..2>	5 29
E85_HS	DP_85D	DP_TX	DP_TBTSNK0_ML_C_N<3..2>	5 29
	DP_85D	DP_TX	DP_E85SNK_ML_P<3..0>	27 29
	DP_85D	DP_TX	DP_E85SNK_ML_N<3..0>	27 29
	DP_85D	DP_TX	E85_HS_DP_ML0_P	27 29
	DP_85D	DP_TX	E85_HS_DP_ML0_N	27 29
	DP_85D	DP_TX	E85_HS_DP_ML1_P	27 29
	DP_85D	DP_TX	E85_HS_DP_ML1_N	27 29
E85_LS	E85_LS_85D	E85_LS	E85_LS_P<2..1>	26 29 64
E85_LS	E85_LS_85D	E85_LS	E85_LS_N<2..1>	26 29 64
	E85_LS_85D	E85_LS	E85_LS_MISSION_P	26
	E85_LS_85D	E85_LS	E85_LS_MISSION_N	26
		E85_CC	E85_CC1	29 60 64
		E85_CC	E85_CC2	29 60 64

Note: 80ohm constraints are actually 85ohm

SYNC_MASTER=DEV_MLB SYNC_DATE=04/17/2014

PAGE TITLE

PCH Constraints 2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DATA2OTHERMEM	*	=8x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=3x_DIELECTRIC	?
MEM_CMD2CMD	*	=3x_DIELECTRIC	?
MEM_CMD2CTRL	*	=3x_DIELECTRIC	?
MEM_CTRL2CTRL	*	=3x_DIELECTRIC	?
MEM_CLK2CLK	*	=6x_DIELECTRIC	?
MEM_2OTHERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	10000
MEM_2GND	*	=2x_DIELECTRIC	10000
MEM_2OTHER	*	=6x_DIELECTRIC	?

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_70D	MEM_TERM	MEM_70D
MEM_40S	MEM_TERM	MEM_50S

Note: changed MEM_TERM physical rule to MEM_70D from MEM_73D temporarily

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DATA_0	*	MEM_DQS2OWNDATA
MEM_A_DQS_1	MEM_A_DATA_1	*	MEM_DQS2OWNDATA
MEM_A_DQS_2	MEM_A_DATA_2	*	MEM_DQS2OWNDATA
MEM_A_DQS_3	MEM_A_DATA_3	*	MEM_DQS2OWNDATA
MEM_A_DQS_4	MEM_A_DATA_4	*	MEM_DQS2OWNDATA
MEM_A_DQS_5	MEM_A_DATA_5	*	MEM_DQS2OWNDATA
MEM_A_DQS_6	MEM_A_DATA_6	*	MEM_DQS2OWNDATA
MEM_A_DQS_7	MEM_A_DATA_7	*	MEM_DQS2OWNDATA
MEM_B_DQS_0	MEM_B_DATA_0	*	MEM_DQS2OWNDATA
MEM_B_DQS_1	MEM_B_DATA_1	*	MEM_DQS2OWNDATA
MEM_B_DQS_2	MEM_B_DATA_2	*	MEM_DQS2OWNDATA
MEM_B_DQS_3	MEM_B_DATA_3	*	MEM_DQS2OWNDATA
MEM_B_DQS_4	MEM_B_DATA_4	*	MEM_DQS2OWNDATA
MEM_B_DQS_5	MEM_B_DATA_5	*	MEM_DQS2OWNDATA
MEM_B_DQS_6	MEM_B_DATA_6	*	MEM_DQS2OWNDATA
MEM_B_DQS_7	MEM_B_DATA_7	*	MEM_DQS2OWNDATA

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	MEM_*	*	MEM_DATA2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTRL	*	MEM_CMD2CTRL
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*	MEM_*	*	MEM_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	*	*	MEM_2OTHER
MEM_A_DQS_1	*	*	MEM_2OTHER
MEM_A_DQS_2	*	*	MEM_2OTHER
MEM_A_DQS_3	*	*	MEM_2OTHER
MEM_A_DQS_4	*	*	MEM_2OTHER
MEM_A_DQS_5	*	*	MEM_2OTHER
MEM_A_DQS_6	*	*	MEM_2OTHER
MEM_A_DQS_7	*	*	MEM_2OTHER
MEM_B_DQS_0	*	*	MEM_2OTHER
MEM_B_DQS_1	*	*	MEM_2OTHER
MEM_B_DQS_2	*	*	MEM_2OTHER
MEM_B_DQS_3	*	*	MEM_2OTHER
MEM_B_DQS_4	*	*	MEM_2OTHER
MEM_B_DQS_5	*	*	MEM_2OTHER
MEM_B_DQS_6	*	*	MEM_2OTHER
MEM_B_DQS_7	*	*	MEM_2OTHER
MEM_A_DATA_0	*	*	MEM_2OTHER
MEM_A_DATA_1	*	*	MEM_2OTHER
MEM_A_DATA_2	*	*	MEM_2OTHER
MEM_A_DATA_3	*	*	MEM_2OTHER
MEM_A_DATA_4	*	*	MEM_2OTHER
MEM_A_DATA_5	*	*	MEM_2OTHER
MEM_A_DATA_6	*	*	MEM_2OTHER
MEM_A_DATA_7	*	*	MEM_2OTHER
MEM_B_DATA_0	*	*	MEM_2OTHER
MEM_B_DATA_1	*	*	MEM_2OTHER
MEM_B_DATA_2	*	*	MEM_2OTHER
MEM_B_DATA_3	*	*	MEM_2OTHER
MEM_B_DATA_4	*	*	MEM_2OTHER
MEM_B_DATA_5	*	*	MEM_2OTHER
MEM_B_DATA_6	*	*	MEM_2OTHER
MEM_B_DATA_7	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CLK	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DATA_0	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_1	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_2	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_3	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_4	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_5	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_6	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_A_DATA_7	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_0	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_1	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_2	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_3	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_4	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_5	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_6	MEM_*_DATA_*	*	MEM_2OTHERMEM
MEM_B_DATA_7	MEM_*_DATA_*	*	MEM_2OTHERMEM

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM_A_CLK_P<0>	7 20
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM_A_CLK_N<0>	7 20
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM_A_CLK_P<1>	7 20
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM_A_CLK_N<1>	7 20
MEM_A_CS0	MEM_40S	MEM_CTRL	MEM_A_CS_L<0>	7 20
MEM_A_CS1	MEM_40S	MEM_CTRL	MEM_A_CS_L<1>	7 20
MEM_A_ODT	MEM_40S	MEM_CTRL	MEM_A_ODT<0>	7 20 61
MEM_A_CKE0	MEM_40S	MEM_CMD	MEM_A_CKE<1..0>	7 20
MEM_A_CKE1	MEM_40S	MEM_CMD	MEM_A_CKE<3..2>	7 20
MEM_A_CMD0	MEM_40S	MEM_CMD	MEM_A_CAA<9..0>	7 20 61
MEM_A_CMD1	MEM_40S	MEM_CMD	MEM_A_CAB<9..0>	7 20 61
MEM_A_DQ_BYTE0	MEM_40S	MEM_A_DATA_0	MEM_A_DQ<7..0>	7 61
MEM_A_DQ_BYTE1	MEM_40S	MEM_A_DATA_1	MEM_A_DQ<15..8>	7 61
MEM_A_DQ_BYTE2	MEM_40S	MEM_A_DATA_2	MEM_A_DQ<23..16>	7 61
MEM_A_DQ_BYTE3	MEM_40S	MEM_A_DATA_3	MEM_A_DQ<31..24>	7 61
MEM_A_DQ_BYTE4	MEM_40S	MEM_A_DATA_4	MEM_A_DQ<39..32>	7 20 61
MEM_A_DQ_BYTE5	MEM_40S	MEM_A_DATA_5	MEM_A_DQ<47..40>	7 61
MEM_A_DQ_BYTE6	MEM_40S	MEM_A_DATA_6	MEM_A_DQ<55..48>	7 61
MEM_A_DQ_BYTE7	MEM_40S	MEM_A_DATA_7	MEM_A_DQ<63..56>	7 61
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM_A_DQS_P<0>	7 61
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM_A_DQS_N<0>	7 61
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM_A_DQS_P<1>	7 61
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM_A_DQS_N<1>	7 61
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM_A_DQS_P<2>	7 61
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM_A_DQS_N<2>	7 61
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM_A_DQS_P<3>	7 61
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM_A_DQS_N<3>	7 61
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM_A_DQS_P<4>	7 61
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM_A_DQS_N<4>	7 61
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM_A_DQS_P<5>	7 61
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM_A_DQS_N<5>	7 61
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM_A_DQS_P<6>	7 20 61
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM_A_DQS_N<6>	7 20 61
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM_A_DQS_P<7>	7 61
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM_A_DQS_N<7>	7 61
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM_B_CLK_P<0>	7 21 64
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM_B_CLK_N<0>	7 21 64
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM_B_CLK_P<1>	7 21
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM_B_CLK_N<1>	7 21
MEM_B_CS0	MEM_40S	MEM_CTRL	MEM_B_CS_L<0>	7 21 64
MEM_B_CS1	MEM_40S	MEM_CTRL	MEM_B_CS_L<1>	7 21 64
MEM_B_ODT	MEM_40S	MEM_CTRL	MEM_B_ODT<0>	7 21 61
MEM_B_CKE0	MEM_40S	MEM_CMD	MEM_B_CKE<1..0>	7 21 64
MEM_B_CKE1	MEM_40S	MEM_CMD	MEM_B_CKE<3..2>	7 21
MEM_B_CMD0	MEM_40S	MEM_CMD	MEM_B_CAA<9..0>	7 21 61 64
MEM_B_CMD1	MEM_40S	MEM_CMD	MEM_B_CAB<9..0>	7 21 61 64
MEM_B_DQ_BYTE0	MEM_40S	MEM_B_DATA_0	MEM_B_DQ<7..0>	7 61 64
MEM_B_DQ_BYTE1	MEM_40S	MEM_B_DATA_1	MEM_B_DQ<15..8>	7 61
MEM_B_DQ_BYTE2	MEM_40S	MEM_B_DATA_2	MEM_B_DQ<23..16>	7 61
MEM_B_DQ_BYTE3	MEM_40S	MEM_B_DATA_3	MEM_B_DQ<31..24>	7 61
MEM_B_DQ_BYTE4	MEM_40S	MEM_B_DATA_4	MEM_B_DQ<39..32>	7 21 61
MEM_B_DQ_BYTE5	MEM_40S	MEM_B_DATA_5	MEM_B_DQ<47..40>	7 61
MEM_B_DQ_BYTE6	MEM_40S	MEM_B_DATA_6	MEM_B_DQ<55..48>	7 61
MEM_B_DQ_BYTE7	MEM_40S	MEM_B_DATA_7	MEM_B_DQ<63..56>	7 61
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM_B_DQS_P<0>	7 61 64
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM_B_DQS_N<0>	7 61 64
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM_B_DQS_P<1>	7 61
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM_B_DQS_N<1>	7 61
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM_B_DQS_P<2>	7 61
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM_B_DQS_N<2>	7 61
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM_B_DQS_P<3>	7 61
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM_B_DQS_N<3>	7 61
MEM_B_DQS4	MEM_70D	MEM_B_DQS_4	MEM_B_DQS_P<4>	7 61
MEM_B_DQS4	MEM_70D	MEM_B_DQS_4	MEM_B_DQS_N<4>	7 61
MEM_B_DQS5	MEM_70D	MEM_B_DQS_5	MEM_B_DQS_P<5>	7 61
MEM_B_DQS5	MEM_70D	MEM_B_DQS_5	MEM_B_DQS_N<5>	7 61
MEM_B_DQS6	MEM_70D	MEM_B_DQS_6	MEM_B_DQS_P<6>	7 21 61
MEM_B_DQS6	MEM_70D	MEM_B_DQS_6	MEM_B_DQS_N<6>	7 21 61
MEM_B_DQS7	MEM_70D	MEM_B_DQS_7	MEM_B_DQS_P<7>	7 61
MEM_B_DQS7	MEM_70D		MEM_B_DQS_N<7>	7 61
		MEM_PWR	PP1V2_S3	8 10 19 20 21 46 52 59 60
		MEM_PWR	PPVREF_S3_MEM_VREFCA	18 19 20 21 69
		MEM_PWR	PPVREF_S3_MEM_VREFDQ_A	18 19 20
		MEM_PWR	PPVREF_S3_MEM_VREFCA	18 19 20 21 69
		MEM_PWR	PPVREF_S3_MEM_VREFDQ_B	18 19 21

SYNC MASTER=J92 LS MLB SYNC DATE=05/07/2013

PAGE TITLE

Memory Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

NAND BUS CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
NAND_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
NAND_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
NAND_DQS	*	0.100 MM	?
NAND_IO	*	0.100 MM	?
NAND_CMD	*	0.100 MM	?

NAND:

DQS_P/N MAX LENGTH 3"
IO<7..0> SIGNALS SHOULD MATCH +/- 50MIL FROM DQS_P/N
IO<7..0> AND ASSOCIATED DQS_P/N ROUTE ON SAME LAYER. NO MORE THAN 2 VIA TRANSITIONS.
NCE<7..0>,ALE,CLE SHOULD MATCH +/- 250MIL FROM NWE
DQS_P/N & NRE_P/N SHOULD MATCH +/- 100MIL FROM NWE

NAND NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
ANI0_IO	NAND_45S	NAND_IO	ANI0_IO<7..0>	55 58
DP_ANI0_DQS	NAND_85D	NAND_DQS	ANI0_DQS_P	55 58
DP_ANI0_DQS	NAND_85D	NAND_DQS	ANI0_DQS_N	55 58
DP_ANI0_NRE	NAND_85D	NAND_DQS	ANI0_NRE_P	55 58
DP_ANI0_NRE	NAND_85D	NAND_DQS	ANI0_NRE_N	55 58
ANI0_NWE	NAND_45S	NAND_CMD	ANI0_NWE	55 58
ANI0_NCE	NAND_45S	NAND_CMD	ANI0_NCE<3..0>	55 58
ANI0_ALE	NAND_45S	NAND_CMD	ANI0_ALE	55 58
ANI0_CLE	NAND_45S	NAND_CMD	ANI0_CLE	55 58
ANI1_IO	NAND_45S	NAND_IO	ANI1_IO<7..0>	55 58
DP_ANI1_DQS	NAND_85D	NAND_DQS	ANI1_DQS_P	55 58
DP_ANI1_DQS	NAND_85D	NAND_DQS	ANI1_DQS_N	55 58
DP_ANI1_NRE	NAND_85D	NAND_DQS	ANI1_NRE_P	55 58
DP_ANI1_NRE	NAND_85D	NAND_DQS	ANI1_NRE_N	55 58
ANI1_NWE	NAND_45S	NAND_CMD	ANI1_NWE	55 58
ANI1_NCE	NAND_45S	NAND_CMD	ANI1_NCE<3..0>	55 58
ANI1_ALE	NAND_45S	NAND_CMD	ANI1_ALE	55 58
ANI1_CLE	NAND_45S	NAND_CMD	ANI1_CLE	55 58
ANI2_IO	NAND_45S	NAND_IO	ANI2_IO<7..0>	
DP_ANI2_DQS	NAND_85D	NAND_DQS	ANI2_DQS_P	
DP_ANI2_DQS	NAND_85D	NAND_DQS	ANI2_DQS_N	
DP_ANI2_NRE	NAND_85D	NAND_DQS	ANI2_NRE_P	
DP_ANI2_NRE	NAND_85D	NAND_DQS	ANI2_NRE_N	
ANI2_NWE	NAND_45S	NAND_CMD	ANI2_NWE	
ANI2_NCE	NAND_45S	NAND_CMD	ANI2_NCE<3..0>	
ANI2_ALE	NAND_45S	NAND_CMD	ANI2_ALE	
ANI2_CLE	NAND_45S	NAND_CMD	ANI2_CLE	
ANI3_IO	NAND_45S	NAND_IO	ANI3_IO<7..0>	
DP_ANI3_DQS	NAND_85D	NAND_DQS	ANI3_DQS_P	
DP_ANI3_DQS	NAND_85D	NAND_DQS	ANI3_DQS_N	
DP_ANI3_NRE	NAND_85D	NAND_DQS	ANI3_NRE_P	
DP_ANI3_NRE	NAND_85D	NAND_DQS	ANI3_NRE_N	
ANI3_NWE	NAND_45S	NAND_CMD	ANI3_NWE	
ANI3_NCE	NAND_45S	NAND_CMD	ANI3_NCE<3..0>	
ANI3_ALE	NAND_45S	NAND_CMD	ANI3_ALE	
ANI3_CLE	NAND_45S	NAND_CMD	ANI3_CLE	

NAND NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
ANI4_IO	NAND_45S	NAND_IO	ANI4_IO<7..0>	
DP_ANI4_DQS	NAND_85D	NAND_DQS	ANI4_DQS_P	
DP_ANI4_DQS	NAND_85D	NAND_DQS	ANI4_DQS_N	
DP_ANI4_NRE	NAND_85D	NAND_DQS	ANI4_NRE_P	
DP_ANI4_NRE	NAND_85D	NAND_DQS	ANI4_NRE_N	
ANI4_NWE	NAND_45S	NAND_CMD	ANI4_NWE	
ANI4_NCE	NAND_45S	NAND_CMD	ANI4_NCE<3..0>	
ANI4_ALE	NAND_45S	NAND_CMD	ANI4_ALE	
ANI4_CLE	NAND_45S	NAND_CMD	ANI4_CLE	
ANI5_IO	NAND_45S	NAND_IO	ANI5_IO<7..0>	
DP_ANI5_DQS	NAND_85D	NAND_DQS	ANI5_DQS_P	
DP_ANI5_DQS	NAND_85D	NAND_DQS	ANI5_DQS_N	
DP_ANI5_NRE	NAND_85D	NAND_DQS	ANI5_NRE_P	
DP_ANI5_NRE	NAND_85D	NAND_DQS	ANI5_NRE_N	
ANI5_NWE	NAND_45S	NAND_CMD	ANI5_NWE	
ANI5_NCE	NAND_45S	NAND_CMD	ANI5_NCE<3..0>	
ANI5_ALE	NAND_45S	NAND_CMD	ANI5_ALE	
ANI5_CLE	NAND_45S	NAND_CMD	ANI5_CLE	
ANI6_IO	NAND_45S	NAND_IO	ANI6_IO<7..0>	55 58
DP_ANI6_DQS	NAND_85D	NAND_DQS	ANI6_DQS_P	55 58
DP_ANI6_DQS	NAND_85D	NAND_DQS	ANI6_DQS_N	55 58
DP_ANI6_NRE	NAND_85D	NAND_DQS	ANI6_NRE_P	55 58
DP_ANI6_NRE	NAND_85D	NAND_DQS	ANI6_NRE_N	55 58
ANI6_NWE	NAND_45S	NAND_CMD	ANI6_NWE	55 58
ANI6_NCE	NAND_45S	NAND_CMD	ANI6_NCE<3..0>	55 58
ANI6_ALE	NAND_45S	NAND_CMD	ANI6_ALE	55 58
ANI6_CLE	NAND_45S	NAND_CMD	ANI6_CLE	55 58
ANI7_IO	NAND_45S	NAND_IO	ANI7_IO<7..0>	55 58
DP_ANI7_DQS	NAND_85D	NAND_DQS	ANI7_DQS_P	55 58
DP_ANI7_DQS	NAND_85D	NAND_DQS	ANI7_DQS_N	55 58
DP_ANI7_NRE	NAND_85D	NAND_DQS	ANI7_NRE_P	55 58
DP_ANI7_NRE	NAND_85D	NAND_DQS	ANI7_NRE_N	55 58
ANI7_NWE	NAND_45S	NAND_CMD	ANI7_NWE	55 58
ANI7_NCE	NAND_45S	NAND_CMD	ANI7_NCE<3..0>	55 58
ANI7_ALE	NAND_45S	NAND_CMD	ANI7_ALE	55 58
ANI7_CLE	NAND_45S	NAND_CMD	ANI7_CLE	55 58

MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_2OTHER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_2OTHER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	*	=2x_DIELECTRIC	?
S2_DQS2OWNDATA	*	=2x_DIELECTRIC	?
S2_CMD2CMD	*	=2x_DIELECTRIC	?
S2_CMD2CTRL	*	=2x_DIELECTRIC	?
S2_CTRL2CTRL	*	=2x_DIELECTRIC	?
S2_2OTHERMEM	*	=4x_DIELECTRIC	?
S2MEM_2PWR	*	=2x_DIELECTRIC	?
S2MEM_2GND	*	=2x_DIELECTRIC	?
S2MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_DQS2OWNDATA	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_2OTHERMEM	TOP,BOTTOM	=6x_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_2OTHER
S2_MEM_DQS*	*	*	S2MEM_2OTHER
S2_MEM_CMD	*	*	S2MEM_2OTHER
S2_MEM_CTRL	*	*	S2MEM_2OTHER
S2_MEM_CLK	*	*	S2MEM_2OTHER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS2OWNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS2OWNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	24 25
	S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	24 25
	S2_MEM_CNTL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	24 25
I101	S2_MEM_CNTL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	24 25 64
I102		S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT	25
I103	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	24 25
I104	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	24 25
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	24 25
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	24 25
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	24 25
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	24 25
I106	S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	24 25
	S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	24 25
I108	S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	24 25
I107	S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	24 25
I109	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	24 25
I110	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	24 25
I147	S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	24 25 64
	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>	24 25
	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>	24 25
I127	MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	24 25
I128	MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	24 25
I129		MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	25 53
I130		MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	25 53
I134	MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	24 25
I133	MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	24 25
I132		MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	25 53
I131		MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	25 53
I145			S2_MEM_PWR	PP1V35_CAM	24 25
I146			S2_MEM_PWR	PP0V675_CAM_VREF	24 25
I148			S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	25

SYNC MASTER=J92 DEVMLB SYNC DATE=08/01/2013

PAGE TITLE

Camera Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1TO1_DIFFPAIR	*	=STANDARD	0.1 MM	=STANDARD	=STANDARD	0.1 MM	0.1 MM
2TO1_DIFFPAIR	*	=STANDARD	0.2 MM	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SMBUS_SMC_0_S0_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_0_S0_SCL	31 34 53
SMBUS_SMC_0_S0_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_0_S0_SDA	31 34 53
SMBUS_SMC_1_S0_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SCL	14 31 34 36 53 68
SMBUS_SMC_1_S0_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SDA	14 31 34 36 53 68
SMBUS_SMC_2_S3_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_2_G3_SCL	31 34 46 48
SMBUS_SMC_2_S3_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_2_G3_SDA	31 34 46 48
SMBUS_SMC_2_S3_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_2_G3_SCL_R	
SMBUS_SMC_2_S3_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_2_G3_SDA_R	
SMBUS_SMC_3_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_3_SCL	30 31 34 40
SMBUS_SMC_3_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_3_SDA	30 31 34 40
SMBUS_SMC_5_G3_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_5_G3_SCL	31 34 41 43
SMBUS_SMC_5_G3_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_5_G3_SDA	31 34 41 43

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SENSE_DIFFPAIR	1TO1_DIFFPAIR		CHGR_CSI_P	43
SENSE_DIFFPAIR	1TO1_DIFFPAIR		CHGR_CSI_N	43
	1TO1_DIFFPAIR		CHGR_CSI_R_P	43
	1TO1_DIFFPAIR		CHGR_CSI_R_N	43
SENSE_DIFFPAIR	1TO1_DIFFPAIR		CHGR_CSO_P	43
SENSE_DIFFPAIR	1TO1_DIFFPAIR		CHGR_CSO_N	43
	1TO1_DIFFPAIR		CHGR_CSO_R_P	43
	1TO1_DIFFPAIR		CHGR_CSO_R_N	43

SYNC_MASTER=J92_DEVMLB SYNC_DATE=09/11/2013

PAGE TITLE

SMC Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1TO1_45S	*	=1TO1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR
SENSE_1TO1_P2MM	*	=1TO1_DIFFPAIR	0.200 MM	=45_OHM_SE	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR
SENSE_1TO1_P3MM	*	=1TO1_DIFFPAIR	0.300 MM	=45_OHM_SE	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR
THERM_1TO1_45S	*	=1TO1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR
SPKR_DIFFPAIR	*	=1TO1_DIFFPAIR	0.400 MM	0.100 MM	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=1:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	CLK_PCIE	*	GND_P2MM
GND	PCIE*	*	GND_P2MM
GND	SATA*	*	GND_P2MM
GND	USB*	*	GND_P2MM
GND	LVDS*	*	GND_P2MM
SB_POWER	CLK_PCIE	*	PWR_P2MM
SB_POWER	SATA*	*	PWR_P2MM
SB_POWER	SATA*	*	PWR_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	10000
PWR_P2MM	*	0.20 MM	10000

RF Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
RF_50S	*	=50_OHM_SE_RF	=50_OHM_SE_RF	=50_OHM_SE_RF	=50_OHM_SE_RF	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
RF	*	0.15 MM	?

J92 MLB Specific Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_GAIN_P	35 36
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_GAIN_N	35 36
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_COMPUTING_N	35
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_COMPUTING_P	35
I358	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	INLET_THMSNS_D1_P	36
I359	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	INLET_THMSNS_D1_N	36
I378	SENSE_DIFFPAIR	SENSE_1TO1_P3MM	SENSE	ISNS_1V2_S3_P	46
I379	SENSE_DIFFPAIR	SENSE_1TO1_P3MM	SENSE	ISNS_1V2_S3_N	46
I380	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_LCDBKLT_P	49
I381	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_LCDBKLT_N	49
I382	SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	ISNS_1V05_SUS_P	48
I383	SENSE_DIFFPAIR	SENSE_1TO1_P2MM	SENSE	ISNS_1V05_SUS_N	48
I384	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUVR_ISNS1_P	45
I385	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUVR_ISNS1_N	45
I386	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUVR_ISNS2_P	45
I387	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUVR_ISNS2_N	45
I370	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUTHMSNS_D2_P	36
I371	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	CPUTHMSNS_D2_N	36
		1TO1_DIFFPAIR	AUDIO	MAX98300_R_P	
		1TO1_DIFFPAIR	AUDIO	MAX98300_R_N	
	SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_ROUT1_P	39 40
	SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_ROUT1_N	39 40
I373	SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_ROUT2_P	39 40
I372	SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_ROUT2_N	39 40
I374	SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_LOUT1_P	38 40
I375	SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_LOUT1_N	38 40
I376	SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_LOUT2_P	38 40
I377	SPKR_OUT	SPKR_DIFFPAIR	AUDIO	SPKRAMP_LOUT2_N	38 40
			SB_POWER	PP3V3_S5	8 11 13 15 16 17 22 33 37 46 47 51 59 60 75
			SB_POWER	PP3V3_S0	8 11 12 13 15 17 18 23 24 29 32 33 34 35 36 40 46 47 53 60 75
			GND	GND	
I346		RF_50S	RF	RF_A_0_DIPLEXER	22
I347		RF_50S	RF	RF_A_0_MATCH	22
I349		RF_50S	RF	RF_G_0_DIPLEXER	22
I348		RF_50S	RF	RF_G_0_MATCH	22
I350		RF_50S	RF	RF_0_ANT	22
I351		RF_50S	RF	RF_0_ANT_MATCH_T	22
I352		RF_50S	RF	RF_A_1_DIPLEXER	22
I353		RF_50S	RF	RF_A_1_MATCH	22
I355		RF_50S	RF	RF_G_1_DIPLEXER	22
I354		RF_50S	RF	RF_G_1_MATCH	22
I356		RF_50S	RF	RF_1_ANT	22
I357		RF_50S	RF	RF_1_ANT_MATCH_T	22
I390	DP_EXT_ML	DP_80D	DP_TX	DP_EXT_ML_P<1..0>	
I391	DP_EXT_ML	DP_80D	DP_TX	DP_EXT_ML_N<1..0>	
I389		DP_80D	DP_TX	DP_EXT_ML_C_P<1..0>	
I388		DP_80D	DP_TX	DP_EXT_ML_C_N<1..0>	
I392	USB_EXTA	USB_80D	USB	DPRUSB_EXTA_P	
I393	USB_EXTA	USB_80D	USB	DPRUSB_EXTA_N	
I396	USB3_EXTA_RX	USB_80D	USB3_PCH_RX	DPRUSB3_EXTA_D2R_P	
I397	USB3_EXTA_RX	USB_80D	USB3_PCH_RX	DPRUSB3_EXTA_D2R_N	
I395	USB3_EXTA_TX	USB_80D	USB3_PCH_TX	DPRUSB3_EXTA_R2D_P	
I394	USB3_EXTA_TX	USB_80D	USB3_PCH_TX	DPRUSB3_EXTA_R2D_N	

SYNC_MASTER=J92_DEVMLB SYNC_DATE=04/08/2014

PAGE TITLE

Project Specific Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Change List:
<radar://component/XXXXXX> J92 HW EE SCHEMATIC | PROTO 0

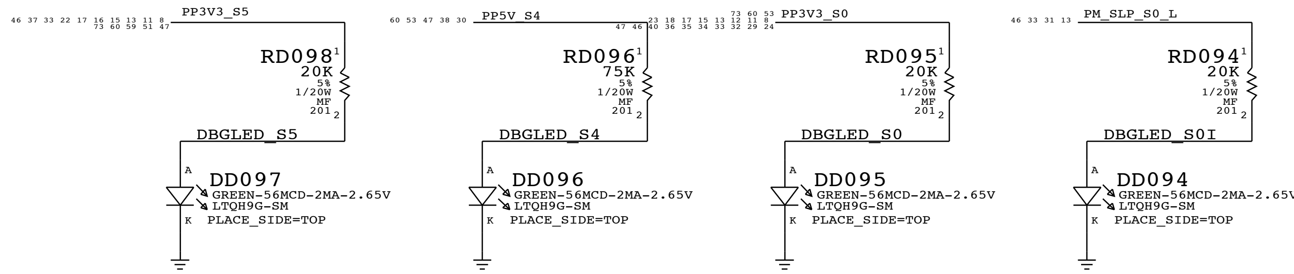
Kismet:
<afp://kismet.apple.com/Kismet-Projects/J92/>

Useful Wiki Links:
Schematic Conventions - <https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions>
Schematic Design Wiki - <https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design>

MobileMac HW Radar:
<radar://component/497591> MobileMac HW | Task
<radar://component/497587> MobileMac HW | Schematic
<radar://component/497585> MobileMac HW | New Bugs
<radar://component/497588> MobileMac HW | Layout
<radar://component/497590> MobileMac HW | Investigation
<radar://component/497589> MobileMac HW | Architecture

Other Info:
Page Allocations - <radar:11791318> 2012 Schematic Page Allocations

Power State Debug LEDs



SYNC_MASTER=MASTER SYNC_DATE=MASTER

PAGE TITLE

Debug Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>